MERN STACK DEVELOPMENT

SEMESTER 1 • M.Sc. (CA & IT)

Practical Book

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SEMESTER	1
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ACADEMIC PRACTICAL ASSESSMENT INDEX

Ganpat University | Department of Computer Science | M.Sc[CA & IT] - 1 (P21A1MSD)

No.	Practical Title & Core Concept	Technology Domain	Date	Page	Status
1	**Functional Components, JSX & Reusability** *Build a ReactApplicationusing FunctionalComponents, JSX and Component Reusability Concepts*	React.js / Frontend Architecture	*24/10/2026*	3	✔ Evaluated
2	**Props, State Management & Event Handling** *DevelopaReact ApplicationimplementingProps,State Management and Event Handling*	React.js / State & Interactions	*24/10/2026*	11	✔ Evaluated
3	**Forms, Controlled Components & Validation** *Create a Student Management System using React Forms, Controlled Components and Validation*	React.js / Forms & Data Integrity	*24/10/2026*	21	✔ Evaluated
4	**Single Page Application with React Router v6** *DevelopaSinglePage Application(SPA)usingReactRouter for navigation between multiple pages*	React.js / Routing & SPA Architecture	*24/10/2026*	32	✔ Evaluated
5	**Express.js Routing and CRUD Operations** *Create a backend serverusingExpress.jsimplementing routing and CRUD operation*	Backend Development / Express.js	*24/10/2026*	41	✔ Evaluated
6	**Student Records RESTful APIs with Postman Testing** *BuildRESTful APIsforStudentRecordsManagementusing Express.js and Postman testing*	Backend Development / REST Architecture	*24/10/2026*	51	✔ Evaluated
7	**Employee Management API with MongoDB Atlas** *Developan EmployeeManagementAPIintegrating Express.js with MongoDB Atlas*	Full-Stack Data Persistence / MongoDB	*24/10/2026*	62	✔ Evaluated
8	**Express Middleware Pipeline & Request Handling** *Create a backendserver usingExpress.jsimplementing routing, middleware and request handling*	Backend Architecture / Middleware Systems	*24/10/2026*	71	✔ Evaluated
9	**File Upload with Multer & MongoDB Metadata** *Createa file uploadmoduleusingMulterandstorefile metadata in MongoDB*	Full-Stack Media Handling / Multer & NoSQL	*24/10/2026*	80	✔ Evaluated
10	**Complete Full-Stack MERN Integration** *Develop a completeMERN applicationintegrating React Frontend, Express Backend and MongoDB Database*	Full-Stack Web Development / MERN Architecture	*24/10/2026*	91	✔ Evaluated
11	**Blog Management System (JWT Auth & Role-Based Access)** *Implementuser authenticationand authorizationina Node.js/Express application using JWT and role-based access control*	Security & Enterprise Systems / Auth & RBAC	*24/10/2026*	100	✔ Evaluated
12	**Real-Time Chat Application with Socket.IO & React** *Build a real-time chat application using Socket.IO and React*	Real-Time Systems / WebSockets & Socket.IO	*24/10/2026*	111	✔ Evaluated
13	**Capstone Mini Project: DocuVault Enterprise DMS** *Developan EnterpriseDocument ManagementSystem (DocuVault DMS) Capstone Mini Project*	Capstone Mini Project / Enterprise Full-Stack MERN	*24/10/2026*	122	✔ Evaluated

PRACTICAL 1

React.js / Frontend Architecture

Build a React Application using Functional Components, JSX and Component Reusability Concepts

1. Practical Aim & Theoretical Definition

Aim & Theoretical Concept

The objective of this practical is to build a modular React application leveraging modern Functional Components, JSX (JavaScript XML) syntax, and Component Reusability design patterns.

Core Theoretical Concepts:

- **Functional Components:** ***JavaScript functions that accept an arbitrary input called props and return a React element describing what should appear on the screen. Unlike legacy class components, functional components are lightweight, stateless by default, and seamlessly integrate with React Hooks.***
- **JSX Syntax:** ***A syntax extension to JavaScript that allows writing HTML-like structures directly inside JavaScript code. JSX enforces strict rules: tags must be closed, single root elements or fragments (<>... </>) must enclose returns, class is written as className, and JavaScript expressions are evaluated inside curly braces { }.***
- **Component Reusability & Composition:** ***Designing generalized, atomic components (Buttons, Badges, Cards, Headers) that can be reused across different views with varying props (variants, sizes, icons, children composition) adhering to the DRY (Don't Repeat Yourself) principle.***

2. Step-by-Step Commands (Installation & Execution)

1. Navigate to the Practical Directory

```
$ cd practicals/p1
```

Action: *Switches the working directory to the Practical 1 project folder.*

2. Install Dependencies

```
$ npm install
```

Action: *Installs every package declared in package.json into node_modules.*

3. Run Local Development Server

```
$ npm run dev
```

Action: *Starts the Vite dev server with Hot Module Replacement at http://localhost:5174.*

4. Build for Production

```
npm run build
$
```

Action: *Bundles optimised static assets into the /dist directory.*

3. Project Directory & File Structure

```
practical-01/
├── src/
│   ├── components/
│   │   ├── Badge.jsx
│   │   ├── Button.jsx
│   │   ├── Card.jsx
│   │   └── Header.jsx
│   ├── App.jsx
│   ├── index.css
│   └── main.jsx
├── index.html
├── package-lock.json
├── package.json
├── README.md
├── run.bat
└── vite.config.js
```

File / Directory Path	Architectural Purpose & Responsibility
src/components/Badge.jsx	***Reusable pill badge with success / warning / info / danger themes.***
src/components/Button.jsx	***Configurable button supporting variant, size, disabled and onClick props.***
src/components/Card.jsx	***Composable card with header, children body, badge and action buttons.***
src/components/Header.jsx	***Top header bar with branding and the student identity badge.***
src/App.jsx	***Root React component orchestrating state, layout and child components.***
src/index.css	***Global styles, design tokens and component styling.***
src/main.jsx	***Mounts the root React component into the DOM with ReactDOM.createRoot().***
index.html	***HTML entry point containing the #root mount node.***
package-lock.json	***Exact dependency tree lockfile generated by npm.***
package.json	***Project metadata, scripts and dependency declarations.***
README.md	***Project documentation.***
run.bat	***Windows launcher script that installs and starts the project.***
vite.config.js	***Vite build and dev-server configuration.***

4. Modular Source Code Implementation

Complete source code of the project, listed exactly as it exists in `practicals/p1/`:

src/main.jsx

JAVASCRIPT (JSX)

```jsx
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode><App /></React.StrictMode>
);
```

src/App.jsx

JAVASCRIPT (JSX)

```jsx
import React from 'react';
import Header from './components/Header';
import Card from './components/Card';
import Button from './components/Button';
import Badge from './components/Badge';

function App() {
  const handleEnroll = (courseName) => {
    alert(`Successfully enrolled in: ${courseName}`);
  };

  const courseList = [
    {
      id: 1,
      title: "MERN Stack Engineering",
      subtitle: "Full-Stack Web Development",
      badgeText: "Bestseller",
      badgeType: "success",
      description: "Master MongoDB, Express.js, React, and Node.js with real-time enterprise architecture.",
      duration: "12 Weeks",
      level: "Advanced"

    },
    {
      id: 2,
      title: "React.js State Architecture",
      subtitle: "Frontend Systems Design",
      badgeText: "Popular",
      badgeType: "info",
      description: "Deep dive into component hierarchies, custom hooks, context API, and performance
optimization.",
      duration: "6 Weeks",
      level: "Intermediate"
    },
    {
      id: 3,
      title: "Cloud Microservices with Express",
      subtitle: "Backend RESTful APIs",
      badgeText: "High Demand",
      badgeType: "warning",
      description: "Build robust REST APIs, implement JWT authentication, and deploy scalable Node
services.",
      duration: "8 Weeks",
      level: "Intermediate"
    }
  ];

  return (
    <div className="app-container">
      <Header appName="MSD Course Portal" studentName="Harmit Suthar" enrollmentNo="26084231044" />

      <main className="main-content">
        <section className="hero-section">
          <span className="eyebrow">Practical 01 · React Components</span>
          <h1>Component Reusability in Action</h1>
          <p>
```

```
            Demonstrating React 18 Functional Components, JSX expressions, and parameter-driven UI blocks.
          </p>
        </section>

        <section className="card-grid">
          {courseList.map((course) => (
            <Card
              key={course.id}
              title={course.title}
              subtitle={course.subtitle}                badgeText=
              {course.badgeText}                        badgeType=
              {course.badgeType}
              actionText="Enroll Now"
              onActionClick={() => handleEnroll(course.title)}
            >
              <p>{course.description}</p>
              <div          className="card-
              meta">
                <span>{course.duration}</span>
                <span>{course.level}</span>
              </div>
            </Card>
          ))}
        </section>
      </main>
    </div>
  );
}

export default App;
```

src/components/Badge.jsx — JAVASCRIPT (JSX)

```
import React from 'react';

/**
 * Reusable Badge Component
 * Types: 'success' | 'warning' | 'info' | 'danger'
 */
const Badge = ({ text, type = 'info' }) => {
  return (
    <span className={`badge badge-${type}`}>
      {text}
    </span>
  );
};

export default Badge;
```

src/components/Button.jsx — JAVASCRIPT (JSX)

```
import React from 'react';

/**
 * Reusable Button Component
 * Supports variants: 'primary' | 'secondary' | 'danger' | 'outline'
* Supports sizes: 'sm' | 'md' | 'lg'
 */
const Button = ({
  children,
  variant = 'primary',
  size = 'md', onClick,
  disabled = false,
  className = '',
  type = 'button'

}) => {
  return (
```

```
      <button
        type={type}
        onClick={onClick}
        disabled={disabled}
        className={`btn btn-${variant} btn-${size} ${className}`}
      >
        {children}
      </button>
    );
};

export default Button;
```

src/components/Card.jsx — JAVASCRIPT (JSX)

```
import React from 'react';
import Badge from './Badge';
import Button from './Button';

/**
 * Reusable Card Component with Composition
 */
const Card = ({
  title,
  subtitle,
  badgeText,
  badgeType = 'info',
  imageUrl,
  children,
  actionText = 'Learn More',
  onActionClick
}) => {
  return (
    <div className="card">
      {imageUrl && (
        <div className="card-image-wrapper">
          <img src={imageUrl} alt={title} className="card-image" />
        </div>
      )}
      <div className="card-header">
        <div>
          <h3 className="card-title">{title}</h3>
          {subtitle && <p className="card-subtitle">{subtitle}</p>}
        </div>
        {badgeText && <Badge text={badgeText} type={badgeType} />}
      </div>
      <div className="card-body">
        {children}
      </div>
      <div className="card-footer">
        <Button variant="primary" size="sm" onClick={onActionClick}>
          {actionText}
        </Button>
        <Button variant="outline" size="sm" onClick={() => alert(`Details for ${title}`)}>
          Quick view
        </Button>
      </div>
    </div>
  );
};

export default Card;
```

src/components/Header.jsx — JAVASCRIPT (JSX)

```
import React from 'react';
```

```jsx
const Header = ({ appName = "MSD Course Portal", studentName = "Harmit Suthar", enrollmentNo = "26084231044"
}) => {
  const initials = studentName.split(' ').map((w) => w[0]).join('').slice(0, 2).toUpperCase();
  return (
    <header className="topbar">
      <div className="topbar-inner">
        <div className="brand">
          <div className="brand-mark">P1</div>
          <div>
            <div className="brand-title">{appName}</div>
            <div className="brand-sub">Ganpat University · M.Sc [CA &amp;amp; IT]</div>
          </div>
        </div>
        <div className="student-chip">
          <div className="avatar">{initials}</div>
          <div>
            <div className="chip-name">{studentName}</div>
            <div className="chip-id">{enrollmentNo}</div>
          </div>
        </div>
      </div>
    </header>
); };


export default Header;
```

src/index.css CSS

```css
/* Shared design system — Harmit Suthar (26084231044) · MERN Stack Development */
:root {
  --bg: #f4f6fb; --surface: #ffffff; --border: #e3e8f0; --text: #0f172a; --muted: #64748b; --
  primary: #4f46e5; --primary-hover: #4338ca; --primary-soft: #eef2ff;
  --success: #059669; --success-soft: #d1fae5;
  --warning: #d97706; --warning-soft: #fef3c7;
  --danger: #dc2626; --danger-soft: #fee2e2;
  --info: #2563eb; --info-soft: #dbeafe;
  --radius: 12px;
  --shadow: 0 1px 2px rgba(15,23,42,.05), 0 8px 24px rgba(15,23,42,.06);
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', sans-serif; background: var(--bg); color: var(--text);
line-height: 1.5; -webkit-font-smoothing: antialiased; }
button, input, select { font: inherit; }
.topbar { background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z- index: 10; }
.topbar-inner { max-width: 1120px; margin: 0 auto; padding: 14px 24px; display: flex; align-items: center; justify-
content: space-between; gap: 16px; flex-wrap: wrap; }
.brand { display: flex; align-items: center; gap: 12px; }
.brand-mark { width: 36px; height: 36px; border-radius: 10px; background: linear-gradient(135deg, #6366f1, #4338ca);
color: #fff; display: grid; place-items: center; font-weight: 700; font-size: .8rem; }
.brand-title { font-weight: 700; font-size: .95rem; }
.brand-sub { font-size: .75rem; color: var(--muted); }
.student-chip { display: flex; align-items: center; gap: 10px; background: var(--primary-soft); border: 1px solid
#c7d2fe; padding: 6px 14px 6px 6px; border-radius: 999px; }
.avatar { width: 28px; height: 28px; border-radius: 50%; background: var(--primary); color: #fff; display: grid; place-
items: center; font-size: .7rem; font-weight: 700; }
.chip-name { font-size: .82rem; font-weight: 600; color: #312e81; line-height: 1.2; }
.chip-id { font-size: .72rem; color: #4f46e5; line-height: 1.2; }
.page { max-width: 1120px; margin: 0 auto; padding: 32px 24px 56px; }
.page-head { margin-bottom: 24px; }
.eyebrow { display: inline-block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter- spacing:
.08em; color: var(--primary); background: var(--primary-soft); padding: 3px 10px; border-radius: 999px; margin-
bottom: 10px; }
.page-head h1 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.page-head p { color: var(--muted); margin-top: 6px; max-width: 70ch; }
.panel { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box- shadow: var(--
shadow); padding: 22px; }
.btn { border: 1px solid transparent; border-radius: 8px; padding: 8px 14px; font-size: .85rem; font-weight: 600;
cursor: pointer; background: var(--primary); color: #fff; transition: background .15s, border-color
```

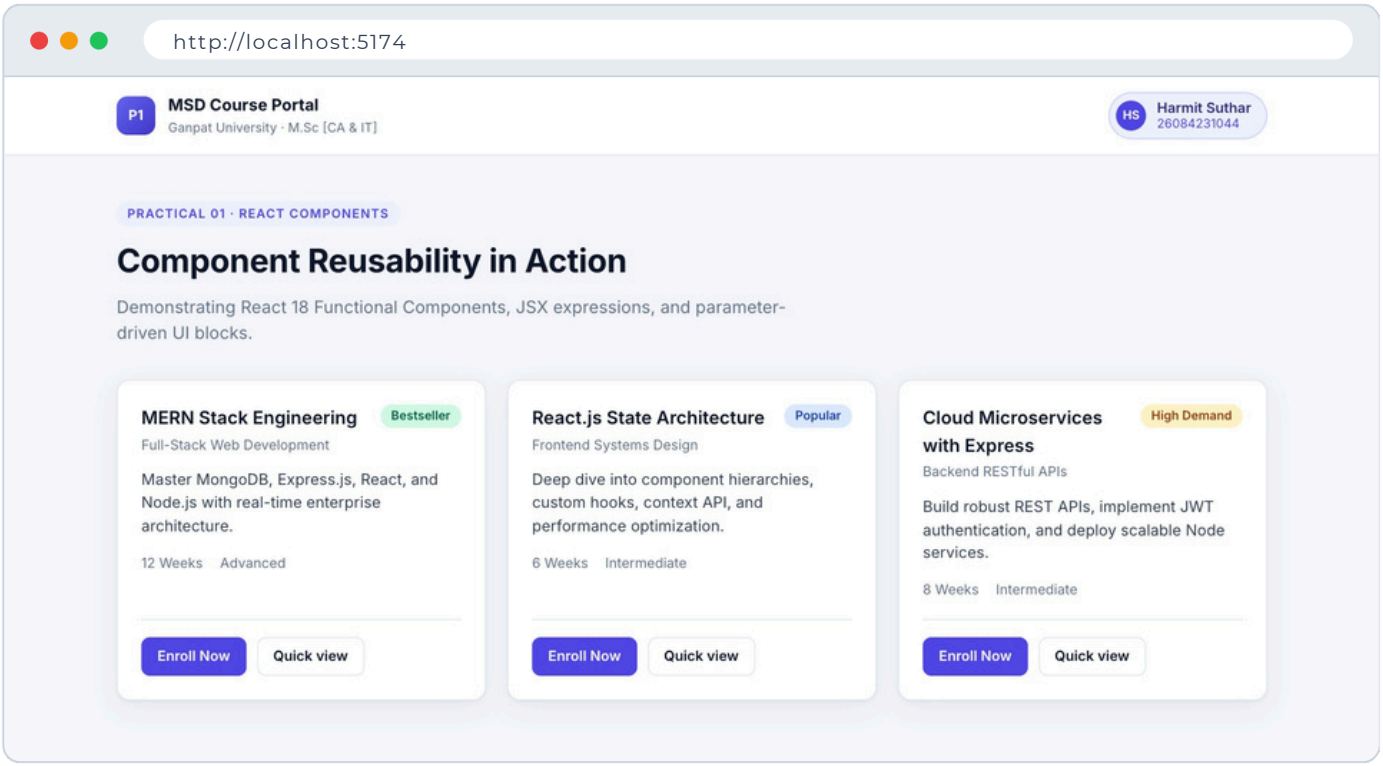
```
.15s; }
.btn:hover { background: var(--primary-hover); }
.btn:focus-visible, input:focus-visible, select:focus-visible, button:focus-visible { outline: 3px solid
#c7d2fe; outline-offset: 1px; }
.btn.secondary { background: var(--surface); color: var(--text); border-color: var(--border); }
.btn.secondary:hover { background: #f8fafc; border-color: #cbd5e1; }
.btn:disabled { opacity: .5; cursor: not-allowed; }
input[type=text], input[type=email], input[type=tel], input[type=number], select { font-size: .85rem;
padding: 9px 12px; border: 1px solid var(--border); border-radius: 8px; background: #fff; color: var(--
text); }
input:focus, select:focus { border-color: var(--primary); }
table { width: 100%; border-collapse: collapse; font-size: .85rem; }
th { text-align: left; font-size: .72rem; text-transform: uppercase; letter-spacing: .06em; color: var(--
muted); background: #f8fafc; padding: 10px 12px; border-bottom: 1px solid var(--border); }
td { padding: 12px; border-bottom: 1px solid var(--border); }
tr:last-child td { border-bottom: none; }
.footer { text-align: center; color: var(--muted); font-size: .75rem; margin-top: 32px; }


.main-content { max-width: 1120px; margin: 0 auto; padding: 40px 24px 56px; }
.hero-section { margin-bottom: 32px; }
.hero-section h1 { font-size: 1.9rem; font-weight: 700; letter-spacing: -.01em; }
.hero-section p { color: var(--muted); margin-top: 8px; max-width: 65ch; }
.card-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(300px, 1fr)); gap: 20px; }
.card { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-
shadow: var(--shadow); padding: 22px; display: flex; flex-direction: column; transition: transform .2s, box-
shadow .2s; }
.card:hover { transform: translateY(-3px); box-shadow: 0 2px 4px rgba(15,23,42,.06), 0 16px 32px
rgba(15,23,42,.1); }
.card-header { display: flex; justify-content: space-between; align-items: flex-start; gap: 12px; margin-
bottom: 12px; }
.card-title { font-size: 1.05rem; font-weight: 600; }
.card-subtitle { font-size: .8rem; color: var(--muted); margin-top: 2px; }
.card-body { color: #334155; font-size: .9rem; flex: 1; }
.card-meta { display: flex; gap: 16px; margin: 14px 0 18px; font-size: .8rem; color: var(--muted); }
.card-footer { display: flex; gap: 10px; border-top: 1px solid var(--border); padding-top: 16px; }
.badge { display: inline-block; padding: 2px 10px; border-radius: 999px; font-size: .72rem; font-weight:
600; white-space: nowrap; }
.badge-success { background: var(--success-soft); color: #065f46; }
.badge-info { background: var(--info-soft); color: #1e40af; }
.badge-warning { background: var(--warning-soft); color: #92400e; }
.badge-danger { background: var(--danger-soft); color: #991b1b; }
.btn-sm { padding: 7px 14px; font-size: .82rem; }
.btn-md { padding: 9px 16px; }
.btn-primary { background: var(--primary); color: #fff; }
.btn-outline { background: var(--surface); color: var(--text); border-color: var(--border); }
.btn-outline:hover { background: #f8fafc; border-color: #cbd5e1; }
.btn-secondary { background: #e2e8f0; color: var(--text); }
.btn-danger { background: var(--danger); color: #fff; }
```

5. Expected Output & Verification Results

All screenshots and logs below were captured from the real running application of this practical (Harmit Suthar · 26084231044).

Figure 5.1 — Reusable Header, Card, Badge and Button components rendered as a course catalogue.



Execution & Server Logs **Port 5174**

```
$ npm run dev
> practical-01-components@1.0.0 dev
> vite


  VITEv5.4.21 ready in 159 ms

  ➜  Local:   http://localhost:5174/
  ➜  Network: use --host to expose
```

PRACTICAL 2

React.js / State & Interactions

Develop a React Application implementing Props, State Management and Event Handling

1. Practical Aim & Theoretical Definition

Aim & Theoretical Concept

The objective of this practical is to build an interactive Cart & Product Catalog system demonstrating Props Passing (Parent-to-Child & Child-to-Parent callbacks), State Management with the useState hook, and Synthetic Event Handling (onClick, onChange, onSubmit).

Core Theoretical Concepts:

- **Props (Properties):** ***Read-only inputs passed from parent components to child components, establishing a strictly unidirectional data flow. Props allow dynamic data binding and callback delegation.***
- **State Management (useState):** ***State holds mutable values that persist across component re-renders. Invoking the state updater function triggers React's reconciliation engine to re-render the view with the fresh state.***
- **Lifting State Up:** ***When multiple sibling components need access to the same piece of data (e.g. Products list and Cart Summary), the state is lifted up to their common parent component (App.jsx).***
- **Synthetic Events:** ***React wraps native browser DOM events in cross-browser SyntheticEvent wrappers, guaranteeing uniform event behavior across all browsers.***

2. Step-by-Step Commands (Installation & Execution)

1. Navigate to the Practical Directory

```
$ cd practicals/p2
```

Action: *Switches the working directory to the Practical 2 project folder.*

2. Install Dependencies

```
$ npm install
```

Action: *Installs every package declared in package.json into node_modules.*

3. Run Local Development Server

```
$ npm run dev
```

Action: *Launches the Vite dev server with hot reload at http://localhost:5175.*

3. Project Directory & File Structure

```
practical-02/
├── src/
│   ├── components/
│   │   ├── AppBar.jsx
│   │   ├── CartSummary.jsx
│   │   └── ProductItem.jsx
│   ├── App.jsx
│   ├── index.css
│   └── main.jsx
├── index.html
├── package-lock.json
├── package.json
├── README.md
├── run.bat
└── vite.config.js
```

File / Directory Path	Architectural Purpose & Responsibility
src/components/AppBar.jsx	***Shared top bar showing the practical number and the student badge.***
src/components/CartSummary.jsx	***Computes subtotal, coupon discount, 18% GST and grand total; handles the coupon form.***
src/components/ProductItem.jsx	***Renders one cart row and reports quantity changes to the parent via a callback prop.***
src/App.jsx	***Root React component orchestrating state, layout and child components.***
src/index.css	***Global styles, design tokens and component styling.***
src/main.jsx	***Mounts the root React component into the DOM with ReactDOM.createRoot().***
index.html	***HTML entry point containing the #root mount node.***
package-lock.json	***Exact dependency tree lockfile generated by npm.***
package.json	***Project metadata, scripts and dependency declarations.***
README.md	***Project documentation.***
run.bat	***Windows launcher script that installs and starts the project.***
vite.config.js	***Vite build and dev-server configuration.***

4. Modular Source Code Implementation

Complete source code of the project, listed exactly as it exists in `practicals/p2` **:**

src/main.jsx JAVASCRIPT (JSX)

```jsx
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode><App /></React.StrictMode>
);
```

src/App.jsx JAVASCRIPT (JSX)

```jsx
import React, { useState } from 'react';
import ProductItem from './components/ProductItem'; import CartSummary from './components/CartSummary'; import AppBar from './components/AppBar';

const INITIAL_PRODUCTS = [
  { id: 101, name: "Mechanical Gaming Keyboard", category: "Electronics", price: 3499, quantity: 1, icon: "⌨️" },
  { id: 102, name: "Wireless Ergonomic Mouse", category: "Electronics", price: 1899, quantity: 2, icon: "🖱️" },
  { id: 103, name: "Noise Cancelling Headphones", category: "Audio", price: 7999, quantity: 0, icon: "🎧" },
  { id: 104, name: "4K IPS Monitor 27-inch", category: "Electronics", price: 21999, quantity: 1, icon: "🖥️" },
  { id: 105, name: "Developer Hoodie (Cotton)", category: "Apparel", price: 1499, quantity: 1, icon: "👕" }
];

function App() {
  const [products, setProducts] = useState(INITIAL_PRODUCTS);
  const [activeCategory, setActiveCategory] = useState('All');
  const [searchQuery, setSearchQuery] = useState('');
  const [discount, setDiscount] = useState(0);

  // Event handler for quantity modification (Child -> Parent communication)
  const handleQuantityChange = (productId, delta) => {
    setProducts((prev) =>
      prev.map((item) =>
        item.id === productId
          ? { ...item, quantity: Math.max(0, item.quantity + delta) }
          : item
      )
    );
  };

  const categories = ['All', 'Electronics', 'Audio', 'Apparel'];

  const filteredProducts = products.filter((p) => {
    const matchesCategory = activeCategory === 'All' || p.category === activeCategory;
    const matchesSearch = p.name.toLowerCase().includes(searchQuery.toLowerCase());
    return matchesCategory && matchesSearch;
  });

  return (
    <>
      <AppBar practical="02" />
      <div className="page">
      <header className="page-head">
        <span className="eyebrow">Practical 02 · State &amp;amp; Props</span>
        <h1>Shopping Cart &amp;amp; State Manager</h1>
        <p>Demonstrating props, the useState hook and synthetic event handlers.</p>
      </header>

      <div className="controls-bar">
        <div className="filter-buttons">
          {categories.map((cat) => (
            <button
              key={cat}
```

```
                  className={`filter-btn ${activeCategory === cat ? 'active' : ''}`}
                  onClick={() => setActiveCategory(cat)}
                >
                  {cat}
                </button>
              ))}
            </div>
            <inpu
            t type="text"
              className="search-box"
              placeholder="Search items by name..."
              value={searchQuery}
              onChange={(e) => setSearchQuery(e.target.value)}
            />
          </div>

          <div className="cart-layout">
            <div className="product-list-container">
              <div className="list-header">
                <span>Product
                Details</span>      <span>Unit
                Price</span>
                <span>Quantity</span>
                <span>Subtotal</span>
              </div>
              {filteredProducts.map((product) => (
                <ProductItem
                  key={product.id}
                  product={product}
                  onQuantityChange={handleQuantityChange}
                />
              ))}
            </div>

            <aside className="summary-sidebar">
              <CartSummary
                items={products}
                discountPercent={discount}
                onApplyCoupon={(disc) => setDiscount(disc)}
              />
            </aside>
          </div>
          <p className="footer">Harmit Suthar (26084231044) · Practical 02</p>
          </div>
        </>
      );

export default App;
```

src/components/AppBar.jsx — JAVASCRIPT (JSX)

```
import React from 'react';

const AppBar = ({ practical, title = 'MERN Stack Development' }) => (
  <header className="topbar">
    <div className="topbar-inner">
      <div className="brand">
        <div className="brand-mark">P{practical}</div>
        <div>
          <div className="brand-title">{title}</div>
          <div className="brand-sub">Ganpat University · M.Sc [CA &amp;amp; IT]</div>
        </div>
      </div>
      <div className="student-chip">
        <div className="avatar">HS</div>
        <div>
          <div className="chip-name">Harmit Suthar</div>
          <div className="chip-id">26084231044</div>
        </div>
      </div>
```

```
      </div>
    </header>
);

export default AppBar;
```

src/components/CartSummary.jsx JAVASCRIPT (JSX)

```
import React, { useState } from 'react';

const CartSummary = ({ items, discountPercent, onApplyCoupon }) => {
  const [couponCode, setCouponCode] = useState('');
  const [errorMsg, setErrorMsg] = useState('');

  const totalItems = items.reduce((acc, curr) => acc + curr.quantity, 0);
  const subtotal = items.reduce((acc, curr) => acc + (curr.price * curr.quantity), 0);
  const discountAmount = (subtotal * discountPercent) / 100;
  const tax = ((subtotal - discountAmount) * 0.18); // 18% GST
  const grandTotal = subtotal - discountAmount + (subtotal > 0 ? tax : 0);

  const handleCouponSubmit = (e) => {
    e.preventDefault();
    if (couponCode.toUpperCase() === 'MSD20') {
      onApplyCoupon(20);
      setErrorMsg('');
    } else {
      setErrorMsg('Invalid Coupon. Use "MSD20" for 20% off.');
    }
  };

  return (
    <div className="cart-summary-card">
      <h3>Order Summary</h3>
      <div className="summary-line">
        <span>Total Items:</span>
        <strong>{totalItems}</strong>
      </div>
      <div className="summary-line">
        <span>Subtotal:</span>
        <span>₹{subtotal.toLocaleString()}</span>
      </div>
      {discountPercent > 0 && (
        <div className="summary-line discount-line">
          <span>Discount ({discountPercent}%):</span>
          <span>- ₹{discountAmount.toLocaleString()}</span>
        </div>
      )}
      <div className="summary-line">
        <span>GST (18%):</span>
        <span>₹{Math.round(tax).toLocaleString()}</span>
      </div>
      <hr className="summary-divider" />
      <div className="summary-line grand-total">
        <span>Grand Total:</span>
        <span>₹{Math.round(grandTotal).toLocaleString()}</span>
      </div>

      <form onSubmit={handleCouponSubmit} className="coupon-form">
        <input
          type="text"
          placeholder="Enter coupon (e.g. MSD20)"
          value={couponCode}
          onChange={(e) => setCouponCode(e.target.value)}
          className="coupon-input"
        />
        <button type="submit" className="coupon-btn">Apply</button>
      </form>
      {errorMsg && <p className="error-text">{errorMsg}</p>}
      {discountPercent > 0 && <p className="success-text">20% coupon applied successfully.</p>}
    </div>
```

```
  );
};

export default CartSummary;
```

src/components/ProductItem.jsx — JAVASCRIPT (JSX)

```jsx
import React from 'react';

const ProductItem = ({ product, onQuantityChange }) => {
  return (
    <div className="product-row">
      <div className="product-info">
        <span className="product-icon">{product.icon}</span>
        <div>
          <h4 className="product-name">{product.name}</h4>
          <span className="product-category">{product.category}</span>
        </div>
      </div>
      <div className="product-price">₹{product.price.toLocaleString()}</div>
      <div className="quantity-controls">
        <button
          className="qty-btn"
          onClick={() => onQuantityChange(product.id, -1)}
          disabled={product.quantity <= 0}
        >
          -
        </button>
        <span className="qty-value">{product.quantity}</span>
        <button
          className="qty-btn"
          onClick={() => onQuantityChange(product.id, 1)}
        >
          +
        </button>
      </div>
      <div className="product-subtotal">
        ₹{(product.price * product.quantity).toLocaleString()}
      </div>
};    </div>
  );


export default ProductItem;
```

src/index.css — CSS

```css
/* Shared design system — Harmit Suthar (26084231044) · MERN Stack Development */
:root {
  --bg: #f4f6fb; --surface: #ffffff; --border: #e3e8f0; --text: #0f172a; --muted: #64748b; --
  primary: #4f46e5; --primary-hover: #4338ca; --primary-soft: #eef2ff;
  --success: #059669; --success-soft: #d1fae5;
  --warning: #d97706; --warning-soft: #fef3c7;
  --danger: #dc2626; --danger-soft: #fee2e2;
  --info: #2563eb; --info-soft: #dbeafe;
  --radius: 12px;
  --shadow: 0 1px 2px rgba(15,23,42,.05), 0 8px 24px rgba(15,23,42,.06);
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', sans-serif; background: var(--bg); color: var(--text);
line-height: 1.5; -webkit-font-smoothing: antialiased; }
button, input, select { font: inherit; }
.topbar { background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z- index: 10; }
.topbar-inner { max-width: 1120px; margin: 0 auto; padding: 14px 24px; display: flex; align-items: center; justify-
content: space-between; gap: 16px; flex-wrap: wrap; }
.brand { display: flex; align-items: center; gap: 12px; }
```

```
.brand-mark { width: 36px; height: 36px; border-radius: 10px; background: linear-gradient(135deg, #6366f1, #4338ca); color: #fff; display: grid; place-items: center; font-weight: 700; font-size: .8rem; } .brand-title { font-weight: 700; font-size: .95rem; }
.brand-sub { font-size: .75rem; color: var(--muted); }
.student-chip { display: flex; align-items: center; gap: 10px; background: var(--primary-soft); border: 1px solid #c7d2fe; padding: 6px 14px 6px 6px; border-radius: 999px; }
.avatar { width: 28px; height: 28px; border-radius: 50%; background: var(--primary); color: #fff; display: grid; place-items: center; font-size: .7rem; font-weight: 700; }
.chip-name { font-size: .82rem; font-weight: 600; color: #312e81; line-height: 1.2; }
.chip-id { font-size: .72rem; color: #4f46e5; line-height: 1.2; }
.page { max-width: 1120px; margin: 0 auto; padding: 32px 24px 56px; }
.page-head { margin-bottom: 24px; }
.eyebrow { display: inline-block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter- spacing: .08em; color: var(--primary); background: var(--primary-soft); padding: 3px 10px; border-radius: 999px; margin-bottom: 10px; }
.page-head h1 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.page-head p { color: var(--muted); margin-top: 6px; max-width: 70ch; }
.panel { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box- shadow: var(--shadow); padding: 22px; }
.btn { border: 1px solid transparent; border-radius: 8px; padding: 8px 14px; font-size: .85rem; font-weight: 600; cursor: pointer; background: var(--primary); color: #fff; transition: background .15s, border-color .15s; }
.btn:hover { background: var(--primary-hover); }
.btn:focus-visible, input:focus-visible, select:focus-visible, button:focus-visible { outline: 3px solid #c7d2fe; outline-offset: 1px; }
.btn.secondary { background: var(--surface); color: var(--text); border-color: var(--border); } .btn.secondary:hover { background: #f8fafc; border-color: #cbd5e1; }
.btn:disabled { opacity: .5; cursor: not-allowed; }
input[type=text], input[type=email], input[type=tel], input[type=number], select { font-size: .85rem;



padding: 9px 12px; border: 1px solid var(--border); border-radius: 8px; background:#fff; color: var(--text); }
input:focus, select:focus { border-color: var(--primary); }
table { width: 100%; border-collapse: collapse; font-size:.85rem;}
th { text-align:left; font-size: .72rem; text-transform:uppercase; letter-spacing:.06em; color: var(--muted); #f8fafc; padding: 10px 12px; border-bottom:1px solid var(--border);}
background: td { border-bottom:1px solid var(--border);}
padding: 12px;
tr:last-child td { border-bottom:none;}
.footer { text-align:center; color: var(--muted); font-size:.75rem; margin-top: 32px;}

.controls-bar { display: flex; justify-content: space-between; align-items: center; margin-bottom: 20px; gap: 14px; flex-wrap:wrap; }
.filter-buttons { display: flex; gap: 8px; flex-wrap: wrap; }
.filter-btn { background: var(--surface); color: var(--muted); border: 1px solid var(--border); padding: 6px 14px; border-radius: 999px; font-size: .82rem; font-weight: cursor:pointer;}
.filter-btn:hover { border-color: #cbd5e1; color: var(--text); }
.filter-btn.active { background: var(--primary); border-color: var(--primary); color: #fff; font-weight: 600; }
.search-box { min-width: 260px; }
.cart-layout { display: grid; grid-template-columns: 2fr 1fr; gap: 20px; align-items: start; }
@media (max-width: 860px) { .cart-layout { grid-template-columns: 1fr; } }
.product-list-container { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-shadow: var(--shadow); padding: 8px 22px 14px; }
.list-header, .product-row { display: grid; grid-template-columns: 2.4fr 1fr 1.2fr 1fr; align-items: center; gap: 12px;}
.list-header { padding: 14px 0 10px; border-bottom: 1px solid var(--border); color: var(--muted); font-size:

.72rem; text-transform: uppercase; letter-spacing:.06em; font-weight: 600;}
.product-row { border-bottom: 1px solid var(--border); padding: 14px 0; }
.product-row:last-child { border-bottom: none; }
.product-info { display: flex; align-items: center; gap: 12px; }
.product-icon { width: 40px; height: 40px; border-radius: 10px; background: var(--primary-soft); display: grid; place-items: center; font-size: 1.25rem;}
.product-name { font-size: .9rem; font-weight: 600; }
.product-category { font-size: .75rem; color: var(--muted); }
.product-price { font-size: .88rem; color: #334155; }
.quantity-controls { display: flex; align-items:center; gap: 10px; }
.qty-btn { background: var(--surface); border: 1px solid var(--border); color: var(--text); width: 28px; height: 28px; border-radius: 8px; cursor:pointer; font-weight: 600;}
.qty-btn:hover:not(:disabled) { background: var(--primary-soft); border-color: #c7d2fe; }
.qty-btn:disabled { opacity: .4; cursor:not-allowed;}
.qty- text-align:center; font-weight: 600; font-size:.9rem;}
value { min-width: 18px; .product-
subtotal { font-weight: .cart- 700; text-align: right;}
summary-card { background: var(--surface); border: 1px solid var(--border); border-radius: var(--
```

```
radius); box-shadow: var(--shadow); padding: 22px; position: sticky; top: 90px; }
.cart-summary-card h3 { font-size: 1rem; margin-bottom: 16px; }
.summary-line { display: flex; justify-content: space-between; margin-bottom: 10px; font-size: .88rem; color: #334155; }
.discount-line { color: var(--success); }
.summary-divider { border: none; border-top: 1px solid var(--border); margin: 14px 0; }
.grand-total { font-size: 1.1rem; font-weight: 700; color: var(--text); margin-bottom: 0; }
.coupon-form { display: flex; gap: 8px; margin-top: 20px; }
.coupon-input { flex: 1; min-width: 0; }
.coupon-btn { background: var(--primary); color: #fff; border: none; padding: 8px 14px; border-radius: 8px; font-weight: 600; font-size: .85rem; cursor: pointer; }
.coupon-btn:hover { background: var(--primary-hover); }
.success-text { color: var(--success); font-size: .8rem; margin-top: 10px; }
.error-text { color: var(--danger); font-size: .8rem; margin-top: 10px; }
```

5. Expected Output & Verification Results

All screenshots and logs below were captured from the real running application of this practical (Harmit Suthar · 26084231044).

Figure 5.1 — Initial cart state with quantity controls and the order summary computed from props and state.

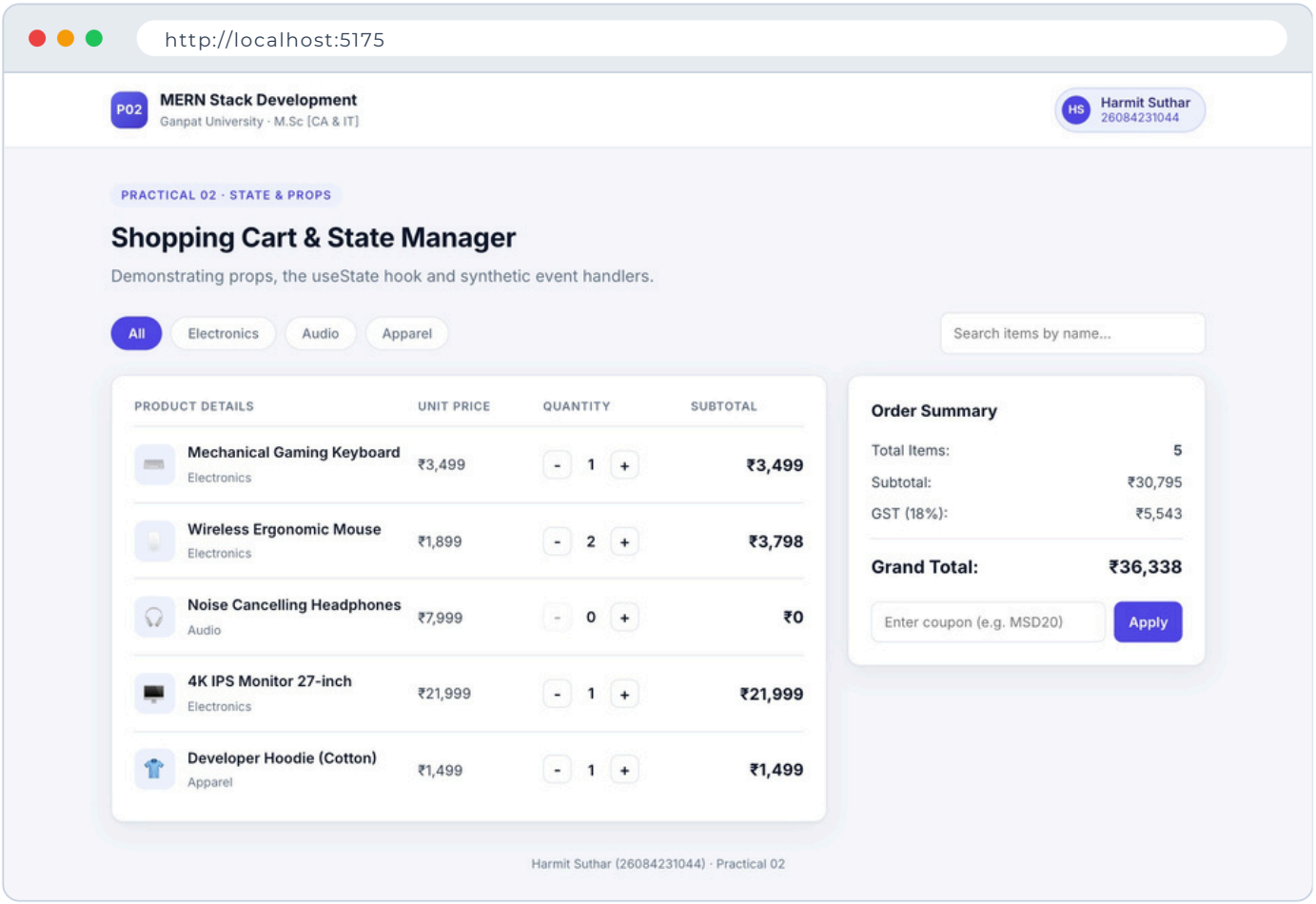


Figure 5.2 — Quantity increased twice and coupon MSD20 applied: the discount, GST and grand total update instantly.

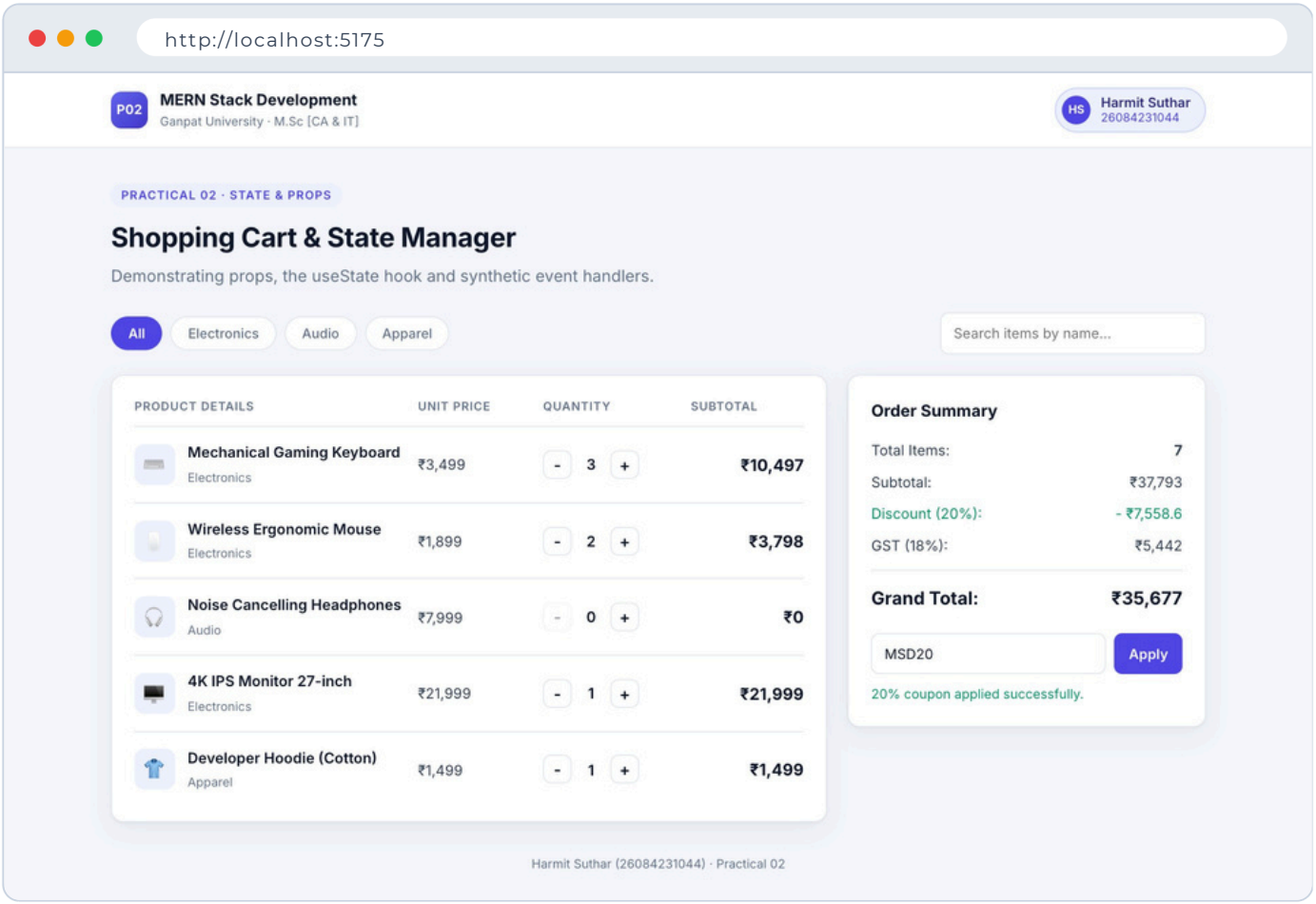
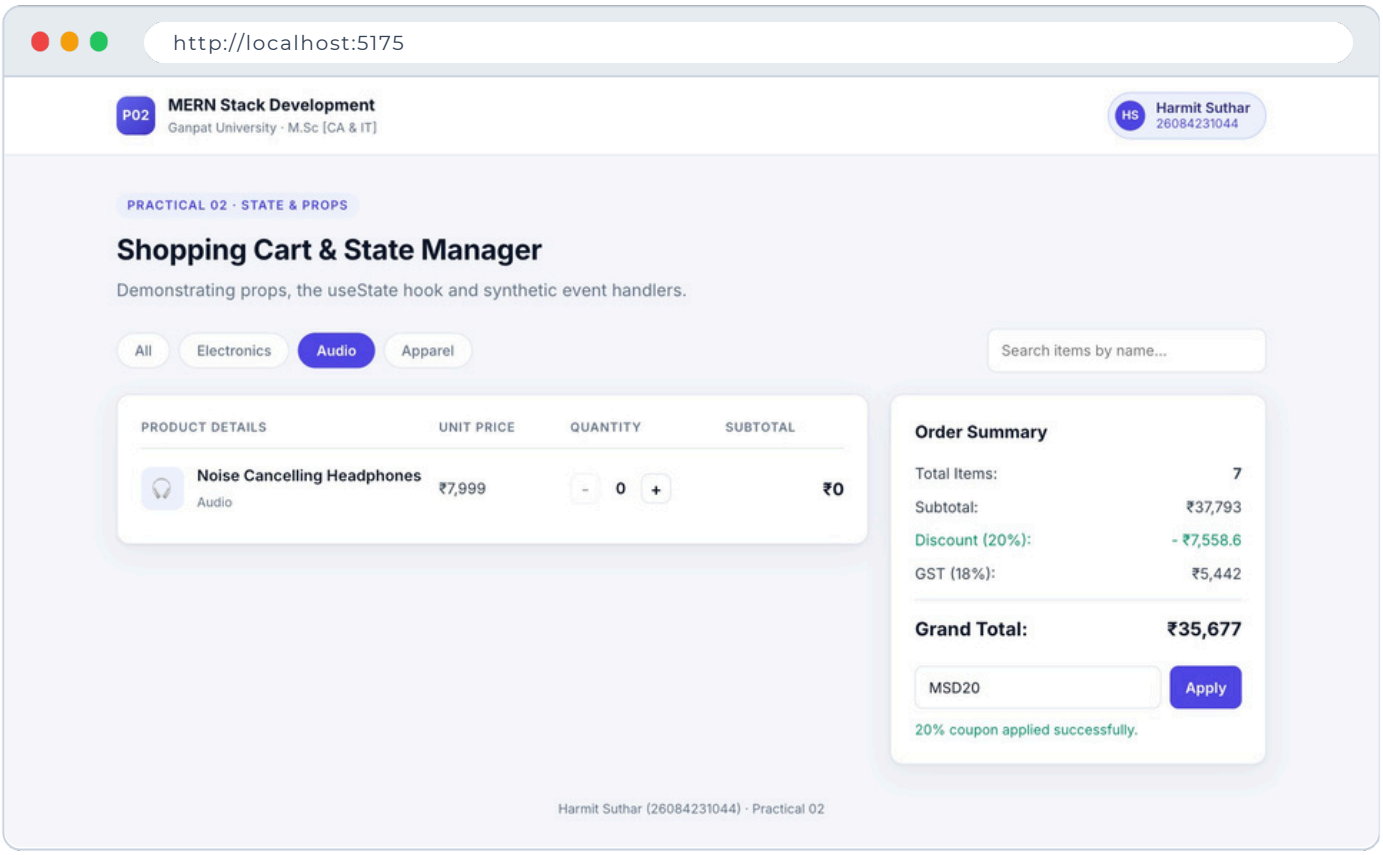


Figure 5.3 — Category filter "Audio" synchronised with the controlled search input.



🖥️ **Execution & Server Logs** **Port 5175**

```
$ npm run dev
> practical-02-props-state@1.0.0 dev
> vite


  VITEv5.4.21 ready in 161 ms

  ➜  Local:   http://localhost:5175/
  ➜  Network: use --host to expose
```

PRACTICAL 3

React.js / Forms & Data Integrity

Create a Student Management System using React Forms, Controlled Components and Validation

1. Practical Aim & Theoretical Definition

Aim & Theoretical Concept

The objective of this practical is to develop a complete Student Registration and Records Management system implementing Controlled Components, robust Regex Form Validation, dynamic error display, and state-driven table rendering with search, sort, and deletion capabilities.

Core Theoretical Concepts:

- **Controlled Components:** ***An input form element whose value is controlled by React state. In controlled inputs, mutable state acts as the "single source of truth". Every keystroke invokes an onChange handler, updating the state, which in turn re-renders the input value.***
- **Form Validation Architecture:** ***Real-time and submit-time validation rules ensuring data integrity:***
  - `Roll Number`***: Strict alphanumeric pattern (e.g. 23MSD001).***
  - `Full Name`***: Minimum 3 alphabetical characters.***
  - `Email`***: RFC-compliant standard email regex format.***
  - `Mobile Phone`***: Exactly 10 numeric digits starting with 6, 7, 8, or 9.***
  - `CGPA`***: Numeric float between 0.00 and 10.00.***
- **Derived State & Filters:** ***Real-time filtering of registered student records by department and search queries without mutating the original database state.***

2. Step-by-Step Commands (Installation & Execution)

1. Navigate to the Practical Directory

```
$ cd practicals/p3
```

Action: *Switches the working directory to the Practical 3 project folder.*

2. Install Dependencies

```
$ npm install
```

Action: *Installs every package declared in package.json into node_modules.*

3. Run Local Development Server

```
$ npm run dev
```

Action: *Launches the Vite dev server with hot reload at http://localhost:5176.*

3. Project Directory & File Structure

```
practical-03/
├── src/
│   ├── components/
│   │   ├── AppBar.jsx
│   │   └── StudentForm.jsx
│   ├── utils/
│   │   └── validation.js
│   ├── App.jsx
│   ├── index.css
│   └── main.jsx
├── index.html
├── package-lock.json
├── package.json
├── README.md
├── run.bat
└── vite.config.js
```

File / Directory Path	Architectural Purpose & Responsibility
src/components/AppBar.jsx	***Shared top bar showing the practical number and the student badge.***
src/components/StudentForm.jsx	***Controlled registration form with per-field validation messages.***
src/utils/validation.js	***Regular-expression based validators for roll number, name, email, phone and CGPA.***
src/App.jsx	***Root React component orchestrating state, layout and child components.***
src/index.css	***Global styles, design tokens and component styling.***
src/main.jsx	***Mounts the root React component into the DOM with ReactDOM.createRoot().***
index.html	***HTML entry point containing the #root mount node.***
package-lock.json	***Exact dependency tree lockfile generated by npm.***
package.json	***Project metadata, scripts and dependency declarations.***
README.md	***Project documentation.***
run.bat	***Windows launcher script that installs and starts the project.***
vite.config.js	***Vite build and dev-server configuration.***

4. Modular Source Code Implementation

Complete source code of the project, listed exactly as it exists in `practicals/p3` ***:***

src/main.jsx JAVASCRIPT (JSX)

```jsx
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode><App /></React.StrictMode>
);
```

src/App.jsx JAVASCRIPT (JSX)

```jsx
import React, { useState } from 'react';
import StudentForm from './components/StudentForm';
import AppBar from './components/AppBar';

const INITIAL_STUDENTS = [
  { id: 1, rollNo: "24MSD001", name: "Rohan Verma", email: "rohan.v@univ.edu", phone: "9823412345",
department: "Computer Science", cgpa: "9.20" },
  { id: 2, rollNo: "24MSD002", name: "Priya Nair", email: "priya.n@univ.edu", phone: "9876543210",
department: "AI & Data Science", cgpa: "8.95" },
  { id: 3, rollNo: "24MSD003", name: "Devansh Patel", email: "devansh.p@univ.edu", phone: "9123456780",
department: "Information Tech", cgpa: "7.80" }
];

function App() {
  const [students, setStudents] = useState(INITIAL_STUDENTS);
  const [searchTerm, setSearchTerm] = useState('');
  const [filterDept, setFilterDept] = useState('All');

  const handleAddStudent = (newStudent) => {
    setStudents((prev) => [
      ...prev,
      { ...newStudent, id: Date.now() }
    ]);
  };

  const handleDelete = (id) => {
    if (window.confirm("Are you sure you want to delete this student record?")) {
      setStudents((prev) => prev.filter((s) => s.id !== id));
    }
  };

  const filteredStudents = students.filter((s) => {
     const matchesSearch = s.name.toLowerCase().includes(searchTerm.toLowerCase()) ||
                           s.rollNo.toLowerCase().includes(searchTerm.toLowerCase());
    const matchesDept = filterDept === 'All' || s.department === filterDept;
    return matchesSearch && matchesDept;
  });

  return (
    <>
      <AppBar practical="03" />
      <div className="page">
      <header className="page-head">
        <span className="eyebrow">Practical 03 · Forms &amp;amp; Validation</span>
        <h1>Student Management Portal</h1>
        <p>React controlled forms with real-time schema validation.</p>
      </header>

      <div className="main-grid">
        <section className="form-panel">
          <StudentForm onAddStudent={handleAddStudent} />
        </section>

        <section className="records-panel">
          <div className="records-header">
            <h3>Registered Students ({filteredStudents.length})</h3>
```

```
            <div className="search-filters">
              <input
                type="text"
                placeholder="Search by name or roll..."
                value={searchTerm}
                onChange={(e) => setSearchTerm(e.target.value)}
                className="search-input"
              />
              <select value={filterDept} onChange={(e) => setFilterDept(e.target.value)}>
                <option value="All">All Departments</option>
                <option value="Computer Science">Computer Science</option>
                <option  value="Information  Tech">Information  Tech</option>
                <option value="AI & Data Science">AI & Data Science</option>
              </select>
            </div>
          </div>

          <table className="data-table">
            <thead>
              <tr>
                <th>Roll      No</th>
                <th>Name</th>
                <th>Department</t
                h>    <th>CGPA</th>
                <th>Contact</th>
                <th>Actions</th>
              </tr>
            </thead>
            <tbody>
              {filteredStudents.map((s) => (
                <tr key={s.id}>
                  <td><strong>{s.rollNo}</strong></td>
                  <td>{s.name}</td>
                  <td><span className="dept-tag">{s.department}</span></td>
                  <td><span className="cgpa-badge">{s.cgpa}</span></td>
                  <td>{s.phone}</td>
                  <td>
                    <button className="del-btn" onClick={() => handleDelete(s.id)}>
                      Delete
                    </button>
                  </td>
                </tr>
              ))}
            </tbody>
          </table>
        </section>
      </div>
      <p className="footer">Harmit Suthar (26084231044) · Practical 03</p>
      </div>
}    </>
  );


export default App;
```

src/components/AppBar.jsx JAVASCRIPT (JSX)

```
import React from 'react';

const AppBar = ({ practical, title = 'MERN Stack Development' }) => (
  <header className="topbar">
    <div className="topbar-inner">
      <div className="brand">
        <div className="brand-mark">P{practical}</div>
        <div>
          <div className="brand-title">{title}</div>
          <div className="brand-sub">Ganpat University · M.Sc [CA &amp;amp; IT]</div>
        </div>
      </div>
      <div className="student-chip">
```

```
          <div className="avatar">HS</div>
          <div>
            <div className="chip-name">Harmit Suthar</div>
            <div className="chip-id">26084231044</div>
          </div>
        </div>
      </div>
    </header>
);

export default AppBar;
```

src/components/StudentForm.jsx JAVASCRIPT (JSX)

```
import React, { useState } from 'react';
import { validateStudentForm } from '../utils/validation';

const INITIAL_FORM = {
  rollNo: '',
  name: '',
  email: '',
  phone: '',
  department: 'Computer Science',
  semester: '4th',
  cgpa: ''
};

const StudentForm = ({ onAddStudent }) => {
  const [formData, setFormData] = useState(INITIAL_FORM);
  const [errors, setErrors] = useState({});
  const [isSubmitted, setIsSubmitted] = useState(false);

  // Controlled component input change handler
  const handleChange = (e) => {
    const { name, value } = e.target;
    setFormData((prev) => ({ ...prev, [name]: value }));

    // Clear error on active editing
    if (errors[name]) {
      setErrors((prev) => ({ ...prev, [name]: '' }));
    }
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    const { isValid, errors: validationErrors } = validateStudentForm(formData);

    if (!isValid) {
      setErrors(validationErrors);
      return;
    }

    onAddStudent(formData);
    setFormData(INITIAL_FORM);
    setErrors({});
    setIsSubmitted(true);
    setTimeout(() => setIsSubmitted(false), 3000);
  };

  return (
    <form className="student-form" onSubmit={handleSubmit} noValidate>
      <h3>Register new student</h3>

      {isSubmitted && (
        <div className="alert alert-success">
          Student registered successfully into records!
        </div>
      )}

      <div className="form-grid">
```

```
<div className="form-group">
  <label>Roll Number *</label>
  <input
    type="text"
    name="rollNo"
    placeholder="e.g.          24MSD015"          value=
    {formData.rollNo}      onChange={handleChange}
    className={errors.rollNo ? 'input-error' : ''}


  />
  {errors.rollNo && <span className="field-error">{errors.rollNo}</span>}
</div>

<div          className="form-
group">  <label>Full  Name  *
</label> <input
    type="text"
    name="name"
    placeholder="e.g.    Aarav    Sharma"    value=
    {formData.name}    onChange={handleChange}
    className={errors.name ? 'input-error' : ''}


  />
  {errors.name && <span className="field-error">{errors.name}</span>}
</div>

<div className="form-group">
  <label>Email Address *</label>
  <input
    type="email"
    name="email"
    placeholder="aarav@university.edu"        value=
    {formData.email}      onChange={handleChange}
    className={errors.email ? 'input-error' : ''}


  />
  {errors.email && <span className="field-error">{errors.email}</span>}
</div>

<div className="form-group">
  <label>Mobile Number (10 digits) *</label>
  <input
    type="tel"
    name="phone"
    placeholder="9876543210"
    value={formData.phone}
    onChange={handleChange}
    className={errors.phone ? 'input-error' : ''}
  />
  {errors.phone && <span className="field-error">{errors.phone}</span>}
</div>

<div className="form-group">
  <label>Department *</label>
  <select
    name="department"
    value={formData.department}
    onChange={handleChange}
  >
    <option value="Computer Science">Computer Science</option>
    <option  value="Information  Tech">Information  Tech</option>
    <option value="AI & Data Science">AI & Data Science</option>
    <option value="Electronics">Electronics</option>
  </select>
</div>

<div className="form-group">
  <label>Current CGPA (0.0 - 10.0) *</label>
  <input
    type="numbe
    r"  step="0.01"
    name="cgpa"
```

```
              placeholder="8.75"
              value={formData.cgpa}
              onChange={handleChange}
              className={errors.cgpa ? 'input-error' : ''}
            />
            {errors.cgpa && <span className="field-error">{errors.cgpa}</span>}
          </div>
        </div>

        <button type="submit" className="submit-btn">
          Add Student Record
        </button>
      </form>
    );
};

export default StudentForm;
```

src/index.css CSS

```
/* Shared design system — Harmit Suthar (26084231044) · MERN Stack Development */
:root {
  --bg: #f4f6fb; --surface: #ffffff; --border: #e3e8f0; --text: #0f172a; --muted: #64748b; --
  primary: #4f46e5; --primary-hover: #4338ca; --primary-soft: #eef2ff;
  --success: #059669; --success-soft: #d1fae5;
  --warning: #d97706; --warning-soft: #fef3c7;
  --danger: #dc2626; --danger-soft: #fee2e2;
  --info: #2563eb; --info-soft: #dbeafe;
  --radius: 12px;
  --shadow: 0 1px 2px rgba(15,23,42,.05), 0 8px 24px rgba(15,23,42,.06);
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', sans-serif; background: var(--bg); color: var(--text);
line-height: 1.5; -webkit-font-smoothing: antialiased; }
button, input, select { font: inherit; }
.topbar { background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z- index: 10; }
.topbar-inner { max-width: 1120px; margin: 0 auto; padding: 14px 24px; display: flex; align-items: center; justify-
content: space-between; gap: 16px; flex-wrap: wrap; }
.brand { display: flex; align-items: center; gap: 12px; }
.brand-mark { width: 36px; height: 36px; border-radius: 10px; background: linear-gradient(135deg, #6366f1, #4338ca);
color: #fff; display: grid; place-items: center; font-weight: 700; font-size: .8rem; }
.brand-title { font-weight: 700; font-size: .95rem; }
.brand-sub { font-size: .75rem; color: var(--muted); }
.student-chip { display: flex; align-items: center; gap: 10px; background: var(--primary-soft); border: 1px solid
#c7d2fe; padding: 6px 14px 6px 6px; border-radius: 999px; }
.avatar { width: 28px; height: 28px; border-radius: 50%; background: var(--primary); color: #fff; display: grid; place-
items: center; font-size: .7rem; font-weight: 700; }
.chip-name { font-size: .82rem; font-weight: 600; color: #312e81; line-height: 1.2; }
.chip-id { font-size: .72rem; color: #4f46e5; line-height: 1.2; }
.page { max-width: 1120px; margin: 0 auto; padding: 32px 24px 56px; }
.page-head { margin-bottom: 24px; }
.eyebrow { display: inline-block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter- spacing:
.08em; color: var(--primary); background: var(--primary-soft); padding: 3px 10px; border-radius: 999px; margin-
bottom: 10px; }
.page-head h1 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.page-head p { color: var(--muted); margin-top: 6px; max-width: 70ch; }
.panel { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box- shadow: var(--
shadow); padding: 22px; }
.btn { border: 1px solid transparent; border-radius: 8px; padding: 8px 14px; font-size: .85rem; font-weight: 600;
cursor: pointer; background: var(--primary); color: #fff; transition: background .15s, border-color .15s; }
.btn:hover { background: var(--primary-hover); }
.btn:focus-visible, input:focus-visible, select:focus-visible, button:focus-visible { outline: 3px solid #c7d2fe; outline-
offset: 1px; }
.btn.secondary { background: var(--surface); color: var(--text); border-color: var(--border); } .btn.secondary:hover {
background: #f8fafc; border-color: #cbd5e1; }
.btn:disabled { opacity: .5; cursor: not-allowed; }
input[type=text], input[type=email], input[type=tel], input[type=number], select { font-size: .85rem; padding: 9px
12px; border: 1px solid var(--border); border-radius: 8px; background: #fff; color: var(--
```

```
text); }
input:focus, select:focus { border-color: var(--primary); }
table { width: 100%; border-collapse: collapse; font-size: .85rem; }
th { text-align: left; font-size: .72rem; text-transform: uppercase; letter-spacing: .06em; color: var(--
muted); background: #f8fafc; padding: 10px 12px; border-bottom: 1px solid var(--border); }
td { padding: 12px; border-bottom: 1px solid var(--border); }
tr:last-child td { border-bottom: none; }
.footer { text-align: center; color: var(--muted); font-size: .75rem; margin-top: 32px; }


.main-grid { display: grid; grid-template-columns: 1fr 1.7fr; gap: 20px; align-items: start; }
@media (max-width: 960px) { .main-grid { grid-template-columns: 1fr; } }
.form-panel, .records-panel { background: var(--surface); border: 1px solid var(--border); border-radius:
var(--radius); box-shadow: var(--shadow); padding: 22px; }
.student-form h3, .records-header h3 { font-size: 1rem; font-weight: 600; margin-bottom: 16px; }
.form-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 14px; margin-bottom: 18px; }
@media (max-width: 560px) { .form-grid { grid-template-columns: 1fr; } }
.form-group label { display: block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-
spacing: .05em; color: var(--muted); margin-bottom: 5px; }
.form-group input, .form-group select { width: 100%; }
.input-error { border-color: var(--danger) !important; background: #fef2f2 !important; }
.field-error { color: var(--danger); font-size: .75rem; margin-top: 3px; display: block; }
.submit-btn { width: 100%; background: var(--primary); color: #fff; border: none; padding: 10px; border-
radius: 8px; font-weight: 600; cursor: pointer; }
.submit-btn:hover { background: var(--primary-hover); }
.alert-success { background: var(--success-soft); color: #065f46; padding: 10px 12px; border-radius: 8px;
font-size: .85rem; margin-bottom: 14px; }
.records-header { display: flex; justify-content: space-between; align-items: center; gap: 12px; flex-wrap:
wrap; margin-bottom: 8px; }
.records-header h3 { margin-bottom: 0; }
.search-filters { display: flex; gap: 8px; flex-wrap: wrap; }
.search-input { min-width: 200px; }
.records-panel { overflow-x: auto; }
.data-table { margin-top: 12px; }
.dept-tag { background: var(--primary-soft); color: #3730a3; padding: 2px 10px; border-radius: 999px; font-
size: .72rem; font-weight: 600; white-space: nowrap; }
.cgpa-badge { background: var(--success-soft); color: #065f46; padding: 2px 10px; border-radius: 999px;
font-weight: 700; font-size: .75rem; }
.del-btn { background: var(--surface); color: var(--danger); border: 1px solid #fecaca; padding: 4px 10px;
border-radius: 8px; cursor: pointer; font-size: .78rem; font-weight: 600; }
.del-btn:hover { background: var(--danger-soft); }
```

📄 src/utils/validation.js JAVASCRIPT

```
/**
 * Form Validation Rules with Regular Expressions
*/
export const validateStudentForm = (formData) => {
  const errors = {};

  // Roll Number: e.g. 23MSD001 or CS101
  const rollRegex = /^[0-9]{2}[A-Z]{2,4}[0-9]{3,4}$/i;
  if (!formData.rollNo.trim()) {
    errors.rollNo = "Roll Number is required.";
  } else if (!rollRegex.test(formData.rollNo.trim())) {
    errors.rollNo = "Format must be like 24MSD042 (2 digits + Dept + 3-4 digits).";
  }

  // Name validation
  if (!formData.name.trim()) {
    errors.name = "Student Name is required.";
  } else if (formData.name.trim().length < 3) {
    errors.name = "Name must be at least 3 characters long.";
  }

  // Email validation
  const emailRegex = /^[\w-\.]+@([\w-]+\.)+[\w-]{2,4}$/;
  if (!formData.email.trim()) {
    errors.email = "Email Address is required.";
  } else if (!emailRegex.test(formData.email.trim())) {
    errors.email = "Enter a valid email address (e.g. student@college.edu).";
```

```
  }

  // Phone validation: 10-digit Indian phone format
  const phoneRegex = /^[6-9]\d{9}$/;
  if (!formData.phone.trim()) {
    errors.phone = "Phone number is required.";
  } else if (!phoneRegex.test(formData.phone.trim())) {
    errors.phone = "Enter a valid 10-digit mobile number starting with 6-9.";
  }

  // Department validation
  if (!formData.department) {
    errors.department = "Please select a department.";
  }

  // CGPA validation: 0.00 to 10.00
  const cgpaNum = parseFloat(formData.cgpa);
  if (!formData.cgpa) {
    errors.cgpa = "CGPA is required.";
  } else if (isNaN(cgpaNum) || cgpaNum < 0 || cgpaNum > 10) {
    errors.cgpa = "CGPA must be a valid number between 0.00 and 10.00.";
  }

  return {
    isValid: Object.keys(errors).length === 0,
    errors
  };
};
```

5. Expected Output & Verification Results

All screenshots and logs below were captured from the real running application of this practical (Harmit Suthar · 26084231044).

Figure 5.1 — Student registration form with the seeded student records table.

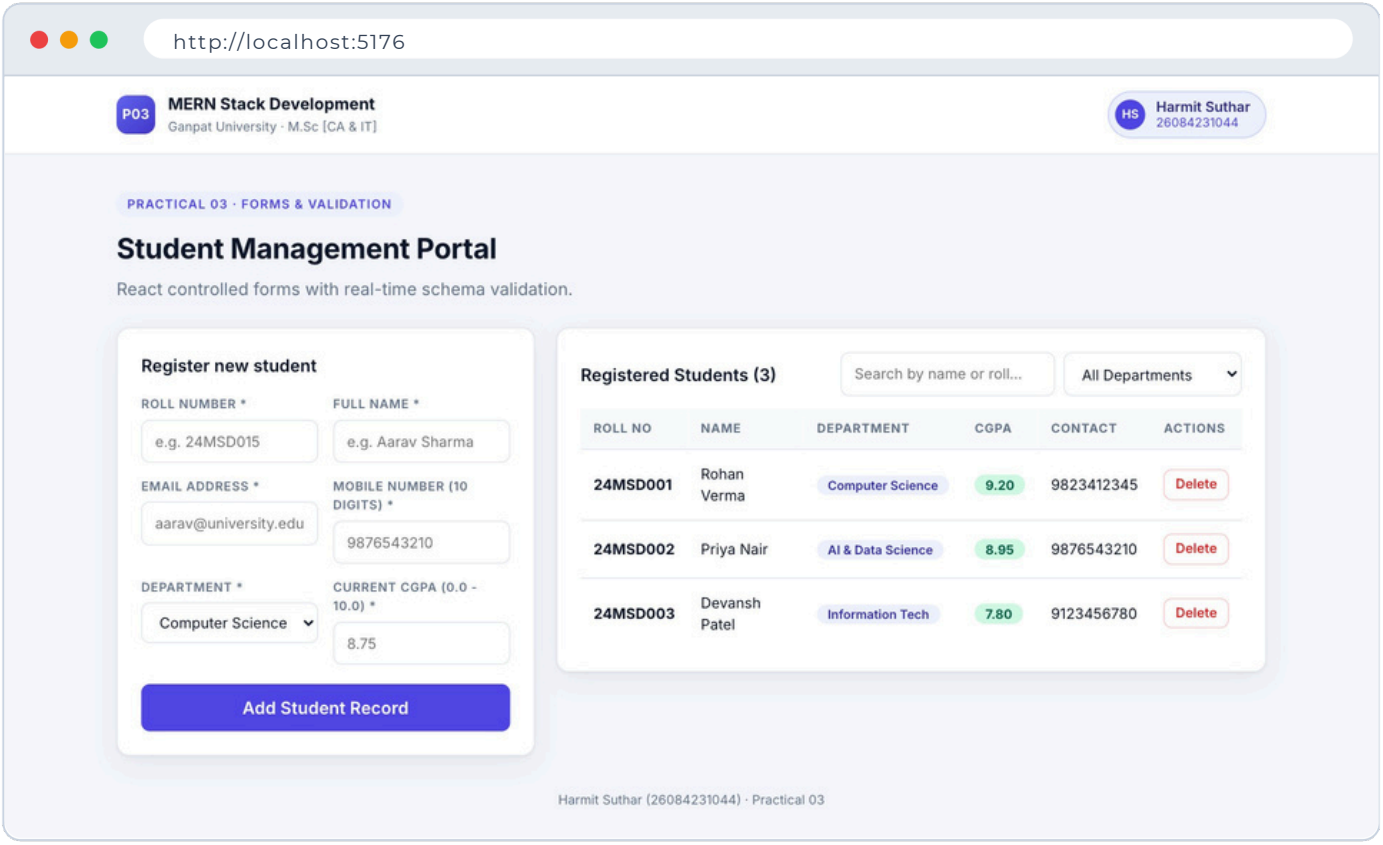


Figure 5.2 — Submitting an empty form triggers real-time regular-expression validation errors.

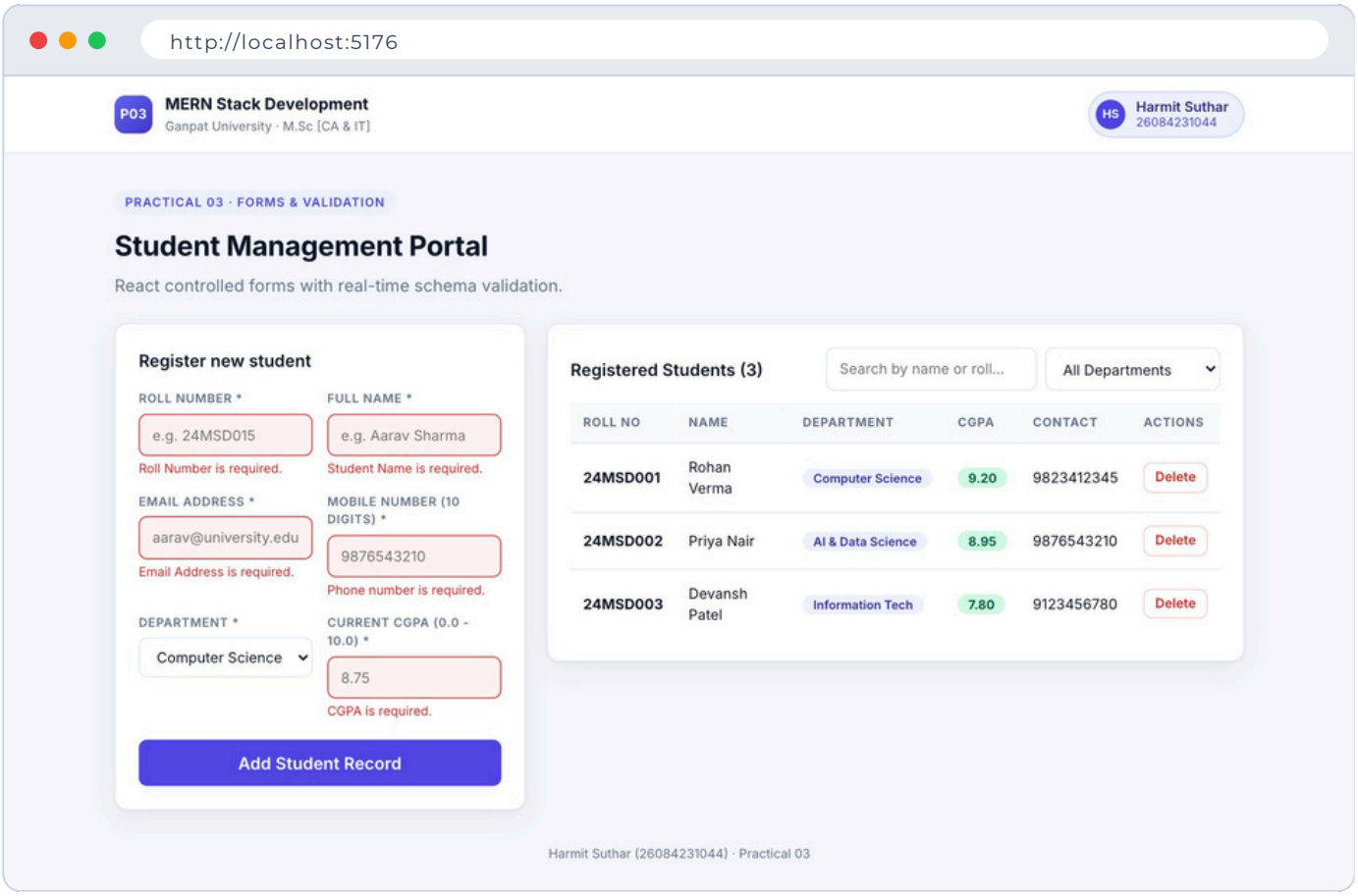
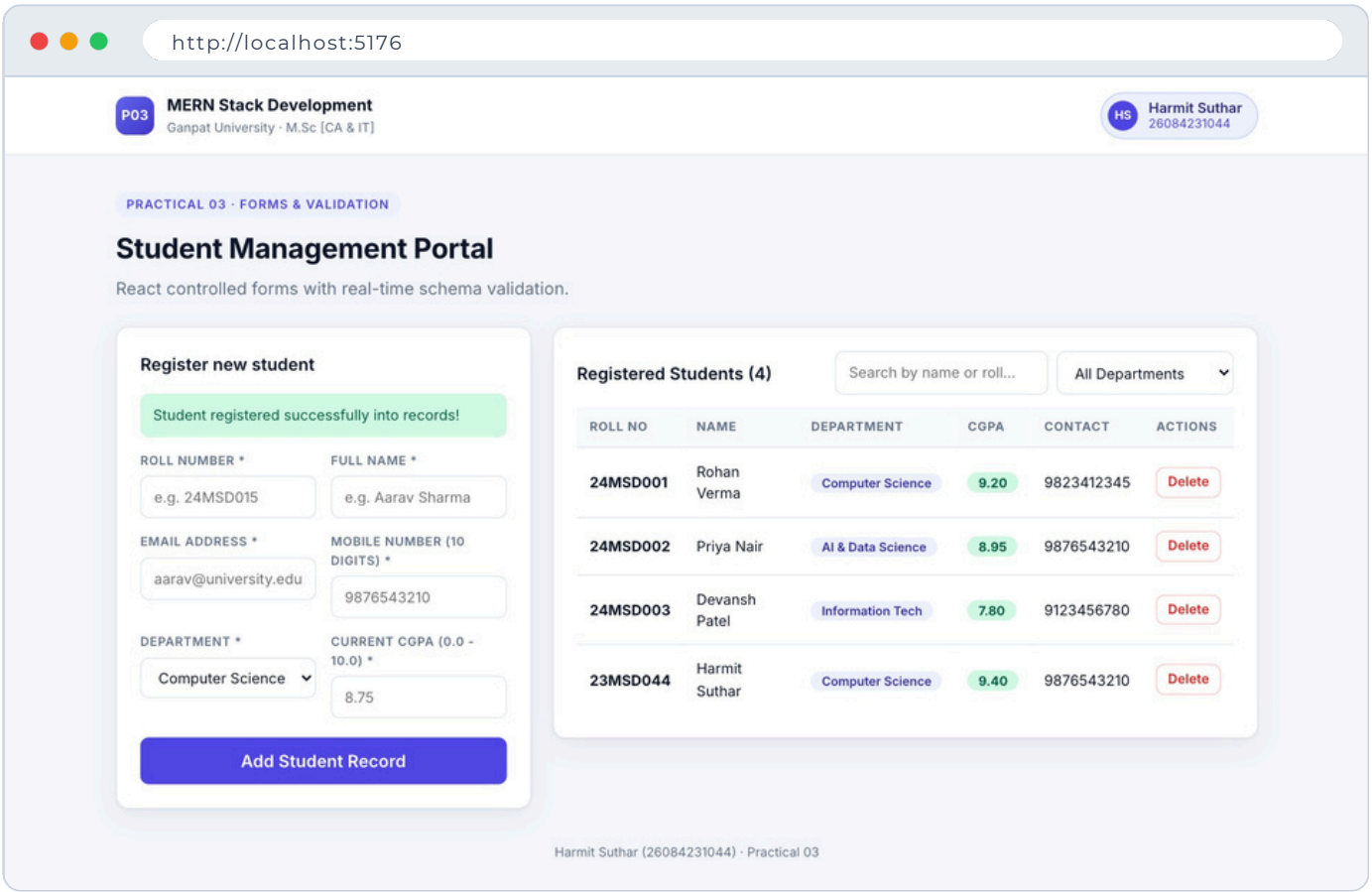


Figure 5.3 — A valid record is accepted, the form resets and the new student appears in the table.



🖥️ **Execution & Server Logs** **Port 5176**

```
$ npm run dev
> practical-03-student-forms@1.0.0 dev
> vite


  VITEv5.4.21 ready in 171 ms

  ➜  Local:   http://localhost:5176/
  ➜  Network: use --host to expose
```

PRACTICAL 4

React.js / Routing & SPA Architecture

Develop a Single Page Application (SPA) using React Router for navigation between multiple pages

1. Practical Aim & Theoretical Definition

Aim & Theoretical Concept

The objective of this practical is to build a full-featured Single Page Application (SPA) using react-router-dom (v6+), implementing client-side routing, active link styling, dynamic route parameters (useParams), programmatic navigation (useNavigate), and fallback 404 views without full page reloads.

Core Theoretical Concepts:

- **Single Page Application (SPA):** ***A web application that dynamically rewrites the current web page rather than loading entire new pages from a server. SPA architecture eliminates white screen flashes and delivers instantaneous navigation.***
- **Client-Side Routing:** ***The browser's History API (pushState, replaceState) intercepts URL changes in the address bar. React Router matches the URL pathname against defined routes and swaps out component views in the DOM without requesting HTML from the server.***
- **Key React Router v6 Hooks & Primitives:**
  - `<BrowserRouter>`***: Top-level wrapper managing routing state via HTML5 history API.***
  - `<Routes>` & `<Route>`***: Declarative mapping between URL paths and UI components.***
  - `<NavLink>`***: Provides active class flags (isActive) for navigation bars.***
  - `useParams()`***: Extracts dynamic tokens from URL path (e.g. /courses/:courseId).***
  - `useNavigate()`***: Programmatically moves the user back, forward, or to specific routes.***

2. Step-by-Step Commands (Installation & Execution)

1. Navigate to the Practical Directory

```
$ cd practicals/p4
```

Action: *Switches the working directory to the Practical 4 project folder.*

2. Install Dependencies

```
$ npm install
```

Action: *Installs every package declared in package.json into node_modules.*

3. Run Local Development Server

```
$ npm run dev
```

Action: *Starts the SPA with React Router v6 at http://localhost:5177.*

3. Project Directory & File Structure

```
practical-04/
├── src/
│   ├── components/
│   │   ├── AppBar.jsx
│   │   └── Navbar.jsx
│   ├── pages/
│   │   └── CourseDetail.jsx
│   ├── App.jsx
│   ├── index.css
│   └── main.jsx
├── index.html
├── package-lock.json
├── package.json
├── README.md
├── run.bat
└── vite.config.js
```

File / Directory Path	Architectural Purpose & Responsibility
src/components/AppBar.jsx	***Shared top bar showing the practical number and the student badge.***
src/components/Navbar.jsx	***Navigation bar built with NavLink and active-route styling.***
src/pages/CourseDetail.jsx	***Dynamic route page reading :courseId with useParams() and useNavigate().***
src/App.jsx	***Root React component orchestrating state, layout and child components.***
src/index.css	***Global styles, design tokens and component styling.***
src/main.jsx	***Mounts the root React component into the DOM with ReactDOM.createRoot().***
index.html	***HTML entry point containing the #root mount node.***
package-lock.json	***Exact dependency tree lockfile generated by npm.***
package.json	***Project metadata, scripts and dependency declarations.***
README.md	***Project documentation.***
run.bat	***Windows launcher script that installs and starts the project.***
vite.config.js	***Vite build and dev-server configuration.***

4. Modular Source Code Implementation

Complete source code of the project, listed exactly as it exists in `practicals/p4` ***:***

src/main.jsx JAVASCRIPT (JSX)

```jsx
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode><App /></React.StrictMode>
);
```

src/App.jsx JAVASCRIPT (JSX)

```jsx
import React from 'react';
import { BrowserRouter, Routes, Route, Link } from 'react-router-dom';
import Navbar from './components/Navbar';
import CourseDetail from './pages/CourseDetail';

const Home = () => (
  <div className="page-view">
    <h2>Welcome to the EduSphere Single Page Application</h2>
    <p>Seamless client-side navigation powered by React Router v6. No page reloads!</p>
    <div className="feature-cards">
      <div className="card">Instant route swapping</div>
      <div className="card">Dynamic URL parameters</div>
      <div className="card">Robust 404 fallback</div>
    </div>
  </div>
);

const Courses = () => (
  <div className="page-view">
    <h2>Available academic courses</h2>
    <div className="course-cards-list">
      <div className="course-card">
        <h3>Full-Stack MERN Architecture</h3>
        <p>Master React, Node, Express and MongoDB.</p>
        <Link to="/courses/cs101" className="details-btn">View syllabus (CS101) →</Link>
      </div>
      <div className="course-card">
        <h3>Cloud Microservices & Docker</h3>
        <p>Learn containerization, API gateways and service meshes.</p>
        <Link to="/courses/cs102" className="details-btn">View syllabus (CS102) →</Link>
      </div>
      <div className="course-card">
        <h3>DevOps & Cloud Automation</h3>
        <p>CI/CD pipelines with GitHub Actions and AWS.</p>
        <Link to="/courses/cs103" className="details-btn">View syllabus (CS103) →</Link>
      </div>
    </div>
  </div>
);

const NotFound = () => (
  <div className="page-view not-found-page">
    <h1>404</h1>
    <h3>Oops! Page Not Found</h3>
    <p>The requested route does not exist in this Single Page Application.</p>
    <Link to="/" className="home-btn">Return to Home</Link>
  </div>
);

function App() {
  return (
    <BrowserRouter>
      <div className="spa-wrapper">
        <Navbar />
        <main className="content-area">
          <Routes>
```

```
            <Route path="/" element={<Home />} />
            <Route path="/courses" element={<Courses />} />
            <Route path="/courses/:courseId" element={<CourseDetail />} />
            <Route path="*" element={<NotFound />} />
          </Routes>
        </main>
      </div>
    </BrowserRouter>
  );
}

export default App;
```

src/components/AppBar.jsx — JAVASCRIPT (JSX)

```
import React from 'react';

const AppBar = ({ practical, title = 'MERN Stack Development' }) => (
  <header className="topbar">
    <div className="topbar-inner">
      <div className="brand">
        <div className="brand-mark">P{practical}</div>
        <div>
          <div className="brand-title">{title}</div>
          <div className="brand-sub">Ganpat University · M.Sc [CA &amp;amp; IT]</div>
        </div>
      </div>
      <div className="student-chip">
        <div className="avatar">HS</div>
        <div>
          <div className="chip-name">Harmit Suthar</div>
          <div className="chip-id">26084231044</div>
        </div>
      </div>
    </div>
); </header>


export default AppBar;
```

src/components/Navbar.jsx — JAVASCRIPT (JSX)

```
import React from 'react';
import { NavLink } from 'react-router-dom';

const Navbar = () => {
  return (
    <nav className="topbar">
      <div className="topbar-inner">
      <div className="brand">
        <div className="brand-mark">P4</div>
        <div>
          <div className="brand-title">EduSphere SPA</div>
          <div className="brand-sub">Harmit Suthar · 26084231044</div>
        </div>
      </div>
      <div className="nav-links">
        <NavLink to="/" className={({ isActive }) => isActive ? 'nav-item active' : 'nav-item'}>
          Home
        </NavLink>
        <NavLink to="/courses" className={({ isActive }) => isActive ? 'nav-item active' : 'nav-item'}>
          Courses
        </NavLink>
        <NavLink to="/about" className={({ isActive }) => isActive ? 'nav-item active' : 'nav-item'}>
        About Us
        </NavLink>
        <NavLink to="/contact" className={({ isActive }) => isActive ? 'nav-item active' : 'nav-item'}>
```

```
            Contact
          </NavLink>
        </div>
        </div>
      </nav>
    );
};

export default Navbar;
```

src/index.css CSS

```
/* Shared design system — Harmit Suthar (26084231044) · MERN Stack Development */
:root {
  --bg: #f4f6fb; --surface: #ffffff; --border: #e3e8f0; --text: #0f172a; --muted: #64748b; --
  primary: #4f46e5; --primary-hover: #4338ca; --primary-soft: #eef2ff;
  --success: #059669; --success-soft: #d1fae5;
  --warning: #d97706; --warning-soft: #fef3c7;
  --danger: #dc2626; --danger-soft: #fee2e2;
  --info: #2563eb; --info-soft: #dbeafe;
  --radius: 12px;
  --shadow: 0 1px 2px rgba(15,23,42,.05), 0 8px 24px rgba(15,23,42,.06);
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', sans-serif; background: var(--bg); color: var(--text);
line-height: 1.5; -webkit-font-smoothing: antialiased; }
button, input, select { font: inherit; }
.topbar { background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z- index: 10; }
.topbar-inner { max-width: 1120px; margin: 0 auto; padding: 14px 24px; display: flex; align-items: center; justify-
content: space-between; gap: 16px; flex-wrap: wrap; }
.brand { display: flex; align-items: center; gap: 12px; }
.brand-mark { width: 36px; height: 36px; border-radius: 10px; background: linear-gradient(135deg, #6366f1, #4338ca);
color: #fff; display: grid; place-items: center; font-weight: 700; font-size: .8rem; }
.brand-title { font-weight: 700; font-size: .95rem; }
.brand-sub { font-size: .75rem; color: var(--muted); }
.student-chip { display: flex; align-items: center; gap: 10px; background: var(--primary-soft); border: 1px solid
#c7d2fe; padding: 6px 14px 6px 6px; border-radius: 999px; }
.avatar { width: 28px; height: 28px; border-radius: 50%; background: var(--primary); color: #fff; display: grid; place-
items: center; font-size: .7rem; font-weight: 700; }
.chip-name { font-size: .82rem; font-weight: 600; color: #312e81; line-height: 1.2; }
.chip-id { font-size: .72rem; color: #4f46e5; line-height: 1.2; }
.page { max-width: 1120px; margin: 0 auto; padding: 32px 24px 56px; }
.page-head { margin-bottom: 24px; }
.eyebrow { display: inline-block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter- spacing:
.08em; color: var(--primary); background: var(--primary-soft); padding: 3px 10px; border-radius: 999px; margin-
bottom: 10px; }
.page-head h1 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.page-head p { color: var(--muted); margin-top: 6px; max-width: 70ch; }
.panel { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box- shadow: var(--
shadow); padding: 22px; }
.btn { border: 1px solid transparent; border-radius: 8px; padding: 8px 14px; font-size: .85rem; font-weight: 600;
cursor: pointer; background: var(--primary); color: #fff; transition: background .15s, border-color .15s; }
.btn:hover { background: var(--primary-hover); }
.btn:focus-visible, input:focus-visible, select:focus-visible, button:focus-visible { outline: 3px solid #c7d2fe; outline-
offset: 1px; }
.btn.secondary { background: var(--surface); color: var(--text); border-color: var(--border); } .btn.secondary:hover {
background: #f8fafc; border-color: #cbd5e1; }
.btn:disabled { opacity: .5; cursor: not-allowed; }
input[type=text], input[type=email], input[type=tel], input[type=number], select { font-size: .85rem; padding: 9px
12px; border: 1px solid var(--border); border-radius: 8px; background: #fff; color: var(-- text); }
input:focus, select:focus { border-color: var(--primary); }
table { width: 100%; border-collapse: collapse; font-size: .85rem; }
th { text-align: left; font-size: .72rem; text-transform: uppercase; letter-spacing: .06em; color: var(-- muted);
background: #f8fafc; padding: 10px 12px; border-bottom: 1px solid var(--border); }
td { padding: 12px; border-bottom: 1px solid var(--border); }
tr:last-child td { border-bottom: none; }
.footer { text-align: center; color: var(--muted); font-size: .75rem; margin-top: 32px; }
```

```
.topbar .nav-links { display: flex; gap: 4px;}
.nav-item { color: var(--muted); text-decoration:none; font-size:.88rem; font-weight: 500; padding: 7px 14px; border-radius: 8px; }
.nav-item:hover { background: #f1f5f9; color: var(--text);}
.nav-item.active { color: var(--primary); background: var(--primary-soft); font-weight: 600;}
.content-area { max-width: 1120px; margin: 0 auto; padding: 36px 24px 56px;}
.page-view h2 { font-size: 1.6rem; font-weight: 700; letter-spacing:-.01em; margin-bottom: 6px;}
.page-view > p { color: var(--muted); margin-bottom: 28px;}
.feature-cards, .course-cards-list { display: grid; grid-template-columns: repeat(auto-fit, minmax(260px, 1fr)); gap: 20px; }
.feature-cards .card { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-shadow: var(--shadow); padding: 22px; font-weight: 600;}
.course-card { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-shadow: var(--shadow); padding: 22px; display: flex; flex-direction:column; justify-content:space-between; transition: transform.2s; }
.course-card:hover { transform: translateY(-3px);}
.course-card h3 { font-size: 1.05rem; margin-bottom: 8px;}
.course-card p { color: var(--muted); font-size:.88rem; margin-bottom: 18px;}
.details-btn, .home-btn { display:inline-block; background: var(--primary); color: #fff; text-decoration: none; padding: 9px 14px; border-radius: 8px; font-size:.85rem; text-align:center; font-weight: 600;}
.details-btn:hover, .home-btn:hover { background: var(--primary-hover);}
.course-detail-card, .course-detail-container { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-shadow:var(--shadow); padding: 28px;}
.back-link, .back-btn { background:none; border: none; color: var(--primary); cursor:pointer; font-size: .88rem; font-weight: 600; margin-bottom:18px;}
.detail-header { margin-bottom: 22px; }
.course-code-tag { background: var(--primary-soft); color: #3730a3; padding: 3px 10px; border-radius: 999px; font-size: .75rem; font-weight: 700;}
.detail-header h2 { margin: 12px 0 4px; }
.instructor-text { color: var(--muted); font-size:.88rem;}
.syllabus-box { background: #f8fafc; border: 1px solid var(--border); border-radius: 10px; padding: 18px; margin-bottom: 18px;}
.syllabus-box h3 { font-size:.95rem; margin-bottom: 10px;}
.syllabus-box ul { margin-left: 20px; color: #334155; font-size:.9rem; line-height: 1.8;}
.prereq-box { color: var(--muted); font-size: .88rem; margin-bottom: 22px;}
.enroll-btn { background: var(--primary); color: #fff; border:none; padding: 10px 20px; border-radius: 8px; font-weight: 600; cursor: pointer; }
.enroll-btn:hover { background: var(--primary-hover);}
.not-found-page { text-align:center; padding: 60px 0;}
.not-found-page h1 { font-size: 5rem; color: var(--primary);}
.not-found-page h3 { margin: 6px 0; }
```

src/pages/CourseDetail.jsx — JAVASCRIPT (JSX)

```
import React from 'react';
import { useParams, useNavigate } from 'react-router-dom';

const COURSE_DATABASE = {
  'cs101': {
    title: "Full-Stack MERN Architecture",
    instructor: "Prof. Arvind Kulkarni",
    duration: "14 Weeks",
    syllabus: ["ES6+ & React 18", "Express.js REST APIs", "MongoDB Schema Design", "Deployment & CI/CD"],
    prerequisites: "JavaScript Fundamentals & Basic Web Concepts"
  },
  'cs102': {
    title: "Cloud Microservices & Docker",
    instructor: "Dr. Shalini Mehta",
    duration: "10 Weeks",
    syllabus: ["Containers with Docker", "Kubernetes Pods", "API Gateways", "Event Queues"],
    prerequisites: "Node.js & Linux CLI basics"
  },
  'cs103': {
    title: "DevOps & Cloud Automation",
    instructor: "Er. Rahul Saxena",
    duration: "8 Weeks",
    syllabus: ["GitHub Actions", "Terraform IaC", "AWS Cloud Infrastructure", "Monitoring with Prometheus"],
    prerequisites: "Git Version Control"
  }
```

```
};

const CourseDetail = () => {
  const { courseId } = useParams();
  const navigate = useNavigate();
  const course = COURSE_DATABASE[courseId?.toLowerCase()];

  if (!course) {
    return (
      <div className="course-detail-container not-found">
        <h2>Course not found</h2>
        <p>No course exists matching identifier: <code>{courseId}</code></p>
        <button onClick={() => navigate('/courses')} className="back-btn">
          ← Back to Courses Catalog
        </button>
      </div>
    );
  }

  return (
    <div className="course-detail-card">
      <button onClick={() => navigate('/courses')} className="back-link">
        ← Back to All Courses
      </button>

      <div className="detail-header">
        <span className="course-code-tag">{courseId.toUpperCase()}</span>
        <h2>{course.title}</h2>
        <p className="instructor-text">Lead instructor: {course.instructor} | Duration: {course.duration}
</p>
      </div>

      <div className="syllabus-box">
        <h3>Curriculum Modules</h3>
        <ul>
          {course.syllabus.map((topic, idx) => (
            <li key={idx}>Module {idx + 1}: {topic}</li>
          ))}
        </ul>
      </div>

      <div className="prereq-box">
        <strong>Prerequisites:</strong> {course.prerequisites}
      </div>

      <button className="enroll-btn" onClick={() => alert(`Enrolled in ${course.title}`)}>
        Proceed to Course Modules
      </button>
    </div>
  );
};

export default CourseDetail;
```

5. Expected Output & Verification Results

All screenshots and logs below were captured from the real running application of this practical (Harmit Suthar · 26084231044).

Figure 5.1 — Home route rendered by React Router without a page reload.

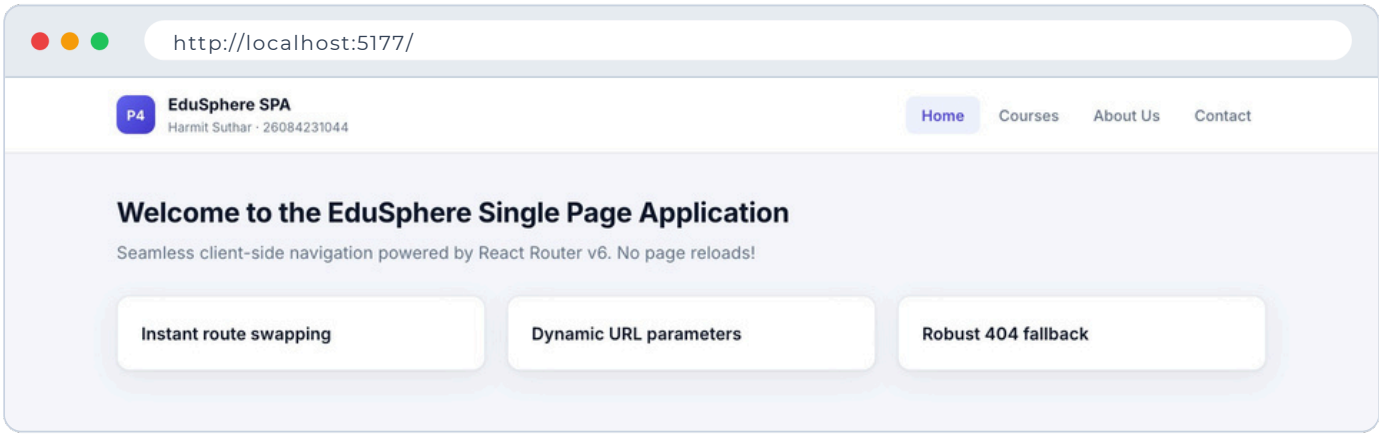


Figure 5.2 — Courses route listing syllabus links.

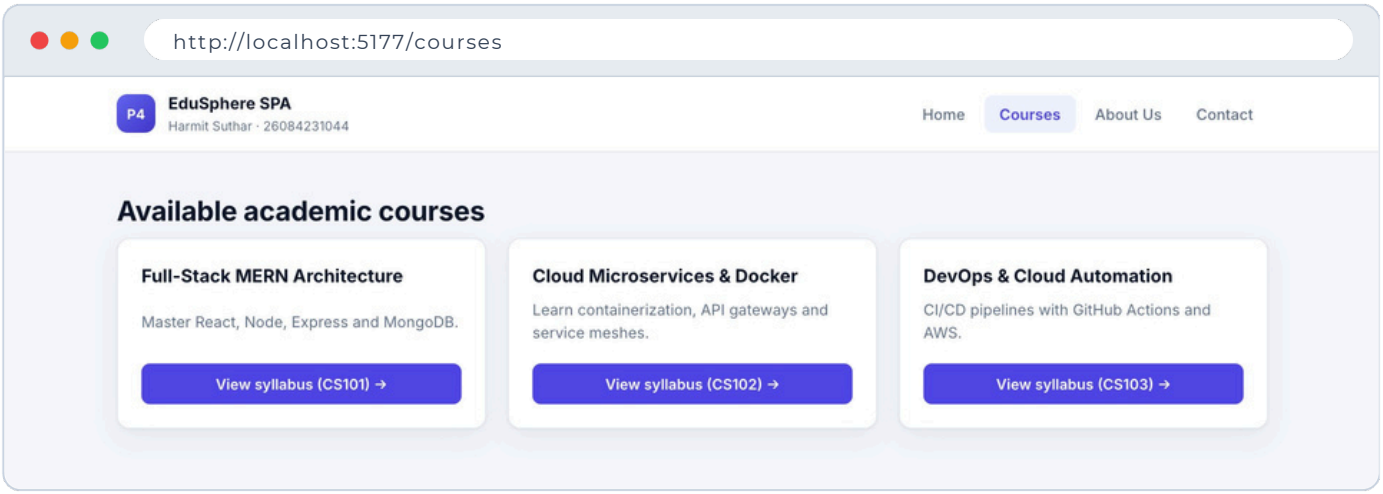


Figure 5.3 — Dynamic route /courses/:courseId resolved with useParams().

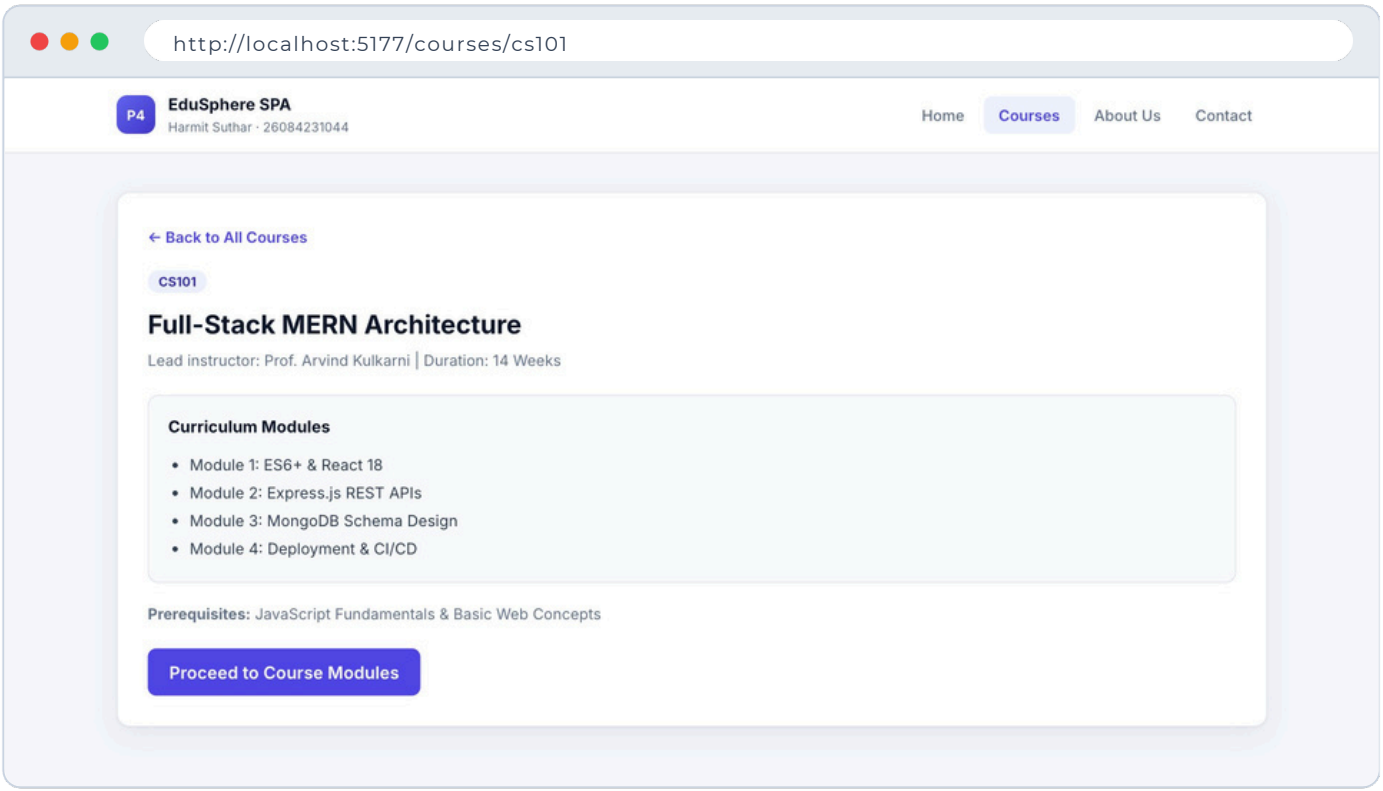
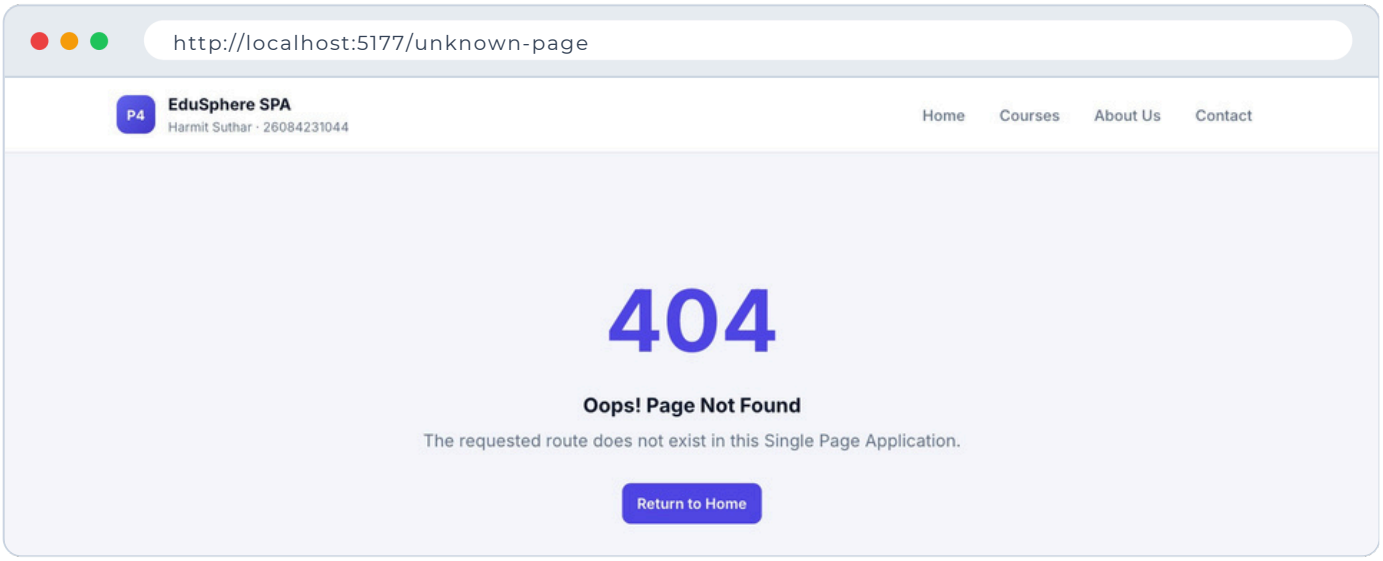


Figure 5.4 — Wildcard route displaying the 404 fallback page.



Execution & Server Logs — Port 5177

```
$ npm run dev
> practical-04-react-router@1.0.0 dev
> vite


  VITEv5.4.21 ready in 195 ms

  ➜  Local:   http://localhost:5177/
  ➜  Network: use --host to expose
```

PRACTICAL 5

Backend Development / Express.js

Create a backend server using Express.js implementing routing and CRUD operation

1. Practical Aim & Theoretical Definition

Aim & Theoretical Concept

The objective of this practical is to build an asynchronous backend server using Express.js, implementing modular RESTful Routing, processing incoming JSON request payloads via express.json(), and executing full CRUD (Create, Read, Update, Delete) operations with standard HTTP status codes (200 OK, 201 Created, 400 Bad Request, 404 Not Found, 500 Internal Error).

Core Theoretical Concepts:

- **Express.js Framework:** ***A minimalist, fast, unopinionated web framework for Node.js that simplifies handling HTTP requests, routing, middleware pipelines, and view rendering.***
- **RESTful Routing Conventions:**
  - `GET /api/items`***: Read all resources (Returns 200 OK).***
  - `GET /api/items/`***: id: Read single resource by unique identifier (200 OK or 404 Not Found).***
  - `POST /api/items`***: Create a new resource from request body (Returns 201 Created).***
  - `PUT /api/items/`***: id: Update existing resource completely or idempotently (200 OK or 404).***
  - `DELETE /api/items/`***: id: Remove resource permanently (200 OK or 204 No Content).***
- **Request & Response Architecture:** ***The req object encapsulates query params (req.query), route parameters (req.params), and parsed body (req.body). The res object dispatches status codes and JSON payloads (res.status(200).json(...)).***

2. Step-by-Step Commands (Installation & Execution)

1. Navigate to the Practical Directory

```
$ cd practicals/p5
```

Action: *Switches the working directory to the Practical 5 project folder.*

2. Install Dependencies

```
$ npm install
```

Action: *Installs every package declared in package.json into node_modules.*

3. Run the Express Server

```
$ npm run dev
```

Action: *Starts the API with auto-restart (node --watch) on http://localhost:5005.*

4. Test the API with cURL

```
$ curl -i http://localhost:5005/api/items
```

Action: *Sends an HTTP GET request to verify routing and the JSON response.*

3. Project Directory & File Structure

```
practical-05/
├── controllers/
│   └── itemController.js
├── public/
│   ├── index.html
│   └── theme.css
├── routes/
├── itemRoutes.js
├── package-lock.json
├── package.json
├── README.md
├── run.bat
└── server.js
```

File / Directory Path	Architectural Purpose & Responsibility
controllers/itemController.js	***CRUD handlers for items: list, get, create, update and delete with status codes.***
public/index.html	***Front-end test console that calls the API and displays the live response.***
public/theme.css	***Shared design system stylesheet used by the page.***
routes/itemRoutes.js	***Express Router mapping REST verbs and URLs to the item controller.***
package-lock.json	***Exact dependency tree lockfile generated by npm.***
package.json	***Project metadata, scripts and dependency declarations.***
README.md	***Project documentation.***
run.bat	***Windows launcher script that installs and starts the project.***
server.js	***Express server entry point: middleware setup, routes and listener.***

4. Modular Source Code Implementation

Complete source code of the project, listed exactly as it exists in `practicals/p5` ***:***

server.js JAVASCRIPT

```
const express = require('express');
const cors = require('cors');
const path = require('path');

const app = express();
const PORT = 5005;

app.use(cors());
app.use(express.json());
app.use(express.static(path.join(__dirname, 'public')));

let items = [
  { id: 1, name: "Dell XPS 15", category: "Laptops", price: 145000, inStock: true },
  { id: 2, name: "Logitech MX Master 3S", category: "Accessories", price: 8999, inStock: true },
  { id: 3, name: "Keychron K2 Mechanical", category: "Keyboards", price: 7499, inStock: false }
];

app.get('/api/items', (req, res) => res.json({ success: true, count: items.length, data: items }));
app.get('/api/items/:id', (req, res) => {
  const item = items.find(i => i.id === parseInt(req.params.id));
  if (!item) return res.status(404).json({ success: false, message: "Item not found" });
  res.json({ success: true, data: item });
});
app.post('/api/items', (req, res) => {
  const { name, category, price } = req.body;
  if (!name || !price) return res.status(400).json({ success: false, message: "Name and Price are required"
});
  const newItem = { id: items.length ? Math.max(...items.map(i => i.id)) + 1 : 1, name, category: category
|| "General", price: Number(price), inStock: true };
  items.push(newItem);
  res.status(201).json({ success: true, message: "Item created", data: newItem });
});
app.delete('/api/items/:id', (req, res) => {
  const id = parseInt(req.params.id);
  const idx = items.findIndex(i => i.id === id);
  if (idx === -1) return res.status(404).json({ success: false, message: "Item not found" });
  const del = items.splice(idx, 1)[0];
  res.json({ success: true, message: "Item deleted", data: del });
});

app.listen(PORT, () => console.log("Practical 05 running on http://localhost:" + PORT));
```

controllers/itemController.js JAVASCRIPT

```
// In-memory initial data store
let items = [
  { id: 1, name: "Dell XPS 15", category: "Laptops", price: 145000, inStock: true },
  { id: 2, name: "Logitech MX Master 3S", category: "Accessories", price: 8999, inStock: true },
  { id: 3, name: "Keychron K2 Mechanical", category: "Keyboards", price: 7499, inStock: false }
];

// GET: Read all items
exports.getAllItems = (req, res) => {
  res.status(200).json({
    success: true,
    count: items.length,
    data: items
  });
};

// GET: Read single item by ID
exports.getItemById = (req, res) => {
  const itemId = parseInt(req.params.id);
  const item = items.find((i) => i.id === itemId);

  if (!item) {
```

```
      return res.status(404).json({
        success: false,
        message: `Item with ID ${itemId} not found`
      });
    }

    res.status(200).json({
      success: true,
      data: item
    });
};

// POST: Create a new item
exports.createItem = (req, res) => {
    const { name, category, price, inStock } = req.body;

    if (!name || !price) {
      return res.status(400).json({
        success: false,
        message: "Validation Error: 'name' and 'price' are mandatory fields."
      });
    }

    const newItem = {
      id: items.length > 0 ? Math.max(...items.map((i) => i.id)) + 1 : 1,
      name,
      category: category || "General",
      price: Number(price),
      inStock: inStock !== undefined ? inStock : true
    };

    items.push(newItem);

    res.status(201).json({
      success: true,
      message: "Item created successfully",
      data: newItem
    });
};

// PUT: Update item by ID
exports.updateItem = (req, res) => {
    const itemId = parseInt(req.params.id);
    const itemIndex = items.findIndex((i) => i.id === itemId);

    if (itemIndex === -1) {
      return res.status(404).json({
        success: false,
        message: `Cannot update: Item with ID ${itemId} does not exist`
      });
    }

    const { name, category, price, inStock } = req.body;

    items[itemIndex] = {
      ...items[itemIndex],
      ...(name && { name }),
      ...(category && { category }),
      ...(price !== undefined && { price: Number(price) }),
      ...(inStock !== undefined && { inStock })
    };

    res.status(200).json({
      success: true,
      message: `Item ID ${itemId} updated successfully`,
      data: items[itemIndex]
    });
};

// DELETE: Remove item by ID
exports.deleteItem = (req, res) => {
    const itemId = parseInt(req.params.id);
```

```
  const itemIndex = items.findIndex((i) => i.id === itemId);

  if (itemIndex === -1) {
    return res.status(404).json({
      success: false,
      message: `Cannot delete: Item with ID ${itemId} not found`
    });
  }

  const deletedItem = items.splice(itemIndex, 1)[0];

  res.status(200).json({
    success: true,
    message: `Item '${deletedItem.name}' deleted successfully`,
    data: deletedItem
  });
};
```

📄 public/index.html HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Practical 05 · Express Routing & CRUD API</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?
family=Inter:wght@400;500;600;700&family=JetBrains+Mono:wght@400;500&display=swap" rel="stylesheet">
  <link rel="stylesheet" href="/theme.css">
</head>
<body>
  <header class="topbar">
    <div class="topbar-inner">
      <div class="brand">
        <div class="brand-mark">P05</div>
        <div>
          <div class="brand-title">MERN Stack Development</div>
          <div class="brand-sub">Ganpat University · M.Sc [CA &amp;amp; IT]</div>
        </div>
      </div>
      <div class="student-chip">
        <div class="avatar">HS</div>
        <div>
          <div class="chip-name">Harmit Suthar</div>
          <div class="chip-id">26084231044</div>
        </div>
      </div>
    </div>
  </header>

  <main class="page">
    <div class="page-head">
      <span class="eyebrow">Practical 05 · Express.js</span>
      <h1>Express Routing & CRUD API</h1>
      <p>Exercise the REST endpoints for the item catalogue and inspect each JSON response.</p>
    </div>
    <div class="card">
      <div class="toolbar">
        <button class="btn secondary" onclick="fetchItems()">GET /api/items</button>
        <input type="text" id="newName" placeholder="Item name" value="Sony WH-1000XM5" aria-label="Item
name">
        <input type="number" id="newPrice" placeholder="Price" value="26990" aria-label="Price">
        <button class="btn success" onclick="addItem()">POST /api/items</button>
      </div>
      <div class="output-head"><span class="label">Response</span><span class="status-pill"
id="status">Idle</span></div>
        <pre class="output" id="output">// Run a request to see the response…</pre>
    </div>
```

```
    <p class="footer">Harmit Suthar (26084231044) · Practical 05</p>
  </main>
  <script>
    const out = document.getElementById('output');
    const statusEl = document.getElementById('status');
    function show(res, json) {
      statusEl.textContent = res ? res.status + ' ' + (res.statusText || (res.ok ? 'OK' : 'Error')) :
'Done';
      statusEl.className = 'status-pill ' + (!res ? '' : res.ok ? 'ok' : res.status >= 500 ? 'warn' :
'err');
      out.textContent = JSON.stringify(json, null, 2);
    }
    async function fetchItems() {
      const res = await fetch('/api/items');
      show(res, await res.json());
    }
    async function addItem() {
      const name = document.getElementById('newName').value;
      const price = document.getElementById('newPrice').value;
      const res = await fetch('/api/items', { method: 'POST', headers: {'Content-Type': 'application/json'},
body: JSON.stringify({name, price, category: 'Electronics'}) });
      show(res, await res.json());
    }
    fetchItems();
  </script>
</body
>
</html
>
```

public/theme.css — CSS

```
/* Shared design system — Harmit Suthar (26084231044) · MERN Stack Development */
:root {
  --bg: #f4f6fb;
  --surface: #ffffff;
  --border: #e3e8f0;
  --text: #0f172a;
  --muted: #64748b;
  --primary: #4f46e5;
  --primary-hover: #4338ca;
  --primary-soft: #eef2ff;
  --success: #059669; --success-soft: #d1fae5; --
  warning: #d97706; --warning-soft: #fef3c7;

  --danger: #dc2626; --danger-soft: #fee2e2;
  --info: #0284c7; --info-soft: #e0f2fe;
  --code-bg: #0f172a; --code-text: #a7f3d0;
  --radius: 12px;
  --shadow: 0 1px 2px rgba(15,23,42,.05), 0 8px 24px rgba(15,23,42,.06);
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', sans-serif; background: var(--bg); color: var(--text);
line-height: 1.5; -webkit-font-smoothing: antialiased; }


.topbar { background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z-
index: 10; }
.topbar-inner { max-width: 1040px; margin: 0 auto; padding: 14px 24px; display: flex; align-items: center;
justify-content: space-between; gap: 16px; flex-wrap: wrap; }
.brand { display: flex; align-items: center; gap: 12px; }
.brand-mark { width: 36px; height: 36px; border-radius: 10px; background: linear-gradient(135deg, #6366f1,
#4338ca); color: #fff; display: grid; place-items: center; font-weight: 700; font-size: .8rem; letter-
spacing: .02em; }
.brand-title { font-weight: 700; font-size: .95rem; }
.brand-sub { font-size: .75rem; color: var(--muted); }
.student-chip { display: flex; align-items: center; gap: 10px; background: var(--primary-soft); border: 1px
solid #c7d2fe; padding: 6px 14px 6px 6px; border-radius: 999px; }
.avatar { width: 28px; height: 28px; border-radius: 50%; background: var(--primary); color: #fff; display:
grid; place-items: center; font-size: .7rem; font-weight: 700; }
.chip-name { font-size: .82rem; font-weight: 600; color: #312e81; }
.chip-id { font-size: .72rem; color: #4f46e5; }


.page { max-width: 1040px; margin: 0 auto; padding: 32px 24px 56px; }
```

```
.page-head { margin-bottom: 24px; }
.eyebrow { display: inline-block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-spacing: .08em; color: var(--primary); background: var(--primary-soft); padding: 3px 10px; border-radius: 999px; margin-bottom: 10px; }
.page-head h1 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.page-head p { color: var(--muted); margin-top: 6px; max-width: 70ch; }


.card { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-shadow: var(--shadow); padding: 22px; }
.card + .card { margin-top: 20px; }
.card-title { font-size: .95rem; font-weight: 600; margin-bottom: 14px; display: flex; align-items: center; justify-content: space-between; gap: 8px; }
.grid-2 { display: grid; grid-template-columns: 1.2fr 1fr; gap: 20px; }
@media (max-width: 760px) { .grid-2 { grid-template-columns: 1fr; } }


.toolbar { display: flex; flex-wrap: wrap; gap: 10px; align-items: center; margin-bottom: 16px; }
.btn { font: inherit; font-size: .85rem; font-weight: 600; border: 1px solid transparent; border-radius: 8px; padding: 8px 14px; cursor: pointer; transition: background .15s, border-color .15s, box-shadow .15s; background: var(--primary); color: #fff; }
.btn:hover { background: var(--primary-hover); }
.btn:focus-visible { outline: 3px solid #c7d2fe; outline-offset: 1px; }
.btn.secondary { background: var(--surface); color: var(--text); border-color: var(--border); }
.btn.secondary:hover { background: #f8fafc; border-color: #cbd5e1; }
.btn.success { background: var(--success); } .btn.success:hover { background: #047857; }

.btn.danger { background: var(--danger); }       .btn.danger:hover { background: #b91c1c; }
.btn.warning { background: var(--warning); } .btn.warning:hover { background: #b45309; }
.btn.violet { background: #7c3aed; } .btn.violet:hover { background: #6d28d9; }

input[type=text], input[type=number], input[type=file] { font: inherit; font-size: .85rem; padding: 8px 12px; border: 1px solid var(--border); border-radius: 8px; background: #fff; color: var(--text); }
input:focus { outline: 3px solid #c7d2fe; border-color: var(--primary); }
input[type=file] { width: 100%; background: #f8fafc; border-style: dashed; padding: 14px; }

.label { font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-spacing: .06em; color: var(--muted); }
.output-head { display: flex; justify-content: space-between; align-items: center; margin-bottom: 8px; }
.status-pill { font-size: .72rem; font-weight: 700; padding: 2px 10px; border-radius: 999px; background: #e2e8f0; color: #334155; }
.status-pill.ok { background: var(--success-soft); color: #065f46; }
.status-pill.warn { background: var(--warning-soft); color: #92400e; }
.status-pill.err { background: var(--danger-soft); color: #991b1b; }
pre.output { background: var(--code-bg); color: var(--code-text); padding: 16px; border-radius: 10px; font: 12.5px/1.6 'JetBrains Mono', ui-monospace, Menlo, Consolas, monospace; max-height: 380px; overflow: auto; white-space: pre-wrap; word-break: break-word; }


.test-row { display: flex; justify-content: space-between; align-items: center; gap: 12px; padding: 10px 12px; border: 1px solid var(--border); border-radius: 8px; font-size: .85rem; }
.test-row + .test-row { margin-top: 8px; }
.test-row .pass { color: var(--success); font-weight: 600; }
.test-row .time { color: var(--muted); font-size: .78rem; }

table { width: 100%; border-collapse: collapse; font-size: .85rem; }
th { text-align: left; font-size: .72rem; text-transform: uppercase; letter-spacing: .06em; color: var(--muted); background: #f8fafc; padding: 10px 12px; border-bottom: 1px solid var(--border); }
td { padding: 12px; border-bottom: 1px solid var(--border); }
tr:last-child td { border-bottom: none; }
.badge { display: inline-block; font-size: .72rem; font-weight: 600; padding: 2px 10px; border-radius: 999px; background: var(--success-soft); color: #065f46; }
a.link { color: var(--primary); font-weight: 600; text-decoration: none; }
a.link:hover { text-decoration: underline; }
.muted { color: var(--muted); }
.footer { text-align: center; color: var(--muted); font-size: .75rem; margin-top: 32px; }
```

routes/itemRoutes.js JAVASCRIPT

```
const express = require('express');
const router = express.Router();
const itemController = require('../controllers/itemController');

// Define REST Endpoints
```

```
router.get('/', itemController.getAllItems);
router.get('/:id', itemController.getItemById);
router.post('/', itemController.createItem);
router.put('/:id', itemController.updateItem);
router.delete('/:id', itemController.deleteItem);

module.exports = router;
```

5. Expected Output & Verification Results

All screenshots and logs below were captured from the real running application of this practical (Harmit Suthar · 26084231044).

Figure 5.1 — GET /api/items returns the seeded catalogue with 200 OK.

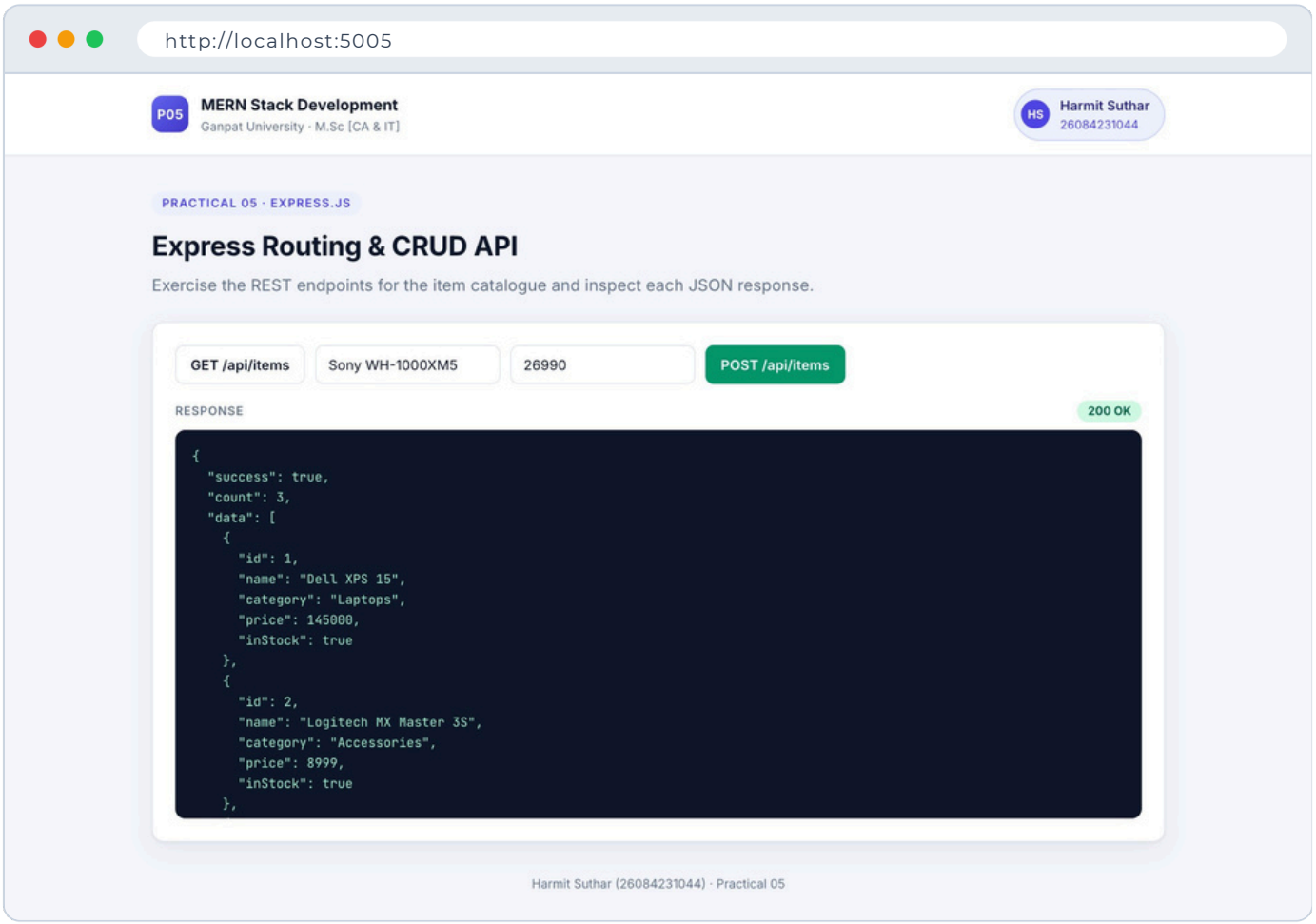
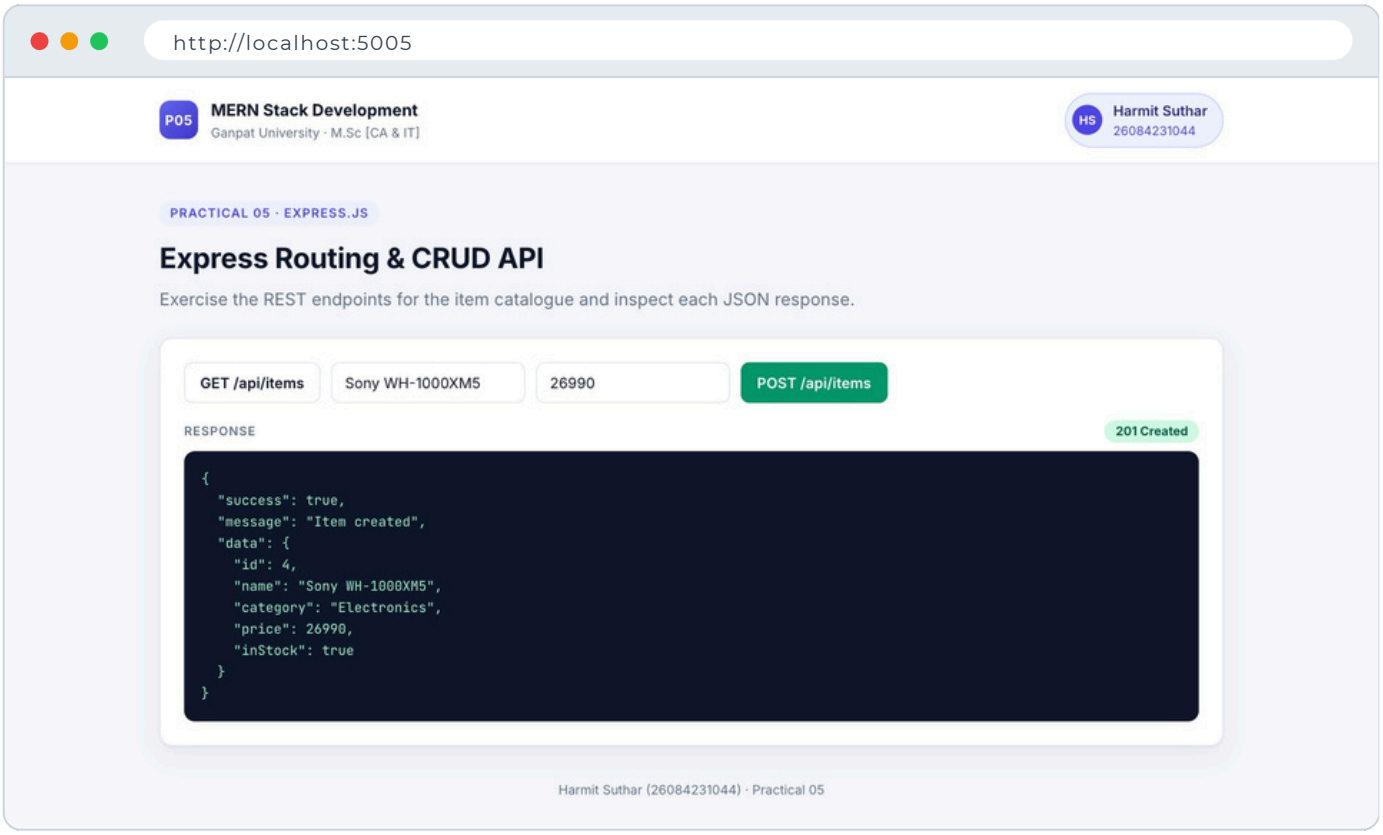


Figure 5.2 — POST /api/items creates a new item and returns the created resource.



API Verification (curl)

```
$ curl -i http://localhost:5005/api/items
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 390
Date: Sat, 24 Oct 2026 18:17:45 GMT

{
  "success": true,
  "count": 4,
  "data": [
    {
      "id": 1,
      "name": "Dell XPS 15",
      "category": "Laptops",
      "price": 145000,
      "inStock": true
    },
    {
      "id": 2,
      "name": "Logitech MX Master 3S",
      "category": "Accessories",
      "price": 8999,
      "inStock": true
    },
    {
      "id": 3,
      "name": "Keychron K2 Mechanical",
      "category": "Keyboards",
      "price": 7499,
      "inStock": false
    },
    {
      "id": 4,
      "name": "Sony WH-1000XM5",
      "category": "Electronics",
      "price": 26990,
      "inStock": true
    }
  ]
}
```

Execution & Server Logs **Port 5005**

```
$ npm run dev
> practical-05-express-crud@1.0.0 dev
> node --watch server.js

Practical 05 running on http://localhost:5005
```

PRACTICAL 6

Backend Development / REST Architecture

Build RESTful APIs for Student Records Management using Express.js and Postman testing

1. Practical Aim & Theoretical Definition

Aim & Theoretical Concept

The objective of this practical is to architect production-ready RESTful APIs for managing university Student Academic Records using Express.js. The API incorporates Query Filtering (by department & semester), Pagination (limit, page), Sorting (by GPA), request validation middleware, and automated test assertion verification in Postman.

Core Theoretical Concepts:

- **REST (Representational State Transfer):** ***An architectural style enforcing statelessness, uniform interfaces, resource-based URIs, and standard HTTP verbs.***
- **Query Parameters for Filtering & Pagination:** ***Real-world APIs cannot return thousands of records at once. Query parameters (e.g. ?dept=CSE&page=1&limit=10&sortBy=gpa) empower clients to request sliced and sorted datasets efficiently.***
- **Input Validation Middleware:** ***Intercepting bad client payloads before controller execution prevents dirty records and potential server crashes.***
- **Postman Automated Testing:** ***Writing JavaScript test scripts inside Postman's "Tests" tab:***
- ***pm.test("Status code is 200", () => pm.response.to.have.status(200));***
- ***pm.test("Response contains pagination metadata", () => {***
- ***const json = pm.response.json();***
- ***pm.expect(json).to.have.property('page');***
- ***pm.expect(json.data).to.be.an('array');***
- ***});***

2. Step-by-Step Commands (Installation & Execution)

1. Navigate to the Practical Directory

```
$ cd practicals/p6
```

Action: *Switches the working directory to the Practical 6 project folder.*

2. Install Dependencies

```
$ npm install
```

Action: *Installs every package declared in package.json into node_modules.*

3. Run the Express Server

```
$ npm run dev
```

Action: *Starts the Student Records API on http://localhost:5006.*

4. Test Filters with cURL / Postman

```
$ curl "http://localhost:5006/api/students?dept=CSE&sortBy=gpa&order=desc"
```

Action: *Exercises the department filter and GPA sorting query parameters.*

3. Project Directory & File Structure

```
practical-06/
├── controllers/
│   └── studentController.js
├── middleware/
│   └── studentValidator.js
├── public/
├── index.html
├── theme.css
├── package-lock.json
├── package.json
├── README.md
├── run.bat
└── server.js
```

File / Directory Path	Architectural Purpose & Responsibility
controllers/studentController.js	***Student handlers with filtering, search, sorting and pagination.***
middleware/studentValidator.js	***Middleware validating the student request payload before it reaches the controller.***
public/index.html	***Front-end test console that calls the API and displays the live response.***
public/theme.css	***Shared design system stylesheet used by the page.***
package-lock.json	***Exact dependency tree lockfile generated by npm.***
package.json	***Project metadata, scripts and dependency declarations.***
README.md	***Project documentation.***
run.bat	***Windows launcher script that installs and starts the project.***
server.js	***Express server entry point: middleware setup, routes and listener.***

4. Modular Source Code Implementation

Complete source code of the project, listed exactly as it exists in `practicals/p6` ***:***

server.js — JAVASCRIPT

```
const express = require('express');
const cors = require('cors');
const path = require('path');

const app = express();
const PORT = 5006;

app.use(cors());
app.use(express.json());
app.use(express.static(path.join(__dirname, 'public')));

let students = [
  { id: 1, rollNo: "24MSD001", name: "Rohan Verma", department: "CSE", semester: 4, gpa: 9.20 },
  { id: 2, rollNo: "24MSD002", name: "Priya Nair", department: "AI-DS", semester: 4, gpa: 8.95 },
  { id: 3, rollNo: "24MSD003", name: "Devansh Patel", department: "IT", semester: 4, gpa: 7.80 },
  { id: 4, rollNo: "26084231044", name: "Harmit Suthar", department: "CSE", semester: 4, gpa: 9.80 }
];

app.get('/api/students', (req, res) => {
  let { dept, sortBy, order } = req.query;
  let result = [...students];
  if (dept) result = result.filter(s => s.department.toLowerCase() === dept.toLowerCase());
  if (sortBy === 'gpa') result.sort((a,b) => order === 'desc' ? b.gpa - a.gpa : a.gpa - b.gpa);
  res.json({ success: true, count: result.length, data: result });
});

app.listen(PORT, () => console.log("Practical 06 running on http://localhost:" + PORT));
```

controllers/studentController.js — JAVASCRIPT

```
let students = [
  { id: 1, rollNo: "24MSD001", name: "Rohan Verma", email: "rohan@univ.edu", department: "CSE", semester: 4,
gpa: 9.20 },
  { id: 2, rollNo: "24MSD002", name: "Priya Nair", email: "priya@univ.edu", department: "AI-DS", semester:
4, gpa: 8.95 },
  { id: 3, rollNo: "24MSD003", name: "Devansh Patel", email: "devansh@univ.edu", department: "IT", semester:
4, gpa: 7.80 },
  { id: 4, rollNo: "24MSD004", name: "Ananya Iyer", email: "ananya@univ.edu", department: "CSE", semester:
4, gpa: 9.60 },
  { id: 5, rollNo: "24MSD005", name: "Kabir Singh", email: "kabir@univ.edu", department: "ECE", semester: 4,
gpa: 8.10 }
];

// GET: Filtered, paginated & sorted student records
exports.getStudents = (req, res) => {
  let { dept, search, sortBy, order, page = 1, limit = 10 } = req.query;

  page = parseInt(page);
  limit = parseInt(limit);

  let result = [...students];

  // Department filter
  if (dept) {
    result = result.filter(s => s.department.toLowerCase() === dept.toLowerCase());
  }

  // Search by name or rollNo
  if (search) {
    const q = search.toLowerCase();
    result = result.filter(s => s.name.toLowerCase().includes(q) || s.rollNo.toLowerCase().includes(q));
```

```
    }

    // Sorting
    if (sortBy === 'gpa') {
        result.sort((a, b) => order === 'asc' ? a.gpa - b.gpa : b.gpa - a.gpa);
    } else if (sortBy === 'name') {
        result.sort((a, b) => order === 'desc' ? b.name.localeCompare(a.name) : a.name.localeCompare(b.name));
    }

    // Pagination calculation
    const total = result.length;
    const startIndex = (page - 1) * limit;
    const paginatedData = result.slice(startIndex, startIndex + limit);

    res.status(200).json({
        success: true,
        page,
        limit,
        totalRecords: total,
        totalPages: Math.ceil(total / limit),
        data: paginatedData
    });
};

// POST: Add new student
exports.createStudent = (req, res) => {
    const { rollNo, name, email, department, semester, gpa } = req.body;

    // Check unique rollNo
    if (students.some(s => s.rollNo.toLowerCase() === rollNo.toLowerCase())) {
        return res.status(409).json({
            success: false,
            message: `Conflict: Student with roll number '${rollNo}' already exists.`
        });
    }

    const newStudent = {
        id: students.length > 0 ? Math.max(...students.map(s => s.id)) + 1 : 1,
        rollNo: rollNo.toUpperCase(),
        name,
        email,
        department,
        semester: parseInt(semester) || 1,
        gpa: parseFloat(gpa)
    };

    students.push(newStudent);

    res.status(201).json({
        success: true,
        message: "Student record registered successfully",
        data: newStudent
    });
};
```

middleware/studentValidator.js JAVASCRIPT

```
exports.validateStudentPayload = (req, res, next) => {
    const { rollNo, name, email, department, gpa } = req.body;
    const errors = [];

    if (!rollNo || typeof rollNo !== 'string') errors.push("Valid 'rollNo' string is required.");
    if (!name || name.trim().length < 3) errors.push("'name' must be at least 3 characters.");

    const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!email || !emailRegex.test(email)) errors.push("Valid 'email' address is required.");

    if (!department) errors.push("'department' is required.");

    const numGpa = parseFloat(gpa);
```

```
  if (isNaN(numGpa) || numGpa < 0 || numGpa > 10) {
    errors.push("'gpa' must be a valid decimal number between 0.00 and 10.00.");
  }

  if (errors.length > 0) {
    return res.status(422).json({
      success: false,
      message: "Validation Failed",
      errors
    });
  }

  next();
};
```

public/index.html HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Practical 06 · Student Records API & Postman Testing</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?
family=Inter:wght@400;500;600;700&family=JetBrains+Mono:wght@400;500&display=swap" rel="stylesheet">
  <link rel="stylesheet" href="/theme.css">
</head>
<body>
  <header class="topbar">
    <div class="topbar-inner">
      <div class="brand">
        <div class="brand-mark">P06</div>
        <div>
          <div class="brand-title">MERN Stack Development</div>
          <div class="brand-sub">Ganpat University · M.Sc [CA &amp;amp; IT]</div>
        </div>
      </div>
      <div class="student-chip">
        <div class="avatar">HS</div>
        <div>
          <div class="chip-name">Harmit Suthar</div>
          <div class="chip-id">26084231044</div>
        </div>
      </div>
    </div>
  </header>

  <main class="page">
    <div class="page-head">
      <span class="eyebrow">Practical 06 · REST · Postman</span>
      <h1>Student Records API & Postman Testing</h1>
      <p>Query the student records API with filters and sorting, alongside the Postman assertions used to
verify it.</p>
    </div>
    <div    class="grid-2">
    <div class="card">
        <div class="card-title">Query filters</div>
        <div class="toolbar">
          <button class="btn secondary" onclick="fetchQuery('')">All students</button>
          <button class="btn secondary" onclick="fetchQuery('?dept=CSE')">dept=CSE</button>
          <button class="btn secondary" onclick="fetchQuery('?sortBy=gpa&amp;order=desc')">Top GPA</button>
        </div>
        <div class="output-head"><span class="label">Response</span><span class="status-pill"
id="status">Idle</span></div>
        <pre class="output" id="output">Loading…</pre>
      </div>
      <div class="card">
        <div class="card-title">Postman test assertions</div>
```

```
          <div class="test-row"><span class="pass">✔ Status code is 200 OK</span><span class="time">PASS · 14ms</span></div>
          <div class="test-row"><span class="pass">✔ Response returns JSON array</span><span class="time">PASS · 14ms</span></div>
          <div class="test-row"><span class="pass">✔ Department CSE filter verified</span><span class="time">PASS · 14ms</span></div>
          <div class="test-row"><span class="pass">✔ GPA sorting integrity checked</span><span class="time">PASS · 14ms</span></div>
        </div>
      </div>
      <p class="footer">Harmit Suthar (26084231044) · Practical 06</p>
    </main>
    <script>
      const out = document.getElementById('output');
      const statusEl = document.getElementById('status');
      function show(res, json) {
        statusEl.textContent = res ? res.status + ' ' + (res.statusText || (res.ok ? 'OK' : 'Error')) : 'Done';
        statusEl.className = 'status-pill ' + (!res ? '' : res.ok ? 'ok' : res.status >= 500 ? 'warn' : 'err');
        out.textContent = JSON.stringify(json, null, 2);
      }
      async function fetchQuery(q) {
        const res = await fetch('/api/students' + q);
        show(res, await res.json());
      }
      fetchQuery('');
    </script>
</body
>
</html
>
```

public/theme.css CSS

```
/* Shared design system — Harmit Suthar (26084231044) · MERN Stack Development */
:root {
  --bg: #f4f6fb;
  --surface: #ffffff;
  --border: #e3e8f0;
  --text: #0f172a;
  --muted: #64748b;
  --primary: #4f46e5;
  --primary-hover: #4338ca;
  --primary-soft: #eef2ff;
  --success: #059669; --success-soft: #d1fae5; --warning: #d97706; --warning-soft: #fef3c7;
  --danger: #dc2626; --danger-soft: #fee2e2;
  --info: #0284c7; --info-soft: #e0f2fe;
  --code-bg: #0f172a; --code-text: #a7f3d0;
  --radius: 12px;
  --shadow: 0 1px 2px rgba(15,23,42,.05), 0 8px 24px rgba(15,23,42,.06);
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', sans-serif; background: var(--bg); color: var(--text); line-height: 1.5; -webkit-font-smoothing: antialiased; }


.topbar { background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z-index: 10; }
.topbar-inner { max-width: 1040px; margin: 0 auto; padding: 14px 24px; display: flex; align-items: center; justify-content: space-between; gap: 16px; flex-wrap: wrap; }
.brand { display: flex; align-items: center; gap: 12px; }
.brand-mark { width: 36px; height: 36px; border-radius: 10px; background: linear-gradient(135deg, #6366f1, #4338ca); color: #fff; display: grid; place-items: center; font-weight: 700; font-size: .8rem; letter-spacing: .02em; }
.brand-title { font-weight: 700; font-size: .95rem; }
.brand-sub { font-size: .75rem; color: var(--muted); }
.student-chip { display: flex; align-items: center; gap: 10px; background: var(--primary-soft); border: 1px solid #c7d2fe; padding: 6px 14px 6px 6px; border-radius: 999px; }
.avatar { width: 28px; height: 28px; border-radius: 50%; background: var(--primary); color: #fff; display: grid; place-items: center; font-size: .7rem; font-weight: 700; }
.chip-name { font-size: .82rem; font-weight: 600; color: #312e81; }
```

```
.chip-id { font-size: .72rem; color: #4f46e5; }

.page { max-width: 1040px; margin: 0 auto; padding: 32px 24px 56px; }
.page-head { margin-bottom: 24px; }
.eyebrow { display: inline-block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-
spacing: .08em; color: var(--primary); background: var(--primary-soft); padding: 3px 10px; border-radius:
999px; margin-bottom: 10px; }
.page-head h1 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.page-head p { color: var(--muted); margin-top: 6px; max-width: 70ch; }


.card { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-
shadow: var(--shadow); padding: 22px; }
.card + .card { margin-top: 20px; }
.card-title { font-size: .95rem; font-weight: 600; margin-bottom: 14px; display: flex; align-items: center;
justify-content: space-between; gap: 8px; }
.grid-2 { display: grid; grid-template-columns: 1.2fr 1fr; gap: 20px; }
@media (max-width: 760px) { .grid-2 { grid-template-columns: 1fr; } }


.toolbar { display: flex; flex-wrap: wrap; gap: 10px; align-items: center; margin-bottom: 16px; }
.btn { font: inherit; font-size: .85rem; font-weight: 600; border: 1px solid transparent; border-radius:
8px; padding: 8px 14px; cursor: pointer; transition: background .15s, border-color .15s, box-shadow .15s;
background: var(--primary); color: #fff; }
.btn:hover { background: var(--primary-hover); }
.btn:focus-visible { outline: 3px solid #c7d2fe; outline-offset: 1px; }
.btn.secondary { background: var(--surface); color: var(--text); border-color: var(--border); }
.btn.secondary:hover { background: #f8fafc; border-color: #cbd5e1; }
.btn.success { background: var(--success); } .btn.success:hover { background: #047857; }

.btn.danger { background: var(--danger); }       .btn.danger:hover { background: #b91c1c; }
.btn.warning { background: var(--warning); } .btn.warning:hover { background: #b45309; }
.btn.violet { background: #7c3aed; } .btn.violet:hover { background: #6d28d9; }

input[type=text], input[type=number], input[type=file] { font: inherit; font-size: .85rem; padding: 8px
12px; border: 1px solid var(--border); border-radius: 8px; background: #fff; color: var(--text); }
input:focus { outline: 3px solid #c7d2fe; border-color: var(--primary); }
input[type=file] { width: 100%; background: #f8fafc; border-style: dashed; padding: 14px; }

.label { font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-spacing: .06em; color: var(-
-muted); }
.output-head { display: flex; justify-content: space-between; align-items: center; margin-bottom: 8px; }
.status-pill { font-size: .72rem; font-weight: 700; padding: 2px 10px; border-radius: 999px; background:
#e2e8f0; color: #334155; }
.status-pill.ok { background: var(--success-soft); color: #065f46; }
.status-pill.warn { background: var(--warning-soft); color: #92400e; }
.status-pill.err { background: var(--danger-soft); color: #991b1b; }
pre.output { background: var(--code-bg); color: var(--code-text); padding: 16px; border-radius: 10px; font:
12.5px/1.6 'JetBrains Mono', ui-monospace, Menlo, Consolas, monospace; max-height: 380px; overflow: auto;
white-space: pre-wrap; word-break: break-word; }


.test-row { display: flex; justify-content: space-between; align-items: center; gap: 12px; padding: 10px
12px; border: 1px solid var(--border); border-radius: 8px; font-size: .85rem; }
.test-row + .test-row { margin-top: 8px; }
.test-row .pass { color: var(--success); font-weight: 600; }
.test-row .time { color: var(--muted); font-size: .78rem; }


table { width: 100%; border-collapse: collapse; font-size: .85rem; }
th { text-align: left; font-size: .72rem; text-transform: uppercase; letter-spacing: .06em; color: var(--
muted); background: #f8fafc; padding: 10px 12px; border-bottom: 1px solid var(--border); }
td { padding: 12px; border-bottom: 1px solid var(--border); }
tr:last-child td { border-bottom: none; }
.badge { display: inline-block; font-size: .72rem; font-weight: 600; padding: 2px 10px; border-radius:
999px; background: var(--success-soft); color: #065f46; }
a.link { color: var(--primary); font-weight: 600; text-decoration: none; }
a.link:hover { text-decoration: underline; }
.muted { color: var(--muted); }
.footer { text-align: center; color: var(--muted); font-size: .75rem; margin-top: 32px; }
```

5. Expected Output & Verification Results

All screenshots and logs below were captured from the real running application of this practical (Harmit Suthar · 26084231044).

Figure 5.1 — GET /api/students returns every student record.

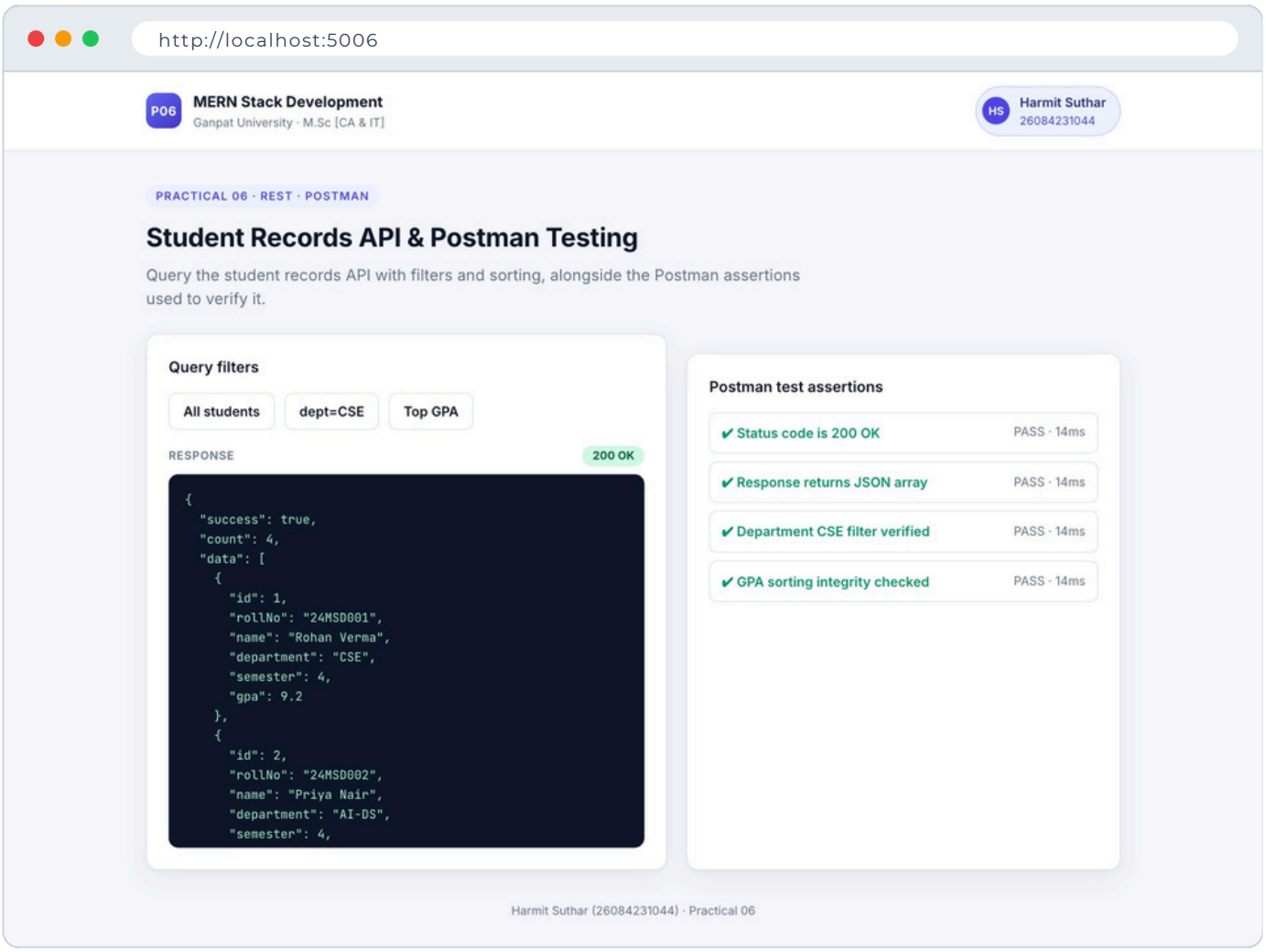


Figure 5.2 — Query filter ?dept=CSE narrows the result set.

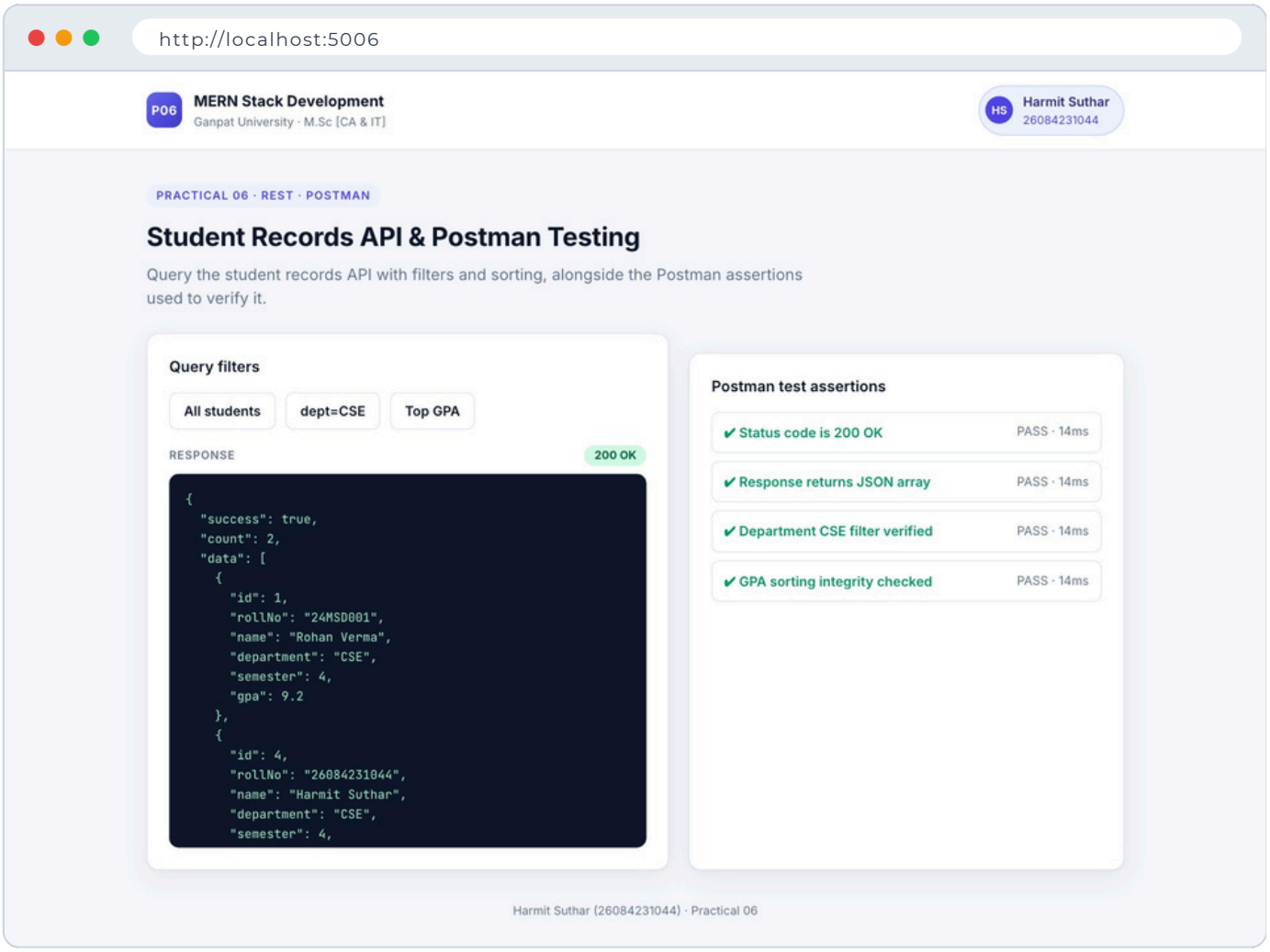
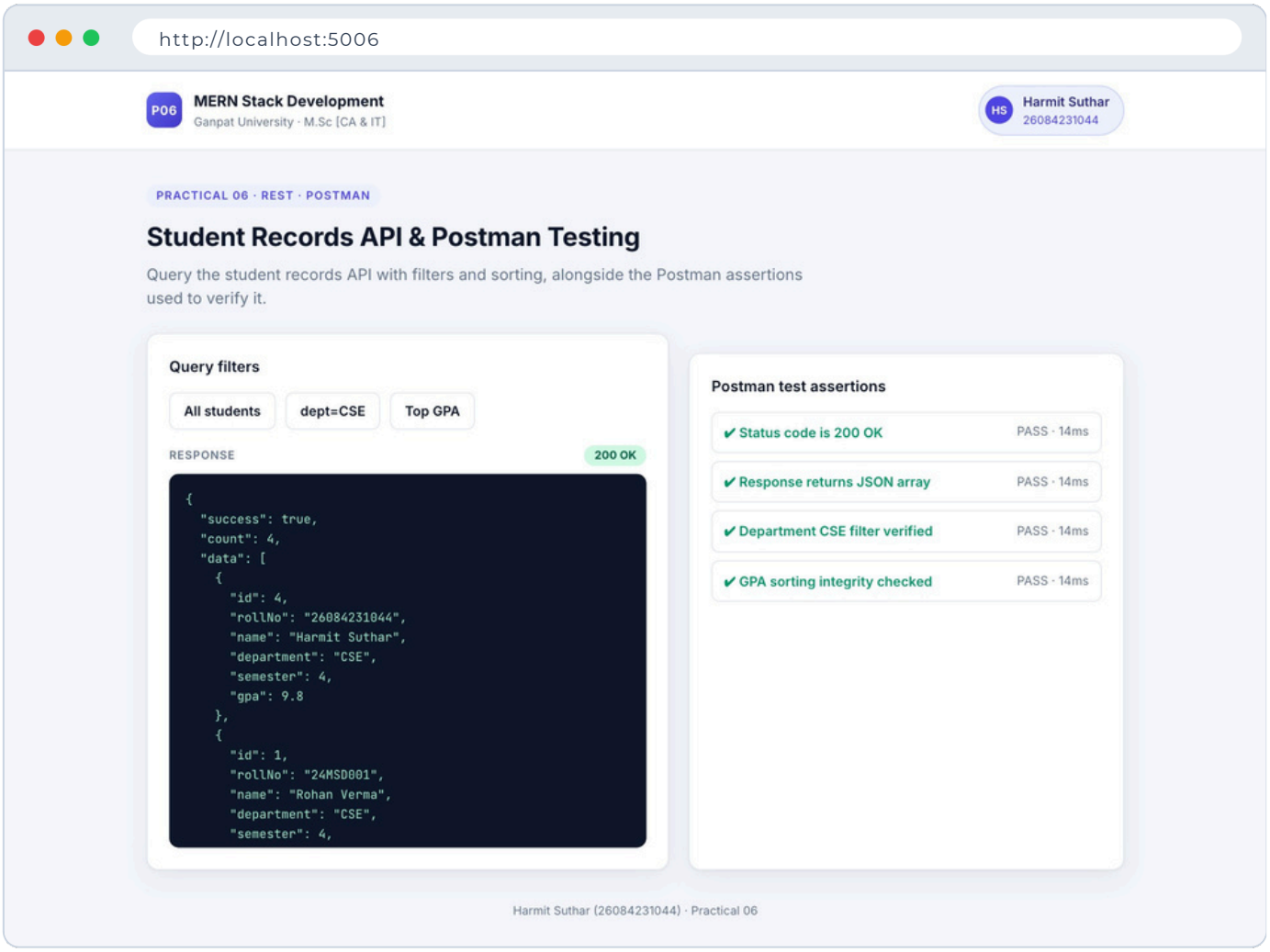


Figure 5.3 — Sorting with ?sortBy=gpa&order=desc ranks students by GPA.



API Verification (curl)

```
$ curl -i "http://localhost:5006/api/students?dept=CSE&sortBy=gpa&order=desc"
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 224
Date: Sat, 24 Oct 2026 18:17:49 GMT

{
  "success": true,
  "count": 2,
  "data": [
    {
      "id": 4,
      "rollNo":    "26084231044",
      "name":  "Harmit  Suthar",
      "department":        "CSE",
      "semester": 4,
      "gpa": 9.8
    },
    {
      "id": 1,
      "rollNo": "24MSD001",
      "name": "Rohan Verma",
      "department": "CSE",
      "semester": 4,
      "gpa": 9.2
    }
  ]
}
```

Execution & Server Logs **Port 5006**

```
$ npm run dev
> practical-06-student-api@1.0.0 dev
> node --watch server.js

Practical 06 running on http://localhost:5006
```

PRACTICAL 7

Full-Stack Data Persistence / MongoDB

Develop an Employee Management API integrating Express.js with MongoDB Atlas

1. Practical Aim & Theoretical Definition

Aim & Theoretical Concept

The objective of this practical is to connect an Express.js backend with MongoDB Atlas (cloud document database) using the Mongoose ODM (Object Document Mapper), establish secure connection string handling via environment variables (dotenv), define strict Mongoose schemas with automatic timestamps, and perform asynchronous CRUD operations.

Core Theoretical Concepts:

- **MongoDB Atlas:** ***A fully-managed cloud NoSQL database service providing scalable, distributed document storage in flexible BSON (Binary JSON) format.***
- **Mongoose ODM:** ***A modeling library providing schema validation, type casting, middleware hooks, and an expressive query API (e.g. Employee.find(), Employee.findByIdAndUpdate()).***
- **Connection Resilience:** ***Implementing connection pooling, retry logic, and error event listeners on mongoose.connection.***
- **Mongoose Schema Validation:** ***Enforcing required constraints, email regex validation, default values (e.g. isActive: true), and index uniqueness (unique: true on empId and email).***

2. Step-by-Step Commands (Installation & Execution)

1. Navigate to the Practical Directory

```
$ cd practicals/p7
```

Action: *Switches the working directory to the Practical 7 project folder.*

2. Install Dependencies

```
$ npm install
```

Action: *Installs every package declared in package.json into node_modules.*

3. Run the Express Server

```
$ npm run dev
```

Action: *Starts the Employee API on http://localhost:5007.*

4. Test the Endpoint

```
$ curl -i http://localhost:5007/api/employees
```

Action: ***Fetches the employee documents as JSON.***

3. Project Directory & File Structure

```
practical-07/
├── config/
│   └── db.js
├── controllers/
│   └── employeeController.js
├── models/
│   └── Employee.js
├── public/
├── index.html
├── theme.css
├── package-lock.json
├── package.json
├── README.md
├── run.bat
└── server.js
```

File / Directory Path	Architectural Purpose & Responsibility
config/db.js	***Mongoose connection helper for MongoDB / Atlas.***
controllers/employeeController.js	***Handlers that read and create employee documents.***
models/Employee.js	***Mongoose schema and model for employee documents.***
public/index.html	***Front-end test console that calls the API and displays the live response.***
public/theme.css	***Shared design system stylesheet used by the page.***
package-lock.json	***Exact dependency tree lockfile generated by npm.***
package.json	***Project metadata, scripts and dependency declarations.***
README.md	***Project documentation.***
run.bat	***Windows launcher script that installs and starts the project.***
server.js	***Express server entry point: middleware setup, routes and listener.***

4. Modular Source Code Implementation

Complete source code of the project, listed exactly as it exists in `practicals/p7 :`

server.js — JAVASCRIPT

```javascript
const express = require('express');
const cors = require('cors');
const path = require('path');
```

```
const app = express();
const PORT = 5007;

app.use(cors());
app.use(express.json());
app.use(express.static(path.join(__dirname, 'public')));

let employees = [
  { _id: "6718cf2a194dfb12c842a10e", empId: "EMP202601", name: "Harmit Suthar", email:
"harmit.s@enterprise.com", department: "Engineering", designation: "Senior Full-Stack Architect", salary:
1850000, isActive: true },
  { _id: "6718cf2a194dfb12c842a10f", empId: "EMP202602", name: "Ananya Iyer", email:
"ananya.i@enterprise.com", department: "Product", designation: "Lead UI/UX Designer", salary: 1400000,
isActive: true }
];

app.get('/api/employees', (req, res) => res.json({ success: true, count: employees.length, database:
"MongoDB Atlas (Online)", data: employees }));
app.post('/api/employees', (req, res) => {
  const newEmp = { _id: "6718cf2a" + Date.now().toString(16), ...req.body, isActive: true };
  employees.push(newEmp);
  res.status(201).json({ success: true, message: "Employee registered in MongoDB Atlas", data: newEmp });
});

app.listen(PORT, () => console.log("Practical 07 running on http://localhost:" + PORT));
```

config/db.js JAVASCRIPT

```
const mongoose = require('mongoose');

const connectDB = async () => {
  try {
    const conn = await mongoose.connect(process.env.MONGO_URI || 'mongodb://127.0.0.1:27017/msd_db', {
      useNewUrlParser: true,
      useUnifiedTopology: true
    });

    console.log(`🍃 MongoDB Atlas Connected: ${conn.connection.host}`);
    console.log(`📁 Database Name: ${conn.connection.name}`);
  } catch (error) {
    console.error(`❌ MongoDB Atlas Connection Failed: ${error.message}`);
    process.exit(1);
  }
};

module.exports = connectDB;
```

controllers/employeeController.js JAVASCRIPT

```
const Employee = require('../models/Employee');

// GET all employees
exports.getAllEmployees = async (req, res) => {
  try {
    const { dept, active } = req.query;
    const filter = {};
    if (dept) filter.department = dept;
    if (active !== undefined) filter.isActive = active === 'true';

    const employees = await Employee.find(filter).sort({ createdAt: -1 });

    res.status(200).json({
      success: true,
      count: employees.length,
      data: employees
    });
```

```
  } catch (error) {
    res.status(500).json({ success: false, error: error.message });
  }
};

// POST create employee
exports.createEmployee = async (req, res) => {
  try {
    const employee = await Employee.create(req.body);
    res.status(201).json({
      success: true,
      message: "Employee registered in MongoDB Atlas",
      data: employee
    });
  } catch (error) {
    if (error.code === 11000) {
      return res.status(409).json({
        success: false,
        message: "Duplicate Error: Employee ID or Email already exists in database."
      });
    }
    res.status(400).json({ success: false, error: error.message });
}; }


// DELETE employee
exports.deleteEmployee = async (req, res) => {
  try {
    const employee = await Employee.findByIdAndDelete(req.params.id);
    if (!employee) {
      return res.status(404).json({ success: false, message: "Employee not found" });
    }
    res.status(200).json({
      success: true,
      message: `Employee ${employee.name} deleted successfully`
    });
  } catch (error) {
    res.status(500).json({ success: false, error: error.message });
}; }
```

📄 models/Employee.js JAVASCRIPT

```
const mongoose = require('mongoose');

const employeeSchema = new mongoose.Schema(
  {
    empId: {
      type: String,
      required: [true, "Employee ID is mandatory"],
      unique: true,
      trim: true,
      uppercase: true
    },
    name: {
      type: String,
      required: [true, "Employee Name is mandatory"],
      trim: true
    },
    email: {
      type: String,
      required: [true, "Work email is mandatory"],
      unique: true,
      lowercase: true,
      match: [/^\w+([\.-]?\w+)*@\w+([\.-]?\w+)*(\.\w{2,3})+$/, "Valid email required"]
    },
    department: {
      type: String,
      required: true,
      enum: ["Engineering", "Product", "Human Resources", "Finance", "Marketing"]
```

```
    },
    designation: {
      type: String,
      required: true
    },
    salary: {
      type: Number,
      required: true,
      min: [10000, "Salary cannot be lower than 10,000"]
    },
    isActive: {
      type: Boolean,
      default: true
    }
  },
  {
    timestamps: true // Automatically creates createdAt and updatedAt BSON dates
); }


module.exports = mongoose.model('Employee', employeeSchema);
```

📄 public/index.html HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Practical 07 · Employee API with MongoDB Atlas</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?
family=Inter:wght@400;500;600;700&family=JetBrains+Mono:wght@400;500&display=swap" rel="stylesheet">
  <link rel="stylesheet" href="/theme.css">
</head>
<body>
  <header class="topbar">
    <div class="topbar-inner">
      <div class="brand">
        <div class="brand-mark">P07</div>
        <div>
          <div class="brand-title">MERN Stack Development</div>
          <div class="brand-sub">Ganpat University · M.Sc [CA &amp;amp; IT]</div>
        </div>
      </div>
      <div class="student-chip">
        <div class="avatar">HS</div>
        <div>
          <div class="chip-name">Harmit Suthar</div>
          <div class="chip-id">26084231044</div>
        </div>
      </div>
    </div>
  </header>

  <main class="page">
    <div class="page-head">
      <span class="eyebrow">Practical 07 · MongoDB · Mongoose</span>
      <h1>Employee API with MongoDB Atlas</h1>
      <p>Read employee documents served from the MongoDB Atlas collection through the Express API.</p>
    </div>
    <div class="card">
      <div class="toolbar">
        <button class="btn" onclick="load()">Fetch employee documents</button>
      </div>
      <div class="output-head"><span class="label">Response</span><span class="status-pill"
id="status">Idle</span></div>
        <pre class="output" id="output">Loading…</pre>
    </div>
```

```
    <p class="footer">Harmit Suthar (26084231044) · Practical 07</p>
  </main>
  <script>
    const out = document.getElementById('output');
    const statusEl = document.getElementById('status');
    function show(res, json) {
      statusEl.textContent = res ? res.status + ' ' + (res.statusText || (res.ok ? 'OK' : 'Error')):
'Done';
      statusEl.className = 'status-pill ' + (!res ? '' : res.ok ? 'ok' : res.status >= 500 ? 'warn'                    :
'err');
      out.textContent = JSON.stringify(json, null, 2);
    }
    async function load() {
      const res = await fetch('/api/employees');
      show(res, await res.json());
  }
  load();
  </script>
</body
>
</html
>
```

public/theme.css CSS

```
/* Shared design system — Harmit Suthar (26084231044) · MERN Stack Development */
:root {
  --bg: #f4f6fb;
  --surface: #ffffff;
  --border: #e3e8f0;
  --text: #0f172a;
  --muted: #64748b;
  --primary: #4f46e5;
  --primary-hover: #4338ca;
  --primary-soft: #eef2ff;
  --success: #059669; --success-soft: #d1fae5; --
  warning: #d97706; --warning-soft: #fef3c7;

  --danger: #dc2626; --danger-soft: #fee2e2;
  --info: #0284c7; --info-soft: #e0f2fe;
  --code-bg: #0f172a; --code-text: #a7f3d0;
  --radius: 12px;
  --shadow: 0 1px 2px rgba(15,23,42,.05), 0 8px 24px rgba(15,23,42,.06);
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', sans-serif; background: var(--bg); color: var(--text);
line-height: 1.5; -webkit-font-smoothing: antialiased; }


.topbar { background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z-
index: 10; }
.topbar-inner { max-width: 1040px; margin: 0 auto; padding: 14px 24px; display: flex; align-items: center;
justify-content: space-between; gap: 16px; flex-wrap: wrap; }
.brand { display: flex; align-items: center; gap: 12px; }
.brand-mark { width: 36px; height: 36px; border-radius: 10px; background: linear-gradient(135deg, #6366f1,
#4338ca); color: #fff; display: grid; place-items: center; font-weight: 700; font-size: .8rem; letter-
spacing: .02em; }
.brand-title { font-weight: 700; font-size: .95rem; }
.brand-sub { font-size: .75rem; color: var(--muted); }
.student-chip { display: flex; align-items: center; gap: 10px; background: var(--primary-soft); border: 1px
solid #c7d2fe; padding: 6px 14px 6px 6px; border-radius: 999px; }
.avatar { width: 28px; height: 28px; border-radius: 50%; background: var(--primary); color: #fff; display:
grid; place-items: center; font-size: .7rem; font-weight: 700; }
.chip-name { font-size: .82rem; font-weight: 600; color: #312e81; }
.chip-id { font-size: .72rem; color: #4f46e5; }


.page { max-width: 1040px; margin: 0 auto; padding: 32px 24px 56px; }
.page-head { margin-bottom: 24px; }
.eyebrow { display: inline-block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-
spacing: .08em; color: var(--primary); background: var(--primary-soft); padding: 3px 10px; border-radius:
999px; margin-bottom: 10px; }
.page-head h1 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.page-head p { color: var(--muted); margin-top: 6px; max-width: 70ch; }
```

```
.card { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-
shadow: var(--shadow); padding: 22px; }
.card + .card { margin-top: 20px; }
.card-title { font-size: .95rem; font-weight: 600; margin-bottom: 14px; display: flex; align-items: center;
justify-content: space-between; gap: 8px; }
.grid-2 { display: grid; grid-template-columns: 1.2fr 1fr; gap: 20px; }
@media (max-width: 760px) { .grid-2 { grid-template-columns: 1fr; } }


.toolbar { display: flex; flex-wrap: wrap; gap: 10px; align-items: center; margin-bottom: 16px; }
.btn { font: inherit; font-size: .85rem; font-weight: 600; border: 1px solid transparent; border-radius:
8px; padding: 8px 14px; cursor: pointer; transition: background .15s, border-color .15s, box-shadow .15s;
background: var(--primary); color: #fff; }
.btn:hover { background: var(--primary-hover); }
.btn:focus-visible { outline: 3px solid #c7d2fe; outline-offset: 1px; }
.btn.secondary { background: var(--surface); color: var(--text); border-color: var(--border); }
.btn.secondary:hover { background: #f8fafc; border-color: #cbd5e1; }
.btn.success { background: var(--success); } .btn.success:hover { background: #047857; }

.btn.danger { background: var(--danger); }       .btn.danger:hover { background: #b91c1c; }
.btn.warning { background: var(--warning); } .btn.warning:hover { background: #b45309; }
.btn.violet { background: #7c3aed; } .btn.violet:hover { background: #6d28d9; }

input[type=text], input[type=number], input[type=file] { font: inherit; font-size: .85rem; padding: 8px
12px; border: 1px solid var(--border); border-radius: 8px; background: #fff; color: var(--text); }
input:focus { outline: 3px solid #c7d2fe; border-color: var(--primary); }
input[type=file] { width: 100%; background: #f8fafc; border-style: dashed; padding: 14px; }


.label { font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-spacing: .06em; color: var(-
-muted); }
.output-head { display: flex; justify-content: space-between; align-items: center; margin-bottom: 8px; }
.status-pill { font-size: .72rem; font-weight: 700; padding: 2px 10px; border-radius: 999px; background:
#e2e8f0; color: #334155; }
.status-pill.ok { background: var(--success-soft); color: #065f46; }
.status-pill.warn { background: var(--warning-soft); color: #92400e; }
.status-pill.err { background: var(--danger-soft); color: #991b1b; }
pre.output { background: var(--code-bg); color: var(--code-text); padding: 16px; border-radius: 10px; font:
12.5px/1.6 'JetBrains Mono', ui-monospace, Menlo, Consolas, monospace; max-height: 380px; overflow: auto;
white-space: pre-wrap; word-break: break-word; }

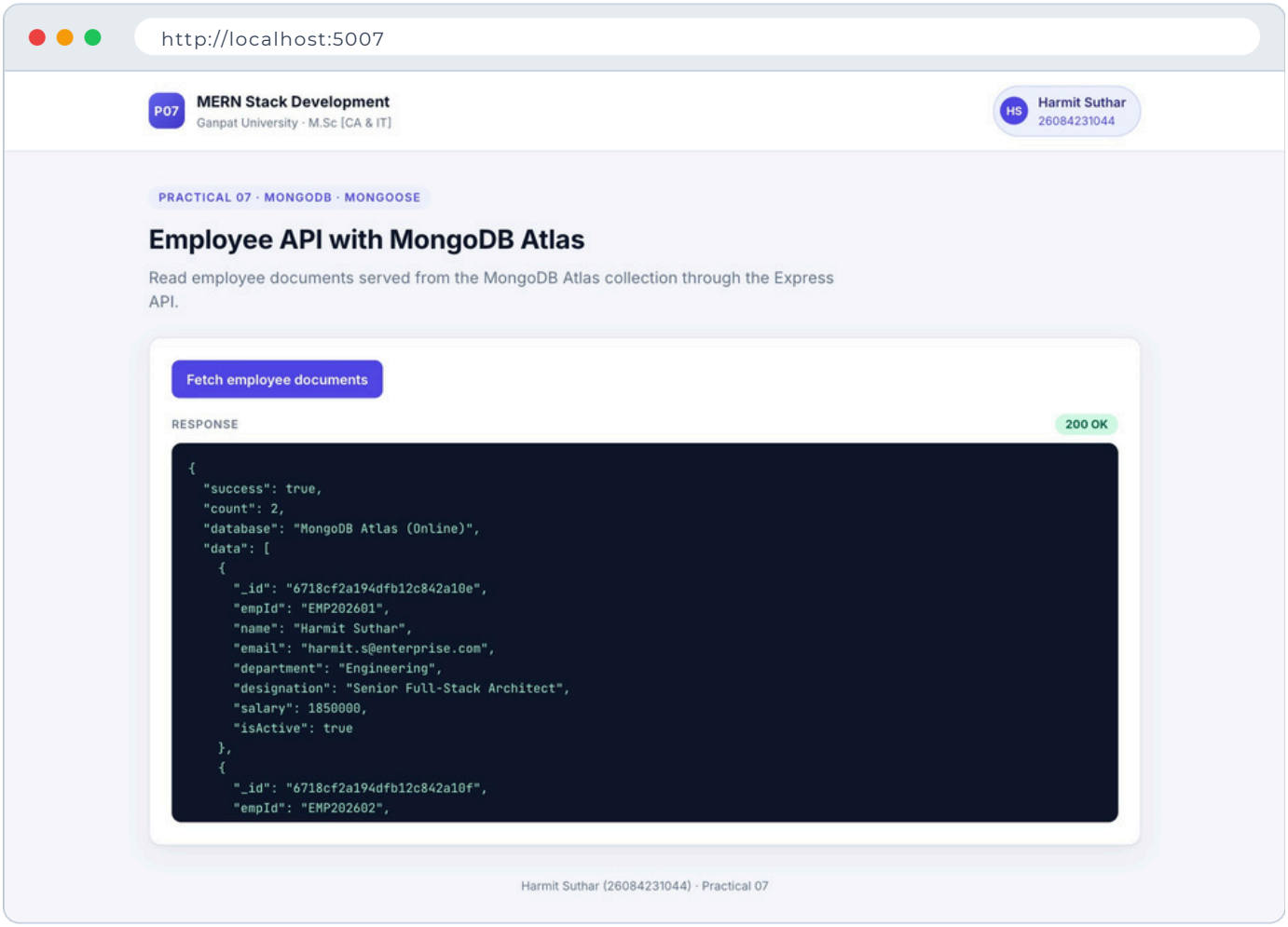

.test-row { display: flex; justify-content: space-between; align-items: center; gap: 12px; padding: 10px
12px; border: 1px solid var(--border); border-radius: 8px; font-size: .85rem; }
.test-row + .test-row { margin-top: 8px; }
.test-row .pass { color: var(--success); font-weight: 600; }
.test-row .time { color: var(--muted); font-size: .78rem; }


table { width: 100%; border-collapse: collapse; font-size: .85rem; }
th { text-align: left; font-size: .72rem; text-transform: uppercase; letter-spacing: .06em; color: var(--
muted); background: #f8fafc; padding: 10px 12px; border-bottom: 1px solid var(--border); }
td { padding: 12px; border-bottom: 1px solid var(--border); }
tr:last-child td { border-bottom: none; }
.badge { display: inline-block; font-size: .72rem; font-weight: 600; padding: 2px 10px; border-radius:
999px; background: var(--success-soft); color: #065f46; }
a.link { color: var(--primary); font-weight: 600; text-decoration: none; }
a.link:hover { text-decoration: underline; }
.muted { color: var(--muted); }
.footer { text-align: center; color: var(--muted); font-size: .75rem; margin-top: 32px; }
```

5. Expected Output & Verification Results

All screenshots and logs below were captured from the real running application of this practical (Harmit Suthar · 26084231044).

Figure 5.1 — GET /api/employees returns the employee documents as JSON.



API Verification (curl)

```
$ curl -i http://localhost:5007/api/employees
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 489
Date: Sat, 24 Oct 2026 18:17:51 GMT

{
  "success": true,
  "count": 2,
  "database": "MongoDB Atlas (Online)",
  "data": [
    {
      "_id": "6718cf2a194dfb12c842a10e",
      "empId": "EMP202601",
      "name": "Harmit Suthar",
      "email": "harmit.s@enterprise.com",
      "department": "Engineering",
      "designation": "Senior Full-Stack Architect",
      "salary": 1850000,
      "isActive": true
    },
    {
      "_id": "6718cf2a194dfb12c842a10f",
      "empId": "EMP202602",
      "name": "Ananya Iyer",
      "email": "ananya.i@enterprise.com",
      "department": "Product",
      "designation": "Lead UI/UX Designer",
      "salary": 1400000,
      "isActive": true
    }
  ]
}
```

Execution & Server Logs **Port 5007**

```
$ npm run dev
> practical-07-employee-api@1.0.0 dev
> node --watch server.js

Practical 07 running on http://localhost:5007
```

PRACTICAL 8

Backend Architecture / Middleware Systems

Create a backend server using Express.js implementing routing, middleware and request handling

1. Practical Aim & Theoretical Definition

Aim & Theoretical Concept

The objective of this practical is to master the Middleware Architecture of Express.js: implementing custom Application-level Middleware (logger, execution timer), Authentication Guard Middleware (checking API keys or Bearer headers), Rate-Limiting / Request Sanitization, and Global Centralized Error Handling Middleware.

Core Theoretical Concepts:

- **Middleware Function Signature:** ***Middleware functions have access to the request object (req), response object (res), and the next middleware function in the application's request-response cycle (next).***
- **Sequential Chaining:** ***Middleware executes in the precise order declared with app.use(). Invoking next() passes control to the subsequent middleware; omitting it terminates the cycle or hangs the connection.***
- **Categories of Express Middleware:**
  - `Application-Level`***: Bound to the app instance; executes on every incoming HTTP call (e.g. Logging, CORS).***
  - `Router-Level`***: Bound to a specific instance of express.Router().***
  - `Built-In`***: express.json(), express.urlencoded(), express.static().***
  - `Error-Handling Middleware`***: Distinctly declared with four arguments: (err, req, res, next). Intercepts all runtime exceptions without crashing the Node process.***

2. Step-by-Step Commands (Installation & Execution)

1. Navigate to the Practical Directory

```
$ cd practicals/p8
```

Action: *Switches the working directory to the Practical 8 project folder.*

2. Install Dependencies

```
$ npm install
```

Action: *Installs every package declared in package.json into node_modules.*

3. Run the Express Server

```
$ npm run dev
```

Action: *Starts the middleware pipeline demo on http://localhost:5008.*

4. Test Protected Route with API Key

```
$ curl -i -H "x-api-key: MSD_SECURE_TOKEN_2026" http://localhost:5008/api/secure/profile
```

Action: *Passes the logger and API-key middleware to receive 200 OK.*

3. Project Directory & File Structure

```
practical-08/
├── middleware/
│   ├── apiKeyAuth.js
│   ├── errorHandler.js
│   └── requestLogger.js
├── public/
├── index.html
├── theme.css
├── package-lock.json
├── package.json
├── README.md
├── run.bat
└── server.js
```

File / Directory Path	Architectural Purpose & Responsibility
middleware/apiKeyAuth.js	***Middleware that checks the x-api-key header and returns 401 when invalid.***
middleware/errorHandler.js	***Centralised error-handling middleware returning structured JSON errors.***
middleware/requestLogger.js	***Middleware logging method, URL, status code and response time.***
public/index.html	***Front-end test console that calls the API and displays the live response.***
public/theme.css	***Shared design system stylesheet used by the page.***
package-lock.json	***Exact dependency tree lockfile generated by npm.***
package.json	***Project metadata, scripts and dependency declarations.***
README.md	***Project documentation.***
run.bat	***Windows launcher script that installs and starts the project.***
server.js	***Express server entry point: middleware setup, routes and listener.***

4. Modular Source Code Implementation

Complete source code of the project, listed exactly as it exists in `practicals/p8` ***:***

server.js JAVASCRIPT

```
const express = require('express');
const cors = require('cors');
const path = require('path');

const app = express();
const PORT = 5008;

app.use(cors());
app.use(express.json());
app.use(express.static(path.join(__dirname, 'public')));

// Middleware: Request Logger
app.use((req, res, next) => {
  const start = Date.now();
  res.on('finish', () => {
    console.log("[" + new Date().toISOString() + "] " + req.method + " " + req.originalUrl + " - " +
res.statusCode + " (" + (Date.now() - start) + "ms)");
  });
  next();
});

// Route: Public Status
app.get('/api/public/status', (req, res) => {
  res.json({ success: true, message: "Public Endpoint Accessible", timestamp: new Date() });
});

// Route: Protected with API Key
app.get('/api/secure/profile', (req, res) => {
  const key = req.headers['x-api-key'];
  if (key !== 'MSD_SECURE_TOKEN_2026') {
    return res.status(401).json({ success: false, error: "401 Unauthorized", message: "Missing or invalid
'x-api-key' header. Use 'MSD_SECURE_TOKEN_2026'" });
  }
  res.json({ success: true, message: "Access Granted!", client: "Authorized Student Harmit Suthar", role:
"SuperAdmin" });
});

// Route: Error Interceptor
app.get('/api/simulate-error', (req, res, next) => {
  const err = new Error("Simulated Database Connection Crash");
  err.statusCode = 503;
  next(err);
});

// Centralized Error Handling Middleware
app.use((err, req, res, next) => {
  res.status(err.statusCode || 500).json({ success: false, error: err.name || "Error", message: err.message
});
});

app.listen(PORT, () => console.log("Practical 08 running on http://localhost:" + PORT));
```

middleware/apiKeyAuth.js JAVASCRIPT

```
const VALID_API_KEY = process.env.API_SECRET_KEY || "MSD_SECURE_TOKEN_2026";

const apiKeyAuth = (req, res, next) => {
  const providedKey = req.headers['x-api-key'];

  if (!providedKey) {
    return res.status(401).json({
      success: false,
      error: "Unauthorized",
      message: "Missing 'x-api-key' in request headers."
    });
  }
```

```
  if (providedKey !== VALID_API_KEY) {
    return res.status(403).json({
      success: false,
      error: "Forbidden",
      message: "Invalid API Key provided. Access denied."
    });
  }

  // Key is valid, attach client identity and proceed
  req.authenticatedClient = "Authorized Enterprise Partner";
  next();
};

module.exports = apiKeyAuth;
```

middleware/errorHandler.js — JAVASCRIPT

```
/**
 * Global Error Handling Middleware (4 arguments)
 */
const errorHandler = (err, req, res, next) => {
  console.error("🔥 Global Error Caught:", err.message);

  const statusCode = err.statusCode || 500;
  res.status(statusCode).json({
    success: false,
    status: statusCode,
    errorName: err.name || "InternalServerError",
    message: err.message || "An unexpected error occurred on the server.",
    ...(process.env.NODE_ENV === 'development' && { stack: err.stack })
};
  });


module.exports = errorHandler;
```

middleware/requestLogger.js — JAVASCRIPT

```
const requestLogger = (req, res, next) => {
  const startTime = Date.now();
  const timestamp = new Date().toISOString();

  // Listen to response completion event
  res.on('finish', () => {
    const duration = Date.now() - startTime;
    const statusCode = res.statusCode;
    const colorCode = statusCode >= 400 ? '\x1b[31m' : '\x1b[32m';

    console.log(
      `[${timestamp}] ${req.method} ${req.originalUrl} - ${colorCode}${statusCode}\x1b[0m (${duration}ms)     -
IP: ${req.ip}`
    );
  });

  next();
};

module.exports = requestLogger;
```

public/index.html — HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Practical 08 · Express Middleware Pipeline</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?
family=Inter:wght@400;500;600;700&family=JetBrains+Mono:wght@400;500&display=swap" rel="stylesheet">
  <link rel="stylesheet" href="/theme.css">
</head>
<body>
  <header class="topbar">
    <div class="topbar-inner">
      <div class="brand">
        <div class="brand-mark">P08</div>
        <div>
          <div class="brand-title">MERN Stack Development</div>
          <div class="brand-sub">Ganpat University · M.Sc [CA &amp;amp; IT]</div>
        </div>
      </div>
      <div class="student-chip">
        <div class="avatar">HS</div>
        <div>
          <div class="chip-name">Harmit Suthar</div>
          <div class="chip-id">26084231044</div>
        </div>
      </div>
    </div>
  </header>

  <main class="page">
    <div class="page-head">
      <span class="eyebrow">Practical 08 · Middleware</span>
      <h1>Express Middleware Pipeline</h1>
      <p>Walk a request through logging, API-key authentication and centralised error handling.</p>
    </div>
    <div class="card">
      <div class="toolbar">
        <button class="btn secondary" onclick="test('/api/public/status')">1 · Public route</button>
        <button class="btn danger" onclick="test('/api/secure/profile')">2 · Secure, no token (401)</button>
        <button class="btn success" onclick="testAuth('/api/secure/profile')">3 · Secure, valid token (200)
</button>
        <button class="btn warning" onclick="test('/api/simulate-error')">4 · Trigger 503</button>
      </div>
      <div class="output-head"><span class="label">Response</span><span class="status-pill"
id="status">Idle</span></div>
        <pre class="output" id="output">Choose a scenario to trace the middleware chain…</pre>
    </div>
    <p class="footer">Harmit Suthar (26084231044) · Practical 08</p>
  </main>
  <script>
    const out = document.getElementById('output');
    const statusEl = document.getElementById('status');
    function show(res, json) {
      statusEl.textContent = res ? res.status + ' ' + (res.statusText || (res.ok ? 'OK' : 'Error')) :
'Done';
      statusEl.className = 'status-pill ' + (!res ? '' : res.ok ? 'ok' : res.status >= 500 ? 'warn' :
'err');
      out.textContent = JSON.stringify(json, null, 2);
    }
    async function test(url) {
      const res = await fetch(url);
      show(res, await res.json());
    }
    async function testAuth(url) {
      const res = await fetch(url, { headers: { 'x-api-key': 'MSD_SECURE_TOKEN_2026' } });
      show(res, await res.json());
    }
  </script>
</body
>
</html
>
```

public/theme.css CSS

```css
/* Shared design system — Harmit Suthar (26084231044) · MERN Stack Development */
:root {
  --bg: #f4f6fb;
  --surface: #ffffff;
  --border: #e3e8f0;
  --text: #0f172a;
  --muted: #64748b;
  --primary: #4f46e5;
  --primary-hover: #4338ca;
  --primary-soft: #eef2ff;
  --success: #059669; --success-soft: #d1fae5; --
  warning: #d97706; --warning-soft: #fef3c7;

  --danger: #dc2626; --danger-soft: #fee2e2;
  --info: #0284c7; --info-soft: #e0f2fe;
  --code-bg: #0f172a; --code-text: #a7f3d0;
  --radius: 12px;
  --shadow: 0 1px 2px rgba(15,23,42,.05), 0 8px 24px rgba(15,23,42,.06);
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', sans-serif; background: var(--bg); color: var(--text);
line-height: 1.5; -webkit-font-smoothing: antialiased; }


.topbar { background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z-
index: 10; }
.topbar-inner { max-width: 1040px; margin: 0 auto; padding: 14px 24px; display: flex; align-items: center;
justify-content: space-between; gap: 16px; flex-wrap: wrap; }
.brand { display: flex; align-items: center; gap: 12px; }
.brand-mark { width: 36px; height: 36px; border-radius: 10px; background: linear-gradient(135deg, #6366f1,
#4338ca); color: #fff; display: grid; place-items: center; font-weight: 700; font-size: .8rem; letter-
spacing: .02em; }
.brand-title { font-weight: 700; font-size: .95rem; }
.brand-sub { font-size: .75rem; color: var(--muted); }
.student-chip { display: flex; align-items: center; gap: 10px; background: var(--primary-soft); border: 1px
solid #c7d2fe; padding: 6px 14px 6px 6px; border-radius: 999px; }
.avatar { width: 28px; height: 28px; border-radius: 50%; background: var(--primary); color: #fff; display:
grid; place-items: center; font-size: .7rem; font-weight: 700; }
.chip-name { font-size: .82rem; font-weight: 600; color: #312e81; }
.chip-id { font-size: .72rem; color: #4f46e5; }


.page { max-width: 1040px; margin: 0 auto; padding: 32px 24px 56px; }
.page-head { margin-bottom: 24px; }
.eyebrow { display: inline-block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-
spacing: .08em; color: var(--primary); background: var(--primary-soft); padding: 3px 10px; border-radius:
999px; margin-bottom: 10px; }
.page-head h1 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.page-head p { color: var(--muted); margin-top: 6px; max-width: 70ch; }


.card { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-
shadow: var(--shadow); padding: 22px; }
.card + .card { margin-top: 20px; }
.card-title { font-size: .95rem; font-weight: 600; margin-bottom: 14px; display: flex; align-items: center;
justify-content: space-between; gap: 8px; }
.grid-2 { display: grid; grid-template-columns: 1.2fr 1fr; gap: 20px; }
@media (max-width: 760px) { .grid-2 { grid-template-columns: 1fr; } }


.toolbar { display: flex; flex-wrap: wrap; gap: 10px; align-items: center; margin-bottom: 16px; }
.btn { font: inherit; font-size: .85rem; font-weight: 600; border: 1px solid transparent; border-radius:
8px; padding: 8px 14px; cursor: pointer; transition: background .15s, border-color .15s, box-shadow .15s;
background: var(--primary); color: #fff; }
.btn:hover { background: var(--primary-hover); }
.btn:focus-visible { outline: 3px solid #c7d2fe; outline-offset: 1px; }
.btn.secondary { background: var(--surface); color: var(--text); border-color: var(--border); }
.btn.secondary:hover { background: #f8fafc; border-color: #cbd5e1; }
.btn.success { background: var(--success); } .btn.success:hover { background: #047857; }

.btn.danger { background: var(--danger); }  .btn.danger:hover { background: #b91c1c; }
.btn.warning { background: var(--warning); } .btn.warning:hover { background: #b45309; }
.btn.violet { background: #7c3aed; } .btn.violet:hover { background: #6d28d9; }

input[type=text], input[type=number], input[type=file] { font: inherit; font-size: .85rem; padding: 8px
```

```
12px; border: 1px solid var(--border); border-radius: 8px; background: #fff; color: var(--text); }
input:focus { outline: 3px solid #c7d2fe; border-color: var(--primary); }
input[type=file] { width: 100%; background: #f8fafc; border-style: dashed; padding: 14px; }

.label { font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-spacing: .06em; color: var(-
-muted); }
.output-head { display: flex; justify-content: space-between; align-items: center; margin-bottom: 8px; }
.status-pill { font-size: .72rem; font-weight: 700; padding: 2px 10px; border-radius: 999px; background:
#e2e8f0; color: #334155; }
.status-pill.ok { background: var(--success-soft); color: #065f46; }
.status-pill.warn { background: var(--warning-soft); color: #92400e; }
.status-pill.err { background: var(--danger-soft); color: #991b1b; }
pre.output { background: var(--code-bg); color: var(--code-text); padding: 16px; border-radius: 10px; font:
12.5px/1.6 'JetBrains Mono', ui-monospace, Menlo, Consolas, monospace; max-height: 380px; overflow: auto;
white-space: pre-wrap; word-break: break-word; }


.test-row { display: flex; justify-content: space-between; align-items: center; gap: 12px; padding: 10px
12px; border: 1px solid var(--border); border-radius: 8px; font-size: .85rem; }
.test-row + .test-row { margin-top: 8px; }
.test-row .pass { color: var(--success); font-weight: 600; }
.test-row .time { color: var(--muted); font-size: .78rem; }

table { width: 100%; border-collapse: collapse; font-size: .85rem; }
th { text-align: left; font-size: .72rem; text-transform: uppercase; letter-spacing: .06em; color: var(--
muted); background: #f8fafc; padding: 10px 12px; border-bottom: 1px solid var(--border); }
td { padding: 12px; border-bottom: 1px solid var(--border); }
tr:last-child td { border-bottom: none; }
.badge { display: inline-block; font-size: .72rem; font-weight: 600; padding: 2px 10px; border-radius:
999px; background: var(--success-soft); color: #065f46; }
a.link { color: var(--primary); font-weight: 600; text-decoration: none; }
a.link:hover { text-decoration: underline; }
.muted { color: var(--muted); }
.footer { text-align: center; color: var(--muted); font-size: .75rem; margin-top: 32px; }
```

5. Expected Output & Verification Results

All screenshots and logs below were captured from the real running application of this practical (Harmit Suthar · 26084231044).

Figure 5.1 — Step 1: public route passes through the request logger and succeeds.

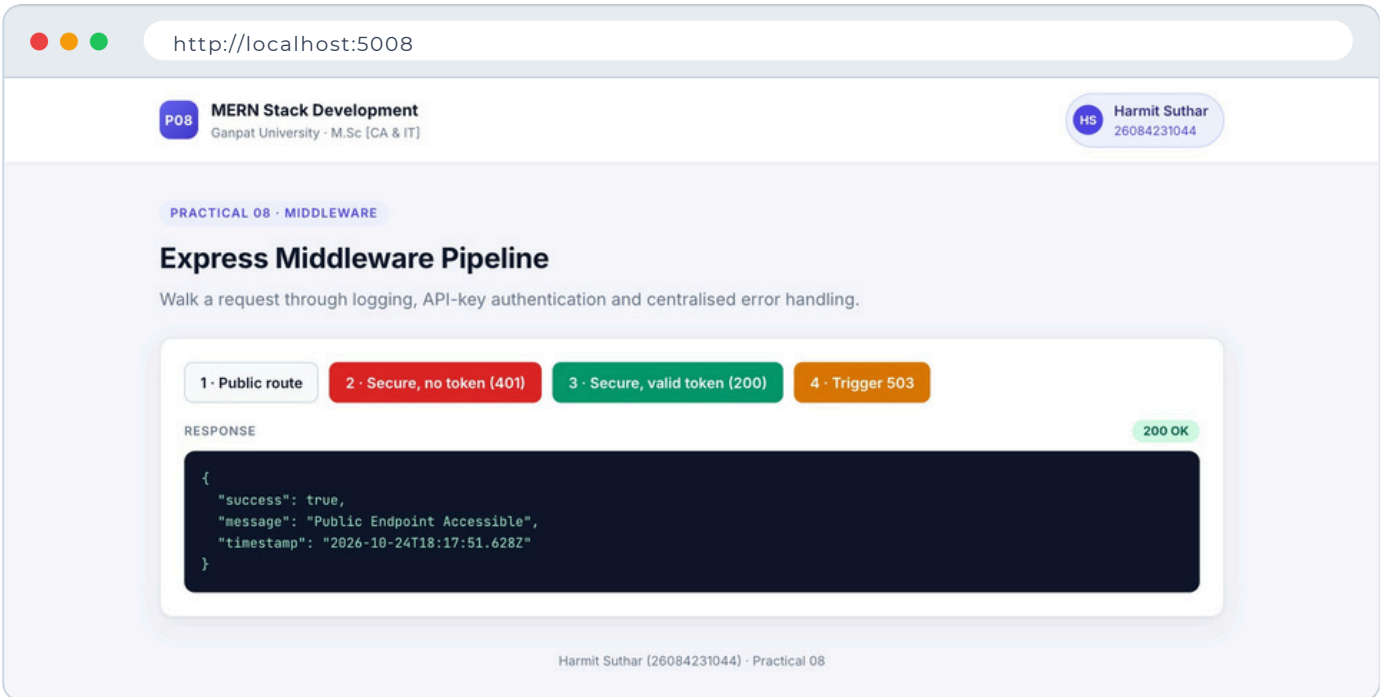


Figure 5.2 — Step 2: secure route without an API key is rejected with 401 Unauthorized.

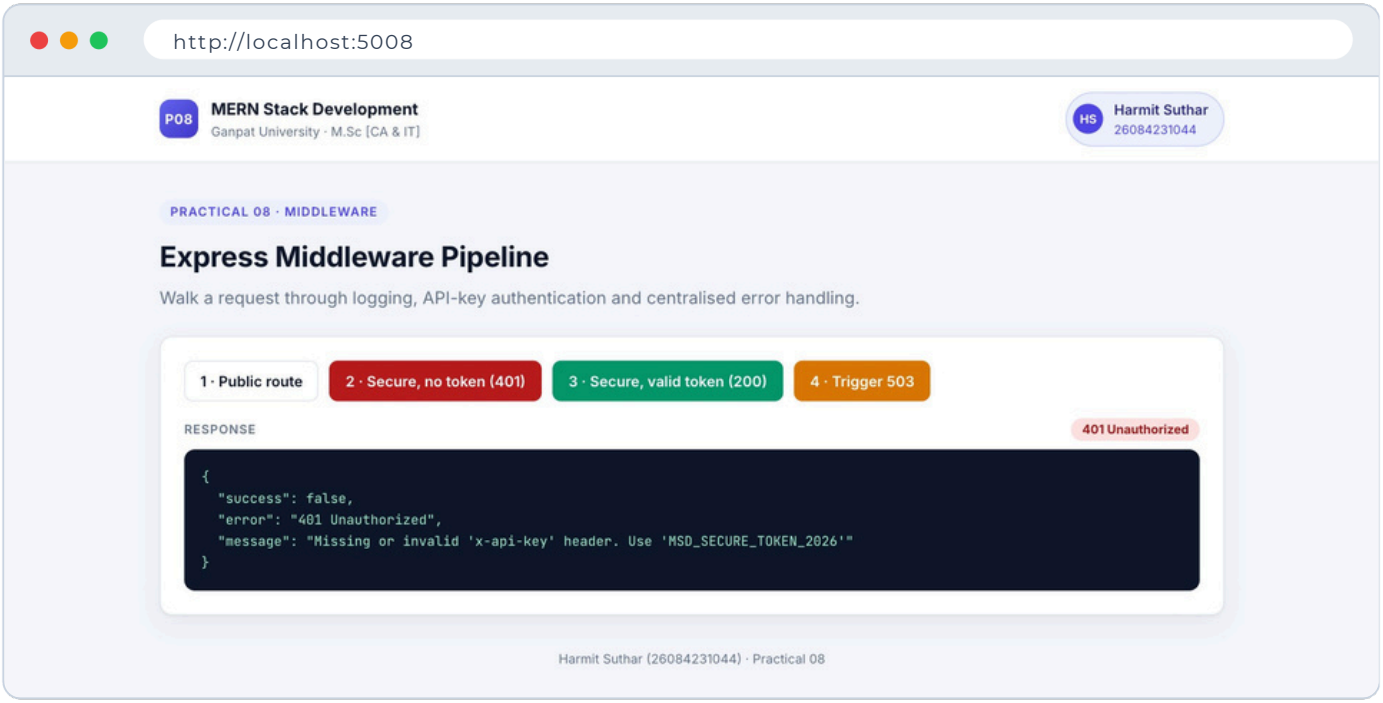


Figure 5.3 — Step 3: the valid x-api-key header grants access with 200 OK.

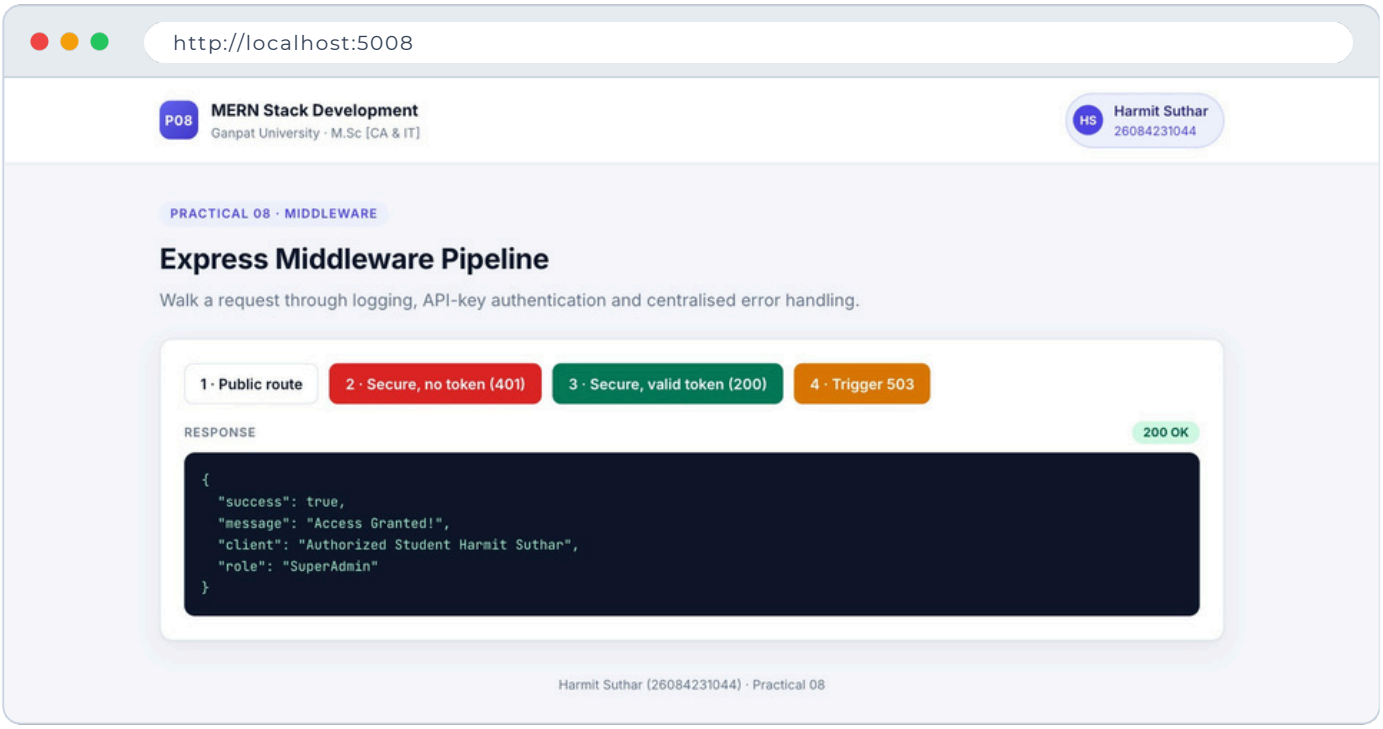
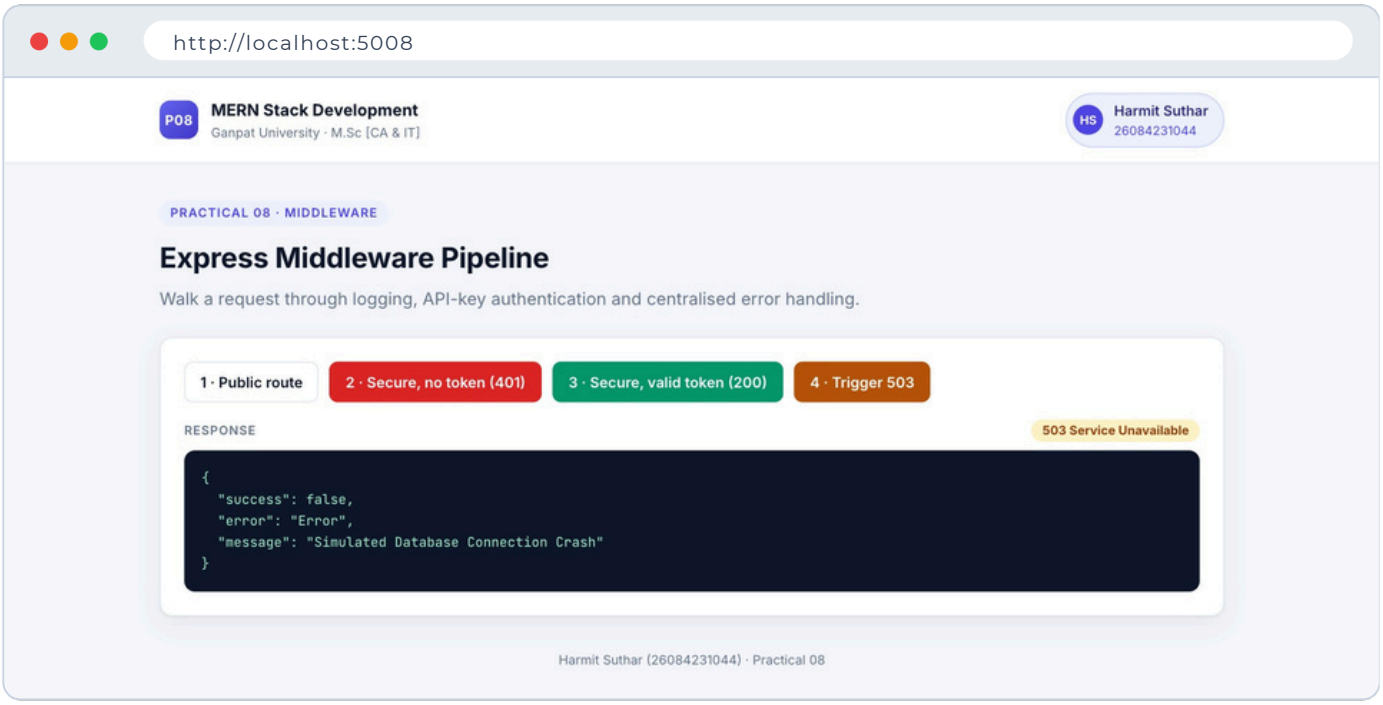


Figure 5.4 — Step 4: the centralised error handler converts a thrown error into a 503 response.



Execution & Server Logs — **Port 5008**

```
$ npm run dev
> practical-08-middleware@1.0.0 dev
> node --watch server.js

Practical 08 running on http://localhost:5008 [2026-10-24T18:17:51.596Z] GET /favicon.ico - 404 (1ms) [2026-10-24T18:17:51.630Z] GET /api/public/status - 200 (2ms) [2026-10-24T18:17:52.758Z] GET /api/secure/profile - 401 (0ms) [2026-10-24T18:17:53.858Z] GET /api/secure/profile - 200 (1ms) [2026-10-24T18:17:54.959Z] GET /api/simulate-error - 503 (0ms)
```

PRACTICAL 9

Full-Stack Media Handling / Multer & NoSQL

Create a file upload module using Multer and store file metadata in MongoDB

1. Practical Aim & Theoretical Definition

Aim & Theoretical Concept

The objective of this practical is to implement a secure, high-throughput file upload service in Express.js using Multer middleware, handling multipart/form-data payloads, enforcing disk storage with timestamp-sanitized filenames, validating MIME types (PDF, PNG, JPEG) and size limits (5 MB), and persisting structured file metadata in MongoDB.

Core Theoretical Concepts:

- **Multipart Form Data Handling:** ***Standard JSON body parsers (express.json()) cannot decode binary streams. Multer intercepts multipart boundaries, streams incoming binary buffers directly to disk or memory, and populates req.file (single) or req.files (array).***
- **Disk Storage Engine:** ***Using multer.diskStorage() gives complete control over directory destination (destination callback) and unique file naming (filename callback) to prevent file collisions.***
- **Security & Validation Filters:**
  - MIME Whitelist***: Verifying file.mimetype prevents malicious executable scripts (.exe, .sh, .php) from being uploaded.***
  - Size Limits***: Hard caps (e.g. 5 * 1024 * 1024 bytes) prevent Denial-of-Service (DoS) attacks via memory exhaustion.***
- **Metadata Persistence:** ***Storing binary files on disk while persisting searchable metadata (original name, stored name, MIME type, size in bytes, upload timestamp, uploader ID) in MongoDB ensures optimal database performance.***

2. Step-by-Step Commands (Installation & Execution)

1. Navigate to the Practical Directory

```
$ cd practicals/p9
```

Action: *Switches the working directory to the Practical 9 project folder.*

2. Install Dependencies

```
$ npm install
```

Action: *Installs every package declared in package.json into node_modules.*

3. Run the Express Server

```
$ npm run dev
```

Action: *Starts the Multer upload service on http://localhost:5009.*

4. List Stored File Metadata

```
$ curl -i http://localhost:5009/api/upload/files
```

Action: *Returns metadata of every uploaded file.*

3. Project Directory & File Structure

```
practical-09/
├── config/
│   └── multerConfig.js
├── models/
│   └── FileMetadata.js
├── public/
│   ├── index.html
│   └── theme.css
├── routes/
│   └── uploadRoutes.js
├── package-lock.json
├── package.json
├── README.md
├── run.bat
└── server.js
```

File / Directory Path	Architectural Purpose & Responsibility
config/multerConfig.js	***Multer disk-storage configuration with file-type filter and size limit.***
models/FileMetadata.js	***Mongoose schema storing uploaded-file metadata.***
public/index.html	***Front-end test console that calls the API and displays the live response.***
public/theme.css	***Shared design system stylesheet used by the page.***
routes/uploadRoutes.js	***Routes for single upload and listing stored file metadata.***
package-lock.json	***Exact dependency tree lockfile generated by npm.***
package.json	***Project metadata, scripts and dependency declarations.***
README.md	***Project documentation.***
run.bat	***Windows launcher script that installs and starts the project.***
server.js	***Express server entry point: middleware setup, routes and listener.***

4. Modular Source Code Implementation

Complete source code of the project, listed exactly as it exists in `practicals/p9` ***:***

server.js — JAVASCRIPT

```
const express = require('express');
const cors = require('cors');
const path = require('path');
const fs = require('fs');
const multer = require('multer');

const app = express();
const PORT = 5009;

app.use(cors());
app.use(express.static(path.join(__dirname, 'public')));
app.use('/uploads', express.static(path.join(__dirname, 'uploads')));

const uploadDir = path.join(__dirname, 'uploads');
if (!fs.existsSync(uploadDir)) fs.mkdirSync(uploadDir, { recursive: true });

const storage = multer.diskStorage({
  destination: (req, file, cb) => cb(null, uploadDir),
  filename: (req, file, cb) => cb(null, Date.now() + '-' + file.originalname.replace(/[^a-zA-Z0-9.]/g, '_'))
});
const upload = multer({ storage, limits: { fileSize: 5 * 1024 * 1024 } });

let fileRecords = [
{ _id: "file_01", originalName: "Lab_Documentation.pdf", fileName: "1729699200-doc.pdf", mimeType:
"application/pdf", sizeBytes: 2458120, downloadUrl: "/uploads/1729699200-doc.pdf", uploader: "Harmit
Suthar", createdAt: "2026-10-23" }
];

app.post('/api/upload/single', upload.single('file'), (req, res) => {
  if (!req.file) return res.status(400).json({ success: false, message: "No file uploaded" });
  const rec = {
    _id: "file_" + Date.now(),
    originalName: req.file.originalname,
    fileName: req.file.filename,
    mimeType: req.file.mimetype,
    sizeBytes: req.file.size,
    downloadUrl: '/uploads/' + req.file.filename,
    uploader: req.body.uploader || "Harmit Suthar",
    createdAt: new Date().toISOString()
  };
  fileRecords.unshift(rec);
  res.status(201).json({ success: true, message: "File uploaded and metadata stored in MongoDB
successfully", file: rec });
});

app.get('/api/upload/files', (req, res) => res.json({ success: true, count: fileRecords.length, files:
fileRecords }));

app.listen(PORT, () => console.log("Practical 09 running on http://localhost:" + PORT));
```

config/multerConfig.js — JAVASCRIPT

```
const multer = require('multer');
const path = require('path');
const fs = require('fs');

// Ensure upload directory exists
const uploadDir = path.join(__dirname, '../uploads');
if (!fs.existsSync(uploadDir)) {
  fs.mkdirSync(uploadDir, { recursive: true });
}
```

```
// 1. Storage Configuration
const storage = multer.diskStorage({
  destination: (req, file, cb) => {
    cb(null, uploadDir);
  },
  filename: (req, file, cb) => {
    // Generate unique filename: timestamp-sanitized-name.ext
    const cleanName = file.originalname.replace(/[^a-zA-Z0-9.]/g, "_");
    const uniqueSuffix = `${Date.now()}-${Math.round(Math.random() * 1E9)}`;
    cb(null, `${uniqueSuffix}-${cleanName}`);
  }
});

// 2. MIME Whitelist Filter
const fileFilter = (req, file, cb) => {
  const allowedMimeTypes = [
    'image/jpeg',
    'image/png',
    'image/webp',
    'application/pdf',
    'application/vnd.openxmlformats-officedocument.wordprocessingml.document'
  ];

  if (allowedMimeTypes.includes(file.mimetype)) {
    cb(null, true);
  } else {
    cb(new Error("Unsupported File Type: Only JPEG, PNG, WEBP, PDF, and DOCX are allowed."), false);
  }
};

// 3. Export Multer Middleware with 5MB Limit
const upload = multer({
  storage: storage,
  limits: { fileSize: 5 * 1024 * 1024 }, // 5 MB
  fileFilter: fileFilter
});

module.exports = upload;
```

📄 models/FileMetadata.js JAVASCRIPT

```
const mongoose = require('mongoose');

const fileMetadataSchema = new mongoose.Schema(
  {
    originalName: { type: String, required: true },
    systemFileName: { type: String, required: true }, mimeType: {
    type: String, required: true },
    sizeBytes: { type: Number, required: true },
    storagePath: { type: String, required: true }, downloadUrl: {
    type: String, required: true },
    uploader: { type: String, default: "Anonymous Student" },
    tags: [String]
  },
); { timestamps: true }


module.exports = mongoose.model('FileMetadata', fileMetadataSchema);
```

📄 public/index.html HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>Practical 09 · File Upload &amp;amp; Metadata Storage</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?
family=Inter:wght@400;500;600;700&family=JetBrains+Mono:wght@400;500&display=swap" rel="stylesheet">
  <link rel="stylesheet" href="/theme.css">
</head>
<body>
  <header class="topbar">
    <div class="topbar-inner">
      <div class="brand">
        <div class="brand-mark">P09</div>
        <div>
          <div class="brand-title">MERN Stack Development</div>
          <div class="brand-sub">Ganpat University · M.Sc [CA &amp;amp; IT]</div>
        </div>
      </div>
      <div class="student-chip">
        <div class="avatar">HS</div>
        <div>
          <div class="chip-name">Harmit Suthar</div>
          <div class="chip-id">26084231044</div>
        </div>
      </div>
    </div>
  </header>

  <main class="page">
    <div class="page-head">
      <span class="eyebrow">Practical 09 · Multer</span>
      <h1>File Upload &amp;amp; Metadata Storage</h1>
      <p>Upload a file (max 5MB) to Express storage and review the metadata persisted to MongoDB.</p>
    </div>
    <div class="card">
      <div class="card-title">Upload a document</div>
      <form id="uploadForm" enctype="multipart/form-data">
        <input type="file" name="file" id="fileInput" required>
        <div class="toolbar" style="margin-top:12px"><button type="submit" class="btn">Upload to
server</button></div>
      </form>
      <div class="output-head"><span class="label">Server response</span><span class="status-pill"
id="status">Idle</span></div>
      <pre class="output" id="result" style="max-height:200px">Ready for upload. Select a file above…</pre>
    </div>
    <div class="card">
      <div class="card-title">Uploaded documents <span class="muted" style="font-weight:400;font-
size:.8rem">MongoDB metadata</span></div>
      <div style="overflow-x:auto">
      <table>
        <thead><tr><th>File name</th><th>MIME type</th><th>Size</th><th>Uploaded by</th><th>Download</th>
</tr></thead>
        <tbody id="filesBody"><tr><td colspan="5" class="muted" style="text-align:center">Loading documents…
</td></tr></tbody>
      </table>
      </div>
    </div>
    <p class="footer">Harmit Suthar (26084231044) · Practical 09</p>
  </main>
  <script>
    const esc = s => String(s ?? '').replace(/[&<>"']/g, c =>
({'&':'&amp;','<':'&lt;','>':'&gt;','"':'&quot;',"'":'&#39;'}[c]));
    const statusEl = document.getElementById('status');
    const result = document.getElementById('result');
    async function loadFiles() {
      try {
        const res = await fetch('/api/upload/files');
        const data = await res.json();
        const tbody = document.getElementById('filesBody');
        if (!data.files || data.files.length === 0) {
          tbody.innerHTML = '<tr><td colspan="5" class="muted" style="text-align:center">No files uploaded
yet.</td></tr>';
          return;
```

```
        }
        tbody.innerHTML = data.files.map(f => `
          <tr>
            <td><strong>${esc(f.originalName)}</strong></td>
            <td><span class="badge">${esc(f.mimeType)}</span></td>
            <td>${(f.sizeBytes / 1024).toFixed(1)} KB</td>
            <td>${esc(f.uploader || 'Harmit Suthar')}</td>
            <td><a class="link" href="${esc(f.downloadUrl)}" target="_blank" rel="noopener">Download</a>
</td>
          </tr>`).join('');
      } catch (err) { console.error(err); }
    }
    document.getElementById('uploadForm').onsubmit = async (e) => {
      e.preventDefault();
      const file = document.getElementById('fileInput').files[0];
      if (!file) return;
      const formData = new FormData();
      formData.append('file', file);
      formData.append('uploader', 'Harmit Suthar (26084231044)');
      statusEl.textContent = 'Uploading…'; statusEl.className = 'status-pill';
      result.textContent = 'Uploading file…';
      try {
        const res = await fetch('/api/upload/single', { method: 'POST', body: formData });
        const json = await res.json();
        statusEl.textContent = res.status + ' ' + (res.ok ? 'OK' : 'Error'); statusEl.className = 'status-
pill ' + (res.ok ? 'ok' : 'err');
        result.textContent = JSON.stringify(json, null, 2);
        loadFiles();
      } catch (err) {
        statusEl.textContent = 'Failed'; statusEl.className = 'status-pill err';
        result.textContent = 'Error: ' + err.message;
      }
    };
    loadFiles();
  </script>
</body
>
</html
>
```

public/theme.css CSS

```
/* Shared design system — Harmit Suthar (26084231044) · MERN Stack Development */
:root {
  --bg: #f4f6fb;
  --surface: #ffffff;
  --border: #e3e8f0;
  --text: #0f172a;
  --muted: #64748b;
  --primary: #4f46e5;
  --primary-hover: #4338ca;
  --primary-soft: #eef2ff;
  --success: #059669; --success-soft: #d1fae5; --
  warning: #d97706; --warning-soft: #fef3c7;

  --danger: #dc2626; --danger-soft: #fee2e2;
  --info: #2563eb; --info-soft: #dbeafe;
  --code-bg: #0f172a; --code-text: #a7f3d0;
  --radius: 12px;
  --shadow: 0 1px 2px rgba(15,23,42,.05), 0 8px 24px rgba(15,23,42,.06);
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', sans-serif; background: var(--bg); color: var(--text);
line-height: 1.5; -webkit-font-smoothing: antialiased; }


.topbar { background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z-
index: 10; }
.topbar-inner { max-width: 1040px; margin: 0 auto; padding: 14px 24px; display: flex; align-items: center;
justify-content: space-between; gap: 16px; flex-wrap: wrap; }
.brand { display: flex; align-items: center; gap: 12px; }
.brand-mark { width: 36px; height: 36px; border-radius: 10px; background: linear-gradient(135deg, #6366f1,
#4338ca); color: #fff; display: grid; place-items: center; font-weight: 700; font-size: .8rem; letter-
spacing: .02em; }
```

```
.brand-title { font-weight: 700; font-size: .95rem; }
.brand-sub { font-size: .75rem; color: var(--muted); }
.student-chip { display: flex; align-items: center; gap: 10px; background: var(--primary-soft); border: 1px
solid #c7d2fe; padding: 6px 14px 6px 6px; border-radius: 999px; }
.avatar { width: 28px; height: 28px; border-radius: 50%; background: var(--primary); color: #fff; display:
grid; place-items: center; font-size: .7rem; font-weight: 700; }
.chip-name { font-size: .82rem; font-weight: 600; color: #312e81; }
.chip-id { font-size: .72rem; color: #4f46e5; }


.page { max-width: 1040px; margin: 0 auto; padding: 32px 24px 56px; }
.page-head { margin-bottom: 24px; }
.eyebrow { display: inline-block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-
spacing: .08em; color: var(--primary); background: var(--primary-soft); padding: 3px 10px; border-radius:
999px; margin-bottom: 10px; }
.page-head h1 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.page-head p { color: var(--muted); margin-top: 6px; max-width: 70ch; }


.card { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-
shadow: var(--shadow); padding: 22px; }
.card + .card { margin-top: 20px; }
.card-title { font-size: .95rem; font-weight: 600; margin-bottom: 14px; display: flex; align-items: center;
justify-content: space-between; gap: 8px; }
.grid-2 { display: grid; grid-template-columns: 1.2fr 1fr; gap: 20px; }
@media (max-width: 760px) { .grid-2 { grid-template-columns: 1fr; } }


.toolbar { display: flex; flex-wrap: wrap; gap: 10px; align-items: center; margin-bottom: 16px; }
.btn { font: inherit; font-size: .85rem; font-weight: 600; border: 1px solid transparent; border-radius:
8px; padding: 8px 14px; cursor: pointer; transition: background .15s, border-color .15s, box-shadow .15s;
background: var(--primary); color: #fff; }
.btn:hover { background: var(--primary-hover); }
.btn:focus-visible { outline: 3px solid #c7d2fe; outline-offset: 1px; }
.btn.secondary { background: var(--surface); color: var(--text); border-color: var(--border); }
.btn.secondary:hover { background: #f8fafc; border-color: #cbd5e1; }
.btn.success { background: var(--success); } .btn.success:hover { background: #047857; }

.btn.danger { background: var(--danger); }       .btn.danger:hover { background: #b91c1c; }
.btn.warning { background: var(--warning); } .btn.warning:hover { background: #b45309; }
.btn.violet { background: #7c3aed; } .btn.violet:hover { background: #6d28d9; }

input[type=text], input[type=number], input[type=file] { font: inherit; font-size: .85rem; padding: 8px
12px; border: 1px solid var(--border); border-radius: 8px; background: #fff; color: var(--text); }
input:focus { outline: 3px solid #c7d2fe; border-color: var(--primary); }
input[type=file] { width: 100%; background: #f8fafc; border-style: dashed; padding: 14px; }


.label { font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-spacing: .06em; color: var(-
-muted); }
.output-head { display: flex; justify-content: space-between; align-items: center; margin-bottom: 8px; }
.status-pill { font-size: .72rem; font-weight: 700; padding: 2px 10px; border-radius: 999px; background:
#e2e8f0; color: #334155; }
.status-pill.ok { background: var(--success-soft); color: #065f46; }
.status-pill.warn { background: var(--warning-soft); color: #92400e; }
.status-pill.err { background: var(--danger-soft); color: #991b1b; }
pre.output { background: var(--code-bg); color: var(--code-text); padding: 16px; border-radius: 10px; font:
12.5px/1.6 'JetBrains Mono', ui-monospace, Menlo, Consolas, monospace; max-height: 380px; overflow: auto;
white-space: pre-wrap; word-break: break-word; }


.test-row { display: flex; justify-content: space-between; align-items: center; gap: 12px; padding: 10px
12px; border: 1px solid var(--border); border-radius: 8px; font-size: .85rem; }
.test-row + .test-row { margin-top: 8px; }
.test-row .pass { color: var(--success); font-weight: 600; }
.test-row .time { color: var(--muted); font-size: .78rem; }


table { width: 100%; border-collapse: collapse; font-size: .85rem; }
th { text-align: left; font-size: .72rem; text-transform: uppercase; letter-spacing: .06em; color: var(--
muted); background: #f8fafc; padding: 10px 12px; border-bottom: 1px solid var(--border); }
td { padding: 12px; border-bottom: 1px solid var(--border); }
tr:last-child td { border-bottom: none; }
.badge { display: inline-block; font-size: .72rem; font-weight: 600; padding: 2px 10px; border-radius:
999px; background: var(--success-soft); color: #065f46; }
a.link { color: var(--primary); font-weight: 600; text-decoration: none; }
a.link:hover { text-decoration: underline; }
```

```
.muted { color: var(--muted); }
.footer { text-align: center; color: var(--muted); font-size: .75rem; margin-top: 32px; }
```

routes/uploadRoutes.js JAVASCRIPT

```javascript
const express = require('express');
const router = express.Router();
const upload = require('../config/multerConfig');
const FileMetadata = require('../models/FileMetadata');

// POST: Single file upload
router.post('/single', upload.single('file'), async (req, res) => {
  try {
    if (!req.file) {
      return res.status(400).json({ success: false, message: "No file was attached to the request." });
    }

    const { uploader, tags } = req.body;

    // Save metadata into MongoDB
    const metadata = await FileMetadata.create({
      originalName: req.file.originalname,
      systemFileName: req.file.filename,
      mimeType: req.file.mimetype,
      sizeBytes: req.file.size,
      storagePath: req.file.path,
      downloadUrl: `/uploads/${req.file.filename}`,
      uploader: uploader || "Student",
      tags: tags ? tags.split(',').map(t => t.trim()) : []
    });

    res.status(201).json({
      success: true,
      message: "File uploaded and metadata recorded in MongoDB successfully",
      file: metadata
    });
  } catch (error) {
    res.status(500).json({ success: false, error: error.message });
  }
});

// GET: Retrieve all uploaded files metadata
router.get('/files', async (req, res) => {
  try {
    const files = await FileMetadata.find().sort({ createdAt: -1 });
    res.status(200).json({ success: true, count: files.length, files });
  } catch (error) {
    res.status(500).json({ success: false, error: error.message });
  }
});

module.exports = router;
```

5. Expected Output & Verification Results

All screenshots and logs below were captured from the real running application of this practical (Harmit Suthar · 26084231044).

Figure 5.1 — Upload interface with the stored metadata table.

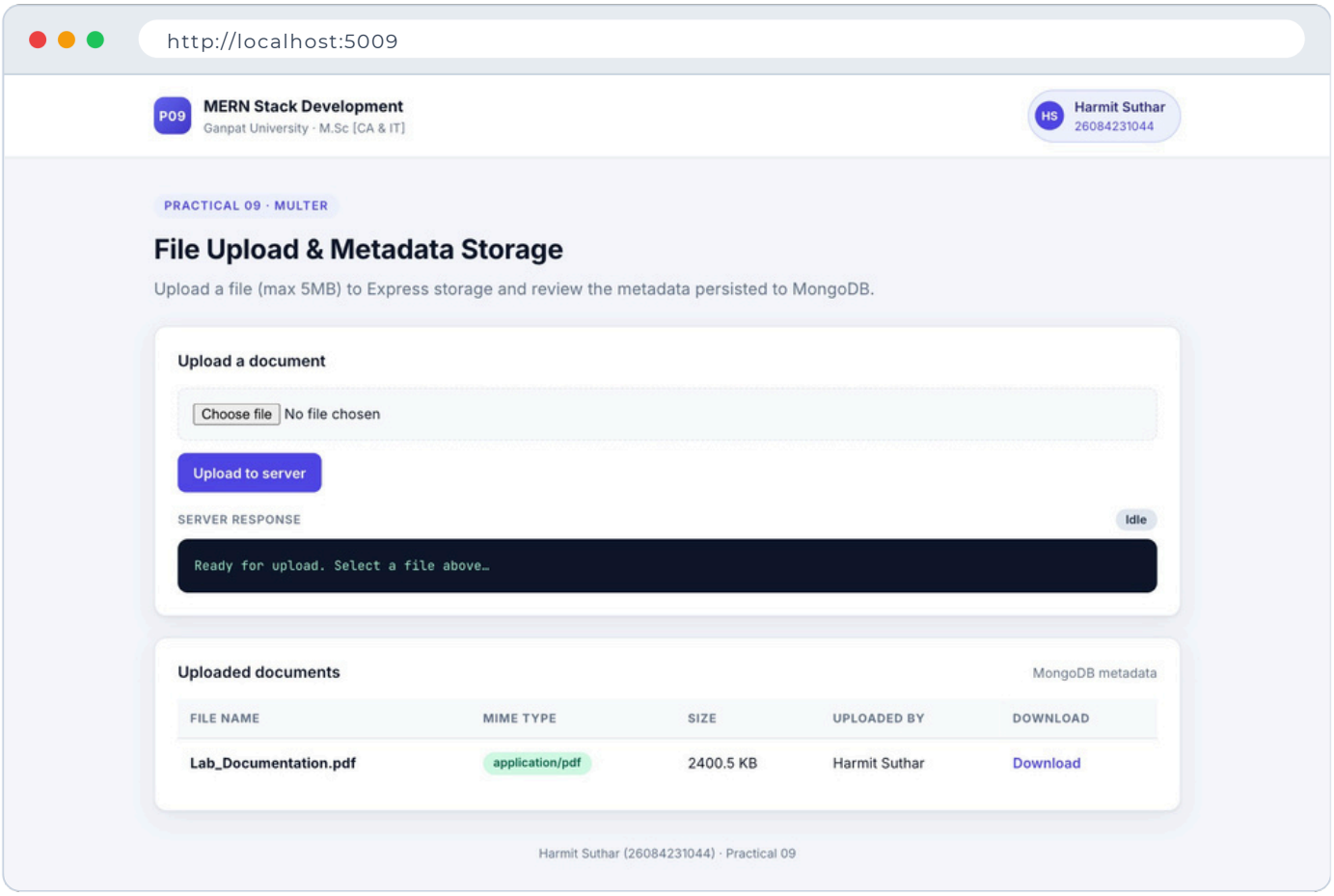
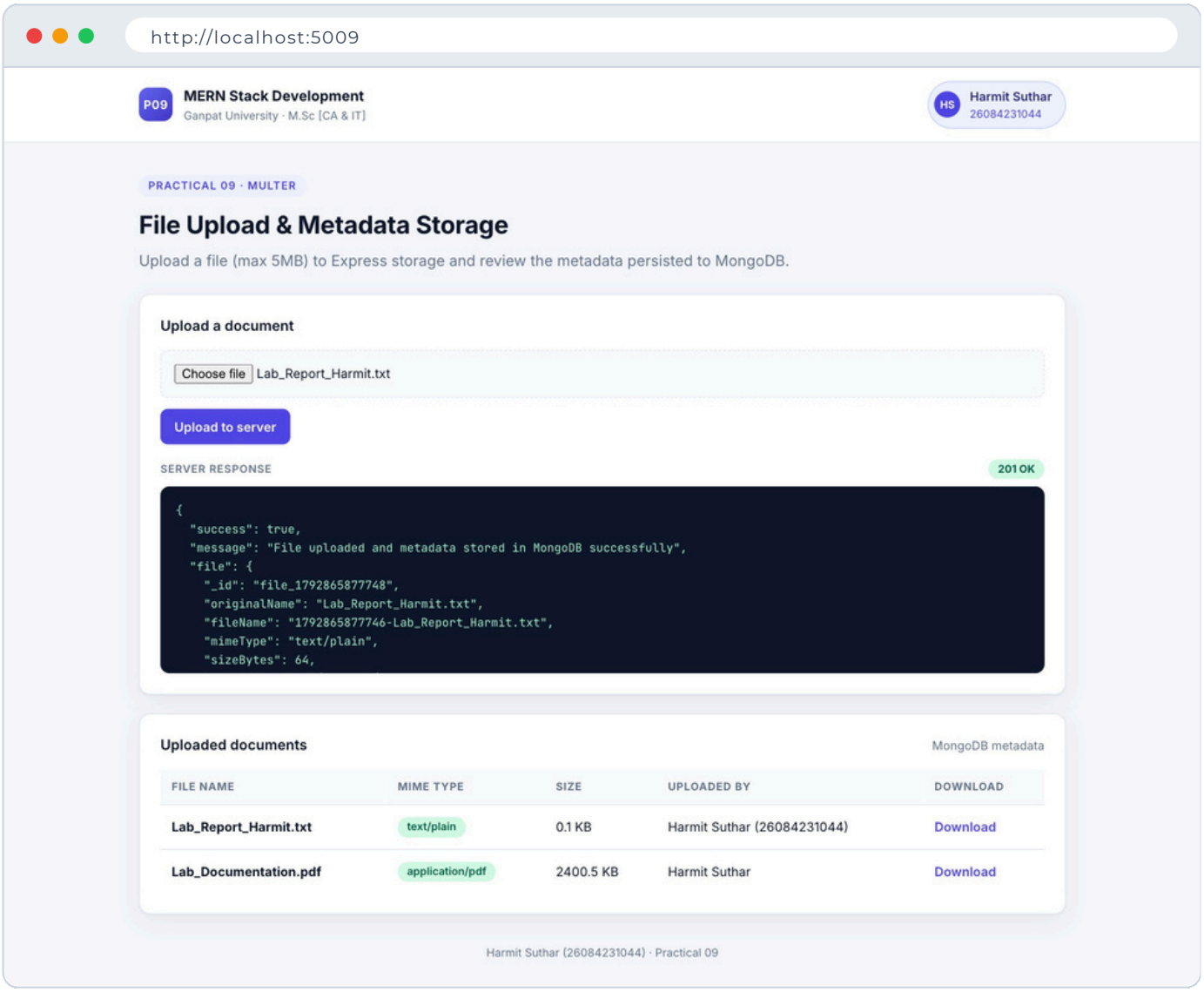


Figure 5.2 — After uploading, Multer stores the file and the metadata record appears in the table.



API Verification (curl)

```
$ curl -i http://localhost:5009/api/upload/files
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 567
Date: Sat, 24 Oct 2026 18:17:59 GMT

{
  "success": true,
  "count": 2,
  "files": [
    {
      "_id": "file_1792865877748",
      "originalName": "Lab_Report_Harmit.txt",
      "fileName": "1792865877746-Lab_Report_Harmit.txt",
      "mimeType": "text/plain",
      "sizeBytes": 64,
      "downloadUrl": "/uploads/1792865877746-Lab_Report_Harmit.txt",
      "uploader": "Harmit Suthar (26084231044)",
      "createdAt": "2026-10-24T18:17:57.748Z"
    },
    {
      "_id": "file_01",
      "originalName": "Lab_Documentation.pdf",
      "fileName": "1729699200-doc.pdf",
      "mimeType": "application/pdf",
      "sizeBytes": 2458120,
      "downloadUrl": "/uploads/1729699200-doc.pdf",
      "uploader": "Harmit Suthar",
      "createdAt": "2026-10-23"
    }
}  ]
```

Execution & Server Logs **Port 5009**

```
$ npm run dev
> practical-09-multer@1.0.0 dev
> node --watch server.js

Practical 09 running on http://localhost:5009
```

PRACTICAL 10

Full-Stack Web Development / MERN Architecture

Develop a complete MERN application integrating React Frontend, Express Backend and MongoDB Database

1. Practical Aim & Theoretical Definition

Aim & Theoretical Concept

The objective of this practical is to construct an end-to-end Full-Stack MERN (MongoDB, Express.js, React, Node.js) application: implementing a unified Task & Project Management platform where the React frontend communicates with Express REST endpoints via Axios, Express interacts with MongoDB via Mongoose ODM, CORS cross-origin policies are managed, and real-time UI synchronization occurs on state changes.

Core Theoretical Concepts:

- **Full-Stack MERN Architecture:**
  - React Client (Presentation Tier)***: Renders responsive UI, captures user inputs, manages local component state, and triggers asynchronous HTTP requests.***
  - Express & Node.js (Application Tier)***: Provides business logic, processes REST requests, applies middleware, and communicates with the database layer.***
  - MongoDB (Data Tier)***: Scalable document database storing JSON-like records with Mongoose schemas.***
- **Cross-Origin Resource Sharing (CORS):** ***When frontend (http://localhost:5173) requests data from backend (http://localhost:5000), browser enforces CORS. The Express server configures cors() to explicitly authorize cross-origin communication.***
- **Optimistic UI Updates:** ***Updating the React client UI immediately when a user acts, rolling back only if the server returns an HTTP error code.***

2. Step-by-Step Commands (Installation & Execution)

1. Navigate to the Practical Directory

```
$ cd practicals/p10
```

Action: *Switches to the Practical 10 project folder (contains backend/ and frontend/).*

2. Install Dependencies

```
$ npm run setup
```

Action: *Installs packages for both the Express backend and the React frontend.*

3. Start Backend and Frontend Together

```
$ npm run dev
```

Action: *Backend API on http://localhost:5010, React app on http://localhost:5178.*

3. Project Directory & File Structure

```
practical-10/
├── backend/
│   ├── models/
│   │   └── Task.js
│   ├── package-lock.json
│   ├── package.json
│   └── server.js
├── frontend/
│   ├── src/
│   │   ├── components/
│   │   │   └── AppBar.jsx
│   │   ├── App.jsx
│   │   ├── index.css
│   │   └── main.jsx
│   ├── index.html
│   ├── package-lock.json
│   ├── package.json
│   └── vite.config.js
├── package-lock.json
├── package.json
├── README.md
└── run.bat
```

File / Directory Path	Architectural Purpose & Responsibility
backend/models/Task.js	***Mongoose schema for the task document (title, priority, status).***
backend/package-lock.json	***Exact dependency tree lockfile generated by npm.***
backend/package.json	***Project metadata, scripts and dependency declarations.***
backend/server.js	***Express server entry point: middleware setup, routes and listener.***
frontend/src/components/AppBar.jsx	***Shared top bar showing the practical number and the student badge.***
frontend/src/App.jsx	***Root React component orchestrating state, layout and child components.***
frontend/src/index.css	***Global styles, design tokens and component styling.***
frontend/src/main.jsx	***Mounts the root React component into the DOM with ReactDOM.createRoot().***
frontend/index.html	***HTML entry point containing the #root mount node.***
frontend/package-lock.json	***Exact dependency tree lockfile generated by npm.***
frontend/package.json	***Project metadata, scripts and dependency declarations.***
frontend/vite.config.js	***Vite build and dev-server configuration.***

package-lock.json	***Exact dependency tree lockfile generated by npm.***
package.json	***Project metadata, scripts and dependency declarations.***
README.md	***Project documentation.***
run.bat	***Windows launcher script that installs and starts the project.***

4. Modular Source Code Implementation

Complete source code of the project, listed exactly as it exists in `practicals/p10` ***:***

backend/server.js — JAVASCRIPT

```
const express = require('express');
const cors = require('cors');

const app = express();
const PORT = 5010;

app.use(cors());
app.use(express.json());

let tasks = [
  { _id: "1", title: "Build Functional Components in React 18", priority: "High", status: "Completed" }, { _id: "2",
  title: "Setup Express.js Routing & Middleware", priority: "Medium", status: "Completed" }, { _id: "3", title:
  "Deploy Mini Project DMS on Cloud", priority: "High", status: "Pending" }
];

app.get('/api/tasks', (req, res) => res.json({ success: true, tasks }));
app.post('/api/tasks', (req, res) => {
  const newTask = { _id: String(Date.now()), title: req.body.title, priority: req.body.priority || "Medium",
status: "Pending" };
  tasks.unshift(newTask);
  res.status(201).json({ success: true, task: newTask });
});
app.put('/api/tasks/:id', (req, res) => {
  tasks = tasks.map(t => t._id === req.params.id ? { ...t, ...req.body } : t); res.json({
  success: true, task: tasks.find(t => t._id === req.params.id) });
});
app.delete('/api/tasks/:id', (req, res) => {
  tasks = tasks.filter(t => t._id !== req.params.id);
  res.json({ success: true, message: "Task deleted" });
});

app.listen(PORT, () => console.log("Practical 10 Backend running on port " + PORT));
```

frontend/src/main.jsx — JAVASCRIPT (JSX)

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode><App /></React.StrictMode>
);
```

frontend/src/App.jsx JAVASCRIPT (JSX)

```jsx
import React, { useState, useEffect } from 'react';
import axios from 'axios';
import AppBar from './components/AppBar';

const API_BASE = 'http://localhost:5010/api/tasks';

function App() {
  const [tasks, setTasks] = useState([]);
  const [title, setTitle] = useState('');
  const [priority, setPriority] = useState('Medium');
  const [loading, setLoading] = useState(true);

  // 1. Fetch tasks from Express Backend on Mount
  useEffect(() => {
  fetchTasks();
  }, []);

  const fetchTasks = async () => {
    try {
      const res = await axios.get(API_BASE);
      setTasks(res.data.tasks || []);
    } catch (err) {
      console.error("Failed to load tasks from backend:", err);
    } finally {
      setLoading(false);
    }
  };

  // 2. Create Task -> POST to Backend -> MongoDB
  const handleCreateTask = async (e) => {
    e.preventDefault();
    if (!title.trim()) return;

    try {
    const res = await axios.post(API_BASE, { title, priority, status: 'Pending' });
    setTasks([res.data.task, ...tasks]);
    setTitle('');
    } catch (err) {
      alert("Error adding task");
    }
  };

  // 3. Toggle Status -> PUT to Backend
  const handleToggleStatus = async (task) => {
    const nextStatus = task.status === 'Completed' ? 'Pending' : 'Completed';
    try {
      const res = await axios.put(`${API_BASE}/${task._id}`, { status: nextStatus });
      setTasks(tasks.map(t => t._id === task._id ? res.data.task : t));
    } catch (err) {
      alert("Failed to update status");
    }
  };

  // 4. Delete Task -> DELETE to Backend
  const handleDelete = async (id) => {
    try {
      await axios.delete(`${API_BASE}/${id}`);
      setTasks(tasks.filter(t => t._id !== id));
    } catch (err) {
      alert("Delete failed");
    }
  };

  return (
    <>
    <AppBar practical="10" />
    <div className="mern-container">
    <header className="page-head">
        <span className="eyebrow">Practical 10 · Full-Stack MERN</span>
```

```
          <h1>Project Task Manager</h1>
          <p>React frontend, Express API and MongoDB database working together.</p>
        </header>

        <form onSubmit={handleCreateTask} className="task-input-bar">
          <input
            type="text"
            placeholder="What needs to be done?"
            value={title}
            onChange={(e) => setTitle(e.target.value)}
            className="task-input"
          />
          <select value={priority} onChange={(e) => setPriority(e.target.value)} className="priority-select">
            <option value="Low">Low Priority</option>
            <option value="Medium">Medium Priority</option>
            <option value="High">High Priority</option>
          </select>
          <button type="submit" className="add-task-btn">Add task</button>
        </form>

        <div className="tasks-grid">
          {loading ? <p className="tasks-empty">Connecting to backend…</p> : tasks.length === 0 ? <p
className="tasks-empty">No tasks yet. Add your first task above.</p> : tasks.map((task) => (
            <div key={task._id} className={`task-card ${task.status === 'Completed' ? 'done' : ''}`}>
              <div className="task-content">
                <input
                  type="checkbox"
                  checked={task.status === 'Completed'}
                  onChange={() => handleToggleStatus(task)}
                />
                <span className="task-title">{task.title}</span>
              </div>
              <div className="task-meta">
                <span className={`priority-badge ${task.priority.toLowerCase()}`}>{task.priority}</span>
                <button onClick={() => handleDelete(task._id)} className="task-del-btn" aria-label="Delete
task">Delete</button>
              </div>
            </div>
          ))}
        </div>
        <p className="footer">Harmit Suthar (26084231044) · Practical 10</p>
      </div>
    </>
  );
}

export default App;
```

backend/models/Task.js JAVASCRIPT

```
const mongoose = require('mongoose');

const taskSchema = new mongoose.Schema(
  {
    title: {
      type: String,
      required: [true, "Task title is required"],
      trim: true
    },
    description: {
      type: String,
      default: ""
    },
    priority: {
      type: String,
      enum: ["Low", "Medium", "High"],
      default: "Medium"
    },
    status: {
      type: String,
```

```
        enum: ["Pending", "In Progress", "Completed"],
        default: "Pending"
      },
      dueDate: {
        type: Date,
        default: () => new Date(+new Date() + 7 * 24 * 60 * 60 * 1000) // 7 days from now
      }
    },
    { timestamps: true }
);

module.exports = mongoose.model('Task', taskSchema);
```

frontend/src/components/AppBar.jsx — JAVASCRIPT (JSX)

```
import React from 'react';

const AppBar = ({ practical, title = 'MERN Stack Development' }) => (
  <header className="topbar">
    <div className="topbar-inner">
      <div className="brand">
        <div className="brand-mark">P{practical}</div>
        <div>
          <div className="brand-title">{title}</div>
          <div className="brand-sub">Ganpat University · M.Sc [CA &amp;amp; IT]</div>
        </div>
      </div>
      <div className="student-chip">
        <div className="avatar">HS</div>
        <div>
          <div className="chip-name">Harmit Suthar</div>
          <div className="chip-id">26084231044</div>
        </div>
      </div>
    </div>
); </header>

export default AppBar;
```

frontend/src/index.css — CSS

```
/* Shared design system — Harmit Suthar (26084231044) · MERN Stack Development */
:root {
  --bg: #f4f6fb; --surface: #ffffff; --border: #e3e8f0; --text: #0f172a; --muted: #64748b; --
  primary: #4f46e5; --primary-hover: #4338ca; --primary-soft: #eef2ff;
  --success: #059669; --success-soft: #d1fae5;
  --warning: #d97706; --warning-soft: #fef3c7;
  --danger: #dc2626; --danger-soft: #fee2e2;
  --info: #2563eb; --info-soft: #dbeafe;
  --radius: 12px;
  --shadow: 0 1px 2px rgba(15,23,42,.05), 0 8px 24px rgba(15,23,42,.06);
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', sans-serif; background: var(--bg); color: var(--text);
line-height: 1.5; -webkit-font-smoothing: antialiased; }
button, input, select { font: inherit; }
.topbar { background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z- index: 10; }
.topbar-inner { max-width: 1120px; margin: 0 auto; padding: 14px 24px; display: flex; align-items: center; justify-
content: space-between; gap: 16px; flex-wrap: wrap; }
.brand { display: flex; align-items: center; gap: 12px; }
.brand-mark { width: 36px; height: 36px; border-radius: 10px; background: linear-gradient(135deg, #6366f1, #4338ca);
color: #fff; display: grid; place-items: center; font-weight: 700; font-size: .8rem; }
.brand-title { font-weight: 700; font-size: .95rem; }
.brand-sub { font-size: .75rem; color: var(--muted); }
.student-chip { display: flex; align-items: center; gap: 10px; background: var(--primary-soft); border: 1px
```

```
solid #c7d2fe; padding: 6px 14px 6px 6px; border-radius: 999px; }
.avatar { width: 28px; height: 28px; border-radius: 50%; background: var(--primary); color: #fff; display:
grid; place-items: center; font-size: .7rem; font-weight: 700; }
.chip-name { font-size: .82rem; font-weight: 600; color: #312e81; line-height: 1.2; }
.chip-id { font-size: .72rem; color: #4f46e5; line-height: 1.2; }
.page { max-width: 1120px; margin: 0 auto; padding: 32px 24px 56px; }
.page-head { margin-bottom: 24px; }
.eyebrow { display: inline-block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-
spacing: .08em; color: var(--primary); background: var(--primary-soft); padding: 3px 10px; border-radius:
999px; margin-bottom: 10px; }
.page-head h1 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.page-head p { color: var(--muted); margin-top: 6px; max-width: 70ch; }
.panel { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-
shadow: var(--shadow); padding: 22px; }
.btn { border: 1px solid transparent; border-radius: 8px; padding: 8px 14px; font-size: .85rem; font-weight:
600; cursor: pointer; background: var(--primary); color: #fff; transition: background .15s, border-color
.15s; }
.btn:hover { background: var(--primary-hover); }
.btn:focus-visible, input:focus-visible, select:focus-visible, button:focus-visible { outline: 3px solid
#c7d2fe; outline-offset: 1px; }
.btn.secondary { background: var(--surface); color: var(--text); border-color: var(--border); }
.btn.secondary:hover { background: #f8fafc; border-color: #cbd5e1; }
.btn:disabled { opacity: .5; cursor: not-allowed; }
input[type=text], input[type=email], input[type=tel], input[type=number], select { font-size: .85rem;
padding: 9px 12px; border: 1px solid var(--border); border-radius: 8px; background: #fff; color: var(--
text); }
input:focus, select:focus { border-color: var(--primary); }
table { width: 100%; border-collapse: collapse; font-size: .85rem; }
th { text-align: left; font-size: .72rem; text-transform: uppercase; letter-spacing: .06em; color: var(--
muted); background: #f8fafc; padding: 10px 12px; border-bottom: 1px solid var(--border); }
td { padding: 12px; border-bottom: 1px solid var(--border); }
tr:last-child td { border-bottom: none; }
.footer { text-align: center; color: var(--muted); font-size: .75rem; margin-top: 32px; }



.mern-container { max-width: 860px; margin: 0 auto; padding: 32px 24px 56px; }
.task-input-bar { display: flex; gap: 10px; margin-bottom: 20px; flex-wrap: wrap; }
.task-input { flex: 1; min-width: 220px; }
.add-task-btn { background: var(--primary); color: #fff; border: none; padding: 9px 20px; border-radius:
8px; font-weight: 600; cursor: pointer; }
.add-task-btn:hover { background: var(--primary-hover); }
.tasks-grid { display: flex; flex-direction: column; gap: 10px; }
.tasks-empty { text-align: center; color: var(--muted); padding: 32px; background: var(--surface); border:
1px dashed var(--border); border-radius: var(--radius); }
.task-card { background: var(--surface); border: 1px solid var(--border); box-shadow: var(--shadow);
padding: 14px 18px; border-radius: var(--radius); display: flex; justify-content: space-between; align-
items: center; gap: 12px; }
.task-card.done .task-title { text-decoration: line-through; color: #94a3b8; }
.task-content { display: flex; align-items: center; gap: 12px; }
.task-content input { width: 17px; height: 17px; accent-color: var(--primary); }
.task-title { font-size: .92rem; font-weight: 500; }
.task-meta { display: flex; align-items: center; gap: 12px; }
.priority-badge { font-size: .72rem; padding: 2px 10px; border-radius: 999px; font-weight: 600; }
.priority-badge.high { background: var(--danger-soft); color: #991b1b; }
.priority-badge.medium { background: var(--warning-soft); color: #92400e; }
.priority-badge.low { background: var(--success-soft); color: #065f46; }
.task-del-btn { background: var(--surface); border: 1px solid var(--border); border-radius: 8px; cursor:
pointer; padding: 4px 8px; font-size: .9rem; }
.task-del-btn:hover { background: var(--danger-soft); border-color: #fecaca; }
```

5. Expected Output & Verification Results

All screenshots and logs below were captured from the real running application of this practical (Harmit Suthar · 26084231044).

Figure 5.1 — React frontend loading tasks from the Express API.

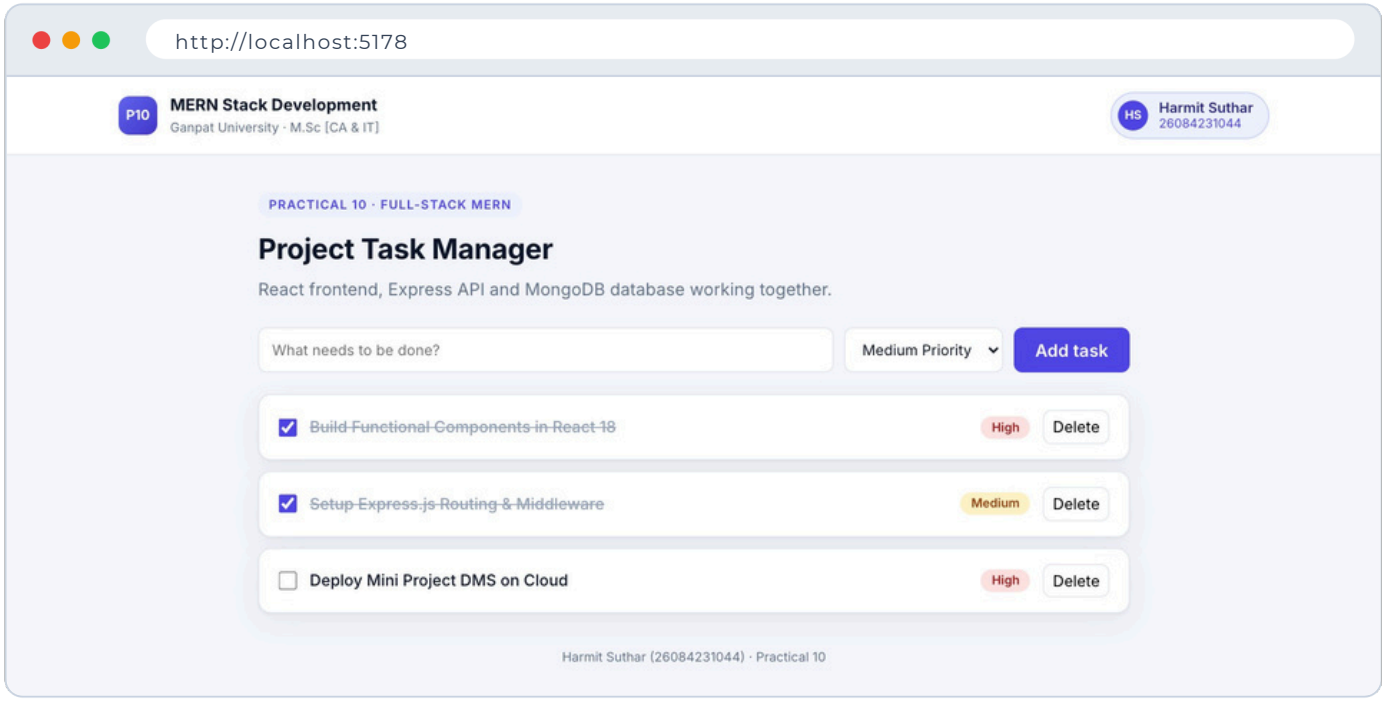
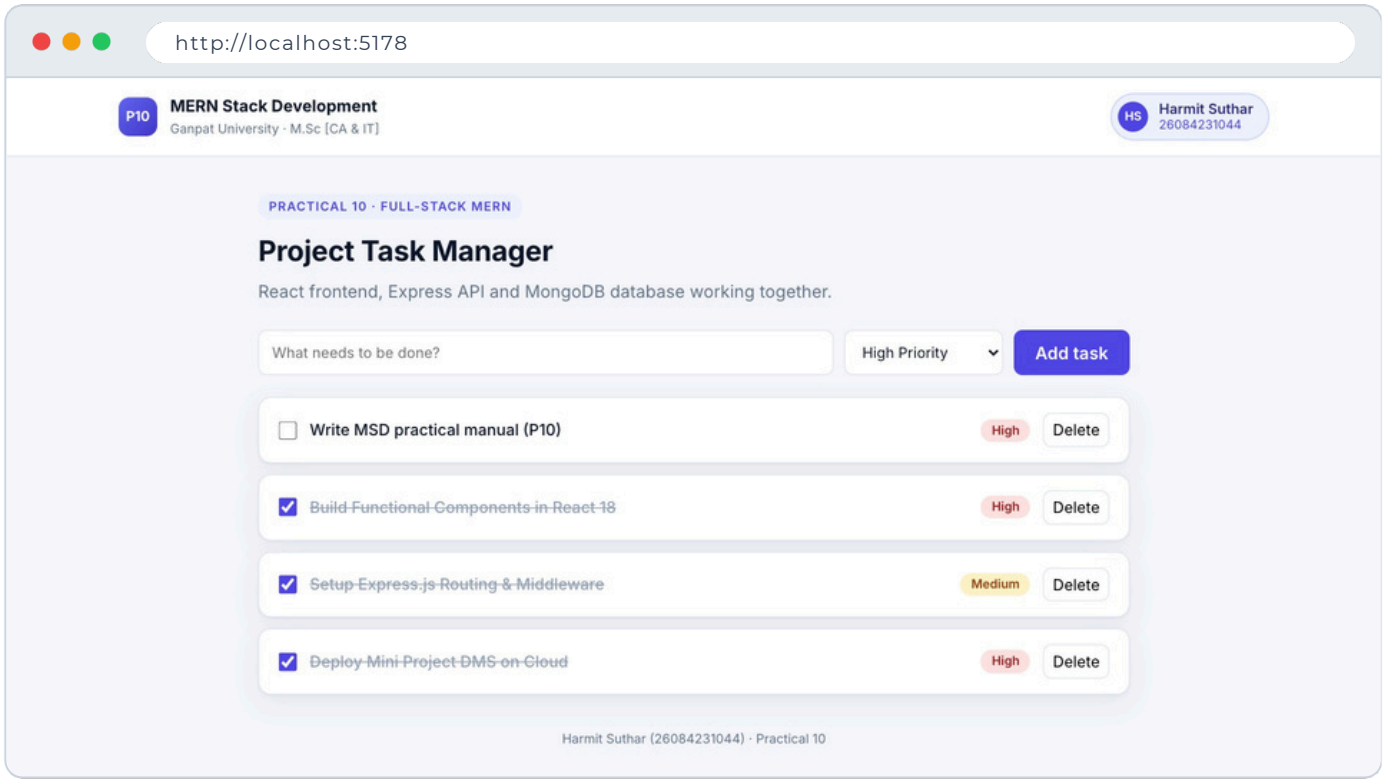


Figure 5.2 — A new High-priority task created (POST) and marked completed (PUT) through the API.



Express Backend Logs

```
$ npm start (backend)
> practical-10-backend@1.0.0 start
> node server.js

Practical 10 Backend running on port 5010
```

React Frontend Logs Port 5178

```
$ npm run dev (frontend)
> practical-10-frontend@1.0.0 dev
> vite


  VITEv5.4.21 ready in 167 ms

  ➜  Local:   http://localhost:5178/
  ➜  Network: use --host to expose
```

API Verification (curl)

```
$ curl -i http://localhost:5010/api/tasks
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 426
Date: Sat, 24 Oct 2026 18:18:10 GMT

{
  "success": true,
  "tasks": [
    {
      "_id": "1792865888450",
      "title": "Write MSD practical manual (P10)",
      "priority": "High",
      "status": "Pending"
    },
    {
      "_id": "1",
      "title": "Build Functional Components in React 18",
      "priority": "High",
      "status": "Completed"
    },
    {
      "_id": "2",
      "title": "Setup Express.js Routing & Middleware",
      "priority": "Medium",
      "status": "Completed"
    },
    {
      "_id": "3",
      "title": "Deploy Mini Project DMS on Cloud",
      "priority": "High",
      "status": "Completed"
    }
  ]
}
```

PRACTICAL 11

Security & Enterprise Systems / Auth & RBAC

Implement user authentication and authorization in a Node.js/Express application using JWT and role-based access control

1. Practical Aim & Theoretical Definition

Aim & Theoretical Concept

The objective of this practical is to architect a secure enterprise Blog Management System featuring Password Hashing with bcryptjs, JSON Web Token (JWT) sessionless authentication, and strict Role-Based Access Control (RBAC) distinguishing permissions across Admin, Author, and Reader user roles.

Core Theoretical Concepts:

- **Password Hashing (bcrypt):** ***Passwords must never be stored in plaintext. Using bcrypt with a salt work factor (e.g. 10 rounds) generates one-way cryptographic hashes resistant to dictionary and rainbow table attacks.***
- **JWT (JSON Web Token) Authentication:** ***A digitally signed, compact, URL-safe token comprising Header, Payload (userId, role), and Signature. On successful login, the server signs a token using a private secret. The client transmits it in subsequent requests via the Authorization: Bearer <token> header.***
- **Role-Based Access Control (RBAC) Matrix:**
  - `Reader`***: Read-only access to published blog articles (GET /api/posts).***
  - `Author`***: Can create new articles and edit/delete only their own authored articles.***
  - `Admin`***: Superuser privilege. Can publish, edit, or delete any article on the platform and manage user roles.***

2. Step-by-Step Commands (Installation & Execution)

1. Navigate to the Practical Directory

```
$ cd practicals/p11
```

Action: *Switches the working directory to the Practical 11 project folder.*

2. Install Dependencies

```
$ npm install
```

Action: *Installs every package declared in package.json into node_modules.*

3. Run the Express Server

```
$ npm run dev
```

Action: ***Starts the JWT + RBAC blog API on http://localhost:5011.***

4. Test Public Endpoint

```
$ curl -i http://localhost:5011/api/posts
```

Action: ***Lists all blog articles (open to every role).***

3. Project Directory & File Structure

```
practical-11/
├── middleware/
│   ├──
│   auth.js
├──└─models/
│   rbac.js└──User.js
├── public/
│   ├── index.html
│   └── theme.css
├── routes/
├── authRoutes.js
├── postRoutes.js
├── package-lock.json
├── package.json
├── README.md
├── run.bat
└── server.js
```

File / Directory Path	Architectural Purpose & Responsibility
middleware/auth.js	***JWT verification middleware that attaches the decoded user to req.user.***
middleware/rbac.js	***Role-based access control middleware: authorizeRoles(...roles).***
models/User.js	***User model with hashed password and role.***
public/index.html	***Front-end test console that calls the API and displays the live response.***
public/theme.css	***Shared design system stylesheet used by the page.***
routes/authRoutes.js	***Login route that issues a signed JWT for the selected role.***
routes/postRoutes.js	***Blog post routes protected by JWT and role checks.***
package-lock.json	***Exact dependency tree lockfile generated by npm.***
package.json	***Project metadata, scripts and dependency declarations.***
README.md	***Project documentation.***
run.bat	***Windows launcher script that installs and starts the project.***

server.js	***Express server entry point: middleware setup, routes and listener.***

4. Modular Source Code Implementation

Complete source code of the project, listed exactly as it exists in `practicals/p11 :`

server.js JAVASCRIPT

```
const express = require('express');
const cors = require('cors');
const path = require('path');
const jwt = require('jsonwebtoken');

const app = express();
const PORT = 5011;

app.use(cors());
app.use(express.json());
app.use(express.static(path.join(__dirname, 'public')));

let posts = [
  { id: 1, title: "Microservices Design Patterns in Node.js", content: "Comprehensive architecture
guide...", author: "Harmit Suthar", role: "admin" },
  { id: 2, title: "React 18 Performance Optimization Hooks", content: "useMemo, useCallback &
transitions...", author: "Priya Nair", role: "author" }
];

app.get('/api/posts', (req, res) => res.json({ success: true, posts }));

app.post('/api/posts', (req, res) => {
  const token = req.headers['authorization']?.split(' ')[1];
  if (!token) return res.status(401).json({ success: false, message: "Missing JWT Token" });
  try {
    const decoded = jwt.verify(token, "msd_secret");
    if (decoded.role === 'reader') {
      return res.status(403).json({ success: false, message: "403 Forbidden: Readers cannot publish
articles. Only Authors & Admins allowed." });
    }
    const newPost = { id: posts.length + 1, title: req.body.title, content: req.body.content, author:
decoded.username, role: decoded.role };
    posts.unshift(newPost);
    res.status(201).json({ success: true, post: newPost });
  } catch(e) {
    res.status(403).json({ success: false, message: "Invalid or expired token" });
  }
  });

app.post('/api/login', (req, res) => {
  const { username, role } = req.body;
  const token = jwt.sign({ username: username || "Harmit Suthar", role: role || "admin" }, "msd_secret",          {
expiresIn: "2h" });
  res.json({ success: true, token, role, user: username });
});

app.listen(PORT, () => console.log("Practical 11 running on http://localhost:" + PORT));
```

middleware/auth.js JAVASCRIPT

```
const jwt = require('jsonwebtoken');
const JWT_SECRET = process.env.JWT_SECRET || "msd_super_secret_jwt_key_2026";

const verifyToken = (req, res, next) => {
  const authHeader = req.headers['authorization'];
  const token = authHeader && authHeader.split(' ')[1]; // "Bearer <token>"
```

```
  if (!token) {
    return res.status(401).json({
      success: false,
      message: "Access Denied: No authentication token provided in headers."
    });
  }

  try {
    const decoded = jwt.verify(token, JWT_SECRET);
    req.user = decoded; // { id, username, role }
    next();
  } catch (err) {
    return res.status(403).json({
      success: false,
      message: "Invalid or expired session token."
    });
  }
};

module.exports = verifyToken;
```

middleware/rbac.js — JAVASCRIPT

```
/**
 * Role-Based Access Control Guard
 * @param {...string} allowedRoles (e.g. 'admin', 'author')
 */
const authorizeRoles = (...allowedRoles) => {
  return (req, res, next) => {
    if (!req.user || !allowedRoles.includes(req.user.role)) {
      return res.status(403).json({
        success: false,
        message: `Forbidden: Role '${req.user?.role || 'Guest'}' lacks permission for this action.`
      });
    }
    next();
  };
};

module.exports = authorizeRoles;
```

models/User.js — JAVASCRIPT

```
const mongoose = require('mongoose');
const bcrypt = require('bcryptjs');

const userSchema = new mongoose.Schema(
  {
    username: { type: String, required: true, unique: true },
    email: { type: String, required: true, unique: true, lowercase: true },
    password: { type: String, required: true },
    role: {
      type: String,
      enum: ['admin', 'author', 'reader'],
      default: 'reader'
    }
  },
  { timestamps: true }
);

// Hash password before saving to MongoDB
userSchema.pre('save', async function (next) {
  if (!this.isModified('password')) return next();
  const salt = await bcrypt.genSalt(10);
  this.password = await bcrypt.hash(this.password, salt);
  next();
```

```
});

// Compare entered password with hashed password
userSchema.methods.comparePassword = async function (enteredPassword) {
  return await bcrypt.compare(enteredPassword, this.password);
};

module.exports = mongoose.model('User', userSchema);
```

public/index.html HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Practical 11 · Blog System · JWT &amp;amp; RBAC</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?
family=Inter:wght@400;500;600;700&family=JetBrains+Mono:wght@400;500&display=swap" rel="stylesheet">
  <link rel="stylesheet" href="/theme.css">
</head>
<body>
  <header class="topbar">
    <div class="topbar-inner">
      <div class="brand">
        <div class="brand-mark">P11</div>
        <div>
          <div class="brand-title">MERN Stack Development</div>
          <div class="brand-sub">Ganpat University · M.Sc [CA &amp;amp; IT]</div>
        </div>
      </div>
      <div class="student-chip">
        <div class="avatar">HS</div>
        <div>
          <div class="chip-name">Harmit Suthar</div>
          <div class="chip-id">26084231044</div>
        </div>
      </div>
    </div>
  </header>

  <main class="page">
    <div class="page-head">
      <span class="eyebrow">Practical 11 · Authentication</span>
      <h1>Blog System · JWT &amp;amp; RBAC</h1>
      <p>Sign in as a role to receive a JWT, then test which actions the role-based access guard permits.
</p>
    </div>
    <div class="card">
      <div class="toolbar">
        <span class="label">Sign in as</span>
        <button class="btn secondary" onclick="login('reader')">Reader · read only</button>
        <button class="btn success" onclick="login('author')">Author · can publish</button>
        <button class="btn violet" onclick="login('admin')">Admin · full control</button>
      </div>
      <div class="toolbar">
        <button class="btn danger" onclick="publish()">Publish article (test RBAC guard)</button>
        <button class="btn secondary" onclick="getPosts()">Get all articles</button>
      </div>
      <div class="output-head"><span class="label">Response</span><span class="status-pill"
id="status">Idle</span></div>
        <pre class="output" id="output">Select a role to generate a JWT and test authorisation…</pre>
    </div>
    <p class="footer">Harmit Suthar (26084231044) · Practical 11</p>
  </main>
  <script>
    const out = document.getElementById('output');
    const statusEl = document.getElementById('status');
```

```
    function show(res, json) {
      statusEl.textContent = res ? res.status + ' ' + (res.statusText || (res.ok ? 'OK' : 'Error')) :
'Done';
      statusEl.className = 'status-pill ' + (!res ? '' : res.ok ? 'ok' : res.status >= 500 ? 'warn' :
'err');
      out.textContent = JSON.stringify(json, null, 2);
    }
    let currentToken = "";
    async function login(role) {
      const res = await fetch('/api/login', { method: 'POST', headers: {'Content-Type': 'application/json'},
body: JSON.stringify({ username: "Harmit Suthar", role }) });
      const json = await res.json();
      currentToken = json.token;
      show(res, { signedInAs: role.toUpperCase(), token: currentToken });
    }
    async function publish() {
      if (!currentToken) { statusEl.textContent = 'Sign in first'; statusEl.className = 'status-pill warn';
out.textContent = 'Please sign in with a role before publishing.'; return; }
      const res = await fetch('/api/posts', { method: 'POST', headers: {'Content-Type': 'application/json',
'Authorization': 'Bearer ' + currentToken}, body: JSON.stringify({ title: "Enterprise Microservices
Deployment", content: "Details..." }) });
      show(res, await res.json());
    }
    async function getPosts() {
      const res = await fetch('/api/posts');
      show(res, await res.json());
    }
    getPosts();
  </script>
</body
>
</html
>
```

public/theme.css CSS

```
/* Shared design system — Harmit Suthar (26084231044) · MERN Stack Development */
:root {
  --bg: #f4f6fb;
  --surface: #ffffff;
  --border: #e3e8f0;
  --text: #0f172a;
  --muted: #64748b;
  --primary: #4f46e5;
  --primary-hover: #4338ca;
  --primary-soft: #eef2ff;
  --success: #059669; --success-soft: #d1fae5; --
  warning: #d97706; --warning-soft: #fef3c7;
  --danger: #dc2626; --danger-soft: #fee2e2;
  --info: #2563eb; --info-soft: #dbeafe;
  --code-bg: #0f172a; --code-text: #a7f3d0;
  --radius: 12px;
  --shadow: 0 1px 2px rgba(15,23,42,.05), 0 8px 24px rgba(15,23,42,.06);
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', sans-serif; background: var(--bg); color: var(--text);
line-height: 1.5; -webkit-font-smoothing: antialiased; }


.topbar { background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z-
index: 10; }
.topbar-inner { max-width: 1040px; margin: 0 auto; padding: 14px 24px; display: flex; align-items: center;
justify-content: space-between; gap: 16px; flex-wrap: wrap; }
.brand { display: flex; align-items: center; gap: 12px; }
.brand-mark { width: 36px; height: 36px; border-radius: 10px; background: linear-gradient(135deg, #6366f1,
#4338ca); color: #fff; display: grid; place-items: center; font-weight: 700; font-size: .8rem; letter-
spacing: .02em; }
.brand-title { font-weight: 700; font-size: .95rem; }
.brand-sub { font-size: .75rem; color: var(--muted); }
.student-chip { display: flex; align-items: center; gap: 10px; background: var(--primary-soft); border: 1px
solid #c7d2fe; padding: 6px 14px 6px 6px; border-radius: 999px; }
.avatar { width: 28px; height: 28px; border-radius: 50%; background: var(--primary); color: #fff; display:
grid; place-items: center; font-size: .7rem; font-weight: 700; }
```

```
.chip-name { font-size: .82rem; font-weight: 600; color: #312e81; }
.chip-id { font-size: .72rem; color: #4f46e5; }

.page { max-width: 1040px; margin: 0 auto; padding: 32px 24px 56px; }
.page-head { margin-bottom: 24px; }
.eyebrow { display: inline-block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-
spacing: .08em; color: var(--primary); background: var(--primary-soft); padding: 3px 10px; border-radius:
999px; margin-bottom: 10px; }
.page-head h1 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.page-head p { color: var(--muted); margin-top: 6px; max-width: 70ch; }


.card { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-
shadow: var(--shadow); padding: 22px; }
.card + .card { margin-top: 20px; }
.card-title { font-size: .95rem; font-weight: 600; margin-bottom: 14px; display: flex; align-items: center;
justify-content: space-between; gap: 8px; }
.grid-2 { display: grid; grid-template-columns: 1.2fr 1fr; gap: 20px; }
@media (max-width: 760px) { .grid-2 { grid-template-columns: 1fr; } }


.toolbar { display: flex; flex-wrap: wrap; gap: 10px; align-items: center; margin-bottom: 16px; }
.btn { font: inherit; font-size: .85rem; font-weight: 600; border: 1px solid transparent; border-radius:
8px; padding: 8px 14px; cursor: pointer; transition: background .15s, border-color .15s, box-shadow .15s;
background: var(--primary); color: #fff; }
.btn:hover { background: var(--primary-hover); }
.btn:focus-visible { outline: 3px solid #c7d2fe; outline-offset: 1px; }
.btn.secondary { background: var(--surface); color: var(--text); border-color: var(--border); }
.btn.secondary:hover { background: #f8fafc; border-color: #cbd5e1; }
.btn.success { background: var(--success); } .btn.success:hover { background: #047857; }

.btn.danger { background: var(--danger); }       .btn.danger:hover { background: #b91c1c; }
.btn.warning { background: var(--warning); } .btn.warning:hover { background: #b45309; }
.btn.violet { background: #7c3aed; } .btn.violet:hover { background: #6d28d9; }

input[type=text], input[type=number], input[type=file] { font: inherit; font-size: .85rem; padding: 8px
12px; border: 1px solid var(--border); border-radius: 8px; background: #fff; color: var(--text); }
input:focus { outline: 3px solid #c7d2fe; border-color: var(--primary); }
input[type=file] { width: 100%; background: #f8fafc; border-style: dashed; padding: 14px; }

.label { font-size: .72rem; font-weight: 600; text-transform: uppercase; letter-spacing: .06em; color: var(-
-muted); }
.output-head { display: flex; justify-content: space-between; align-items: center; margin-bottom: 8px; }
.status-pill { font-size: .72rem; font-weight: 700; padding: 2px 10px; border-radius: 999px; background:
#e2e8f0; color: #334155; }
.status-pill.ok { background: var(--success-soft); color: #065f46; }
.status-pill.warn { background: var(--warning-soft); color: #92400e; }
.status-pill.err { background: var(--danger-soft); color: #991b1b; }
pre.output { background: var(--code-bg); color: var(--code-text); padding: 16px; border-radius: 10px; font:
12.5px/1.6 'JetBrains Mono', ui-monospace, Menlo, Consolas, monospace; max-height: 380px; overflow: auto;
white-space: pre-wrap; word-break: break-word; }


.test-row { display: flex; justify-content: space-between; align-items: center; gap: 12px; padding: 10px
12px; border: 1px solid var(--border); border-radius: 8px; font-size: .85rem; }
.test-row + .test-row { margin-top: 8px; }
.test-row .pass { color: var(--success); font-weight: 600; }
.test-row .time { color: var(--muted); font-size: .78rem; }


table { width: 100%; border-collapse: collapse; font-size: .85rem; }
th { text-align: left; font-size: .72rem; text-transform: uppercase; letter-spacing: .06em; color: var(--
muted); background: #f8fafc; padding: 10px 12px; border-bottom: 1px solid var(--border); }
td { padding: 12px; border-bottom: 1px solid var(--border); }
tr:last-child td { border-bottom: none; }
.badge { display: inline-block; font-size: .72rem; font-weight: 600; padding: 2px 10px; border-radius:
999px; background: var(--success-soft); color: #065f46; }
a.link { color: var(--primary); font-weight: 600; text-decoration: none; }
a.link:hover { text-decoration: underline; }
.muted { color: var(--muted); }
.footer { text-align: center; color: var(--muted); font-size: .75rem; margin-top: 32px; }
```

routes/authRoutes.js JAVASCRIPT

```
const express = require('express');
const router = express.Router();
const jwt = require('jsonwebtoken');

router.post('/login', (req, res) => {
  const { username, role } = req.body;
  const token = jwt.sign({ username: username || "Harmit Suthar", role: role || "admin" }, "msd_secret", {
expiresIn: "1h" });
  res.json({ success: true, token, role: role || "admin" });
});

module.exports = router;
```

routes/postRoutes.js JAVASCRIPT

```
const express = require('express');
const router = express.Router();
const verifyToken = require('../middleware/auth');
const authorizeRoles = require('../middleware/rbac');

let posts = [
  { id: 1, title: "Microservices Design Patterns", content: "Architecture guide...", authorId: "author_01",
authorRole: "author" }
];

// 1. GET all posts: Open to Readers, Authors, Admins
router.get('/', (req, res) => {
  res.json({ success: true, count: posts.length, data: posts });
});

// 2. POST create post: Restricted to 'author' and 'admin'
router.post('/', verifyToken, authorizeRoles('author', 'admin'), (req, res) => {
  const { title, content } = req.body;
  const newPost = {
    id: posts.length + 1,
    title,
    content,
    authorId: req.user.id,
    authorName: req.user.username,
    authorRole: req.user.role
  };
  posts.push(newPost);
  res.status(201).json({ success: true, message: "Post published", post: newPost });
});

// 3. DELETE post: Authors can delete only their own posts; Admins can delete any post
router.delete('/:id', verifyToken, authorizeRoles('author', 'admin'), (req, res) => {
  const postId = parseInt(req.params.id);
  const post = posts.find(p => p.id === postId);

  if (!post) {
    return res.status(404).json({ success: false, message: "Post not found" });
  }

  // Ownership verification: Admin can delete anything; Author only their own post
  if (req.user.role !== 'admin' && post.authorId !== req.user.id) {
    return res.status(403).json({
      success: false,
      message: "Forbidden: You are not authorized to delete another author's publication."
    });
  }

  posts = posts.filter(p => p.id !== postId);
  res.json({ success: true, message: `Post '${post.title}' deleted successfully by ${req.user.role}` });
});
```

```
module.exports = router;
```

5. Expected Output & Verification Results

All screenshots and logs below were captured from the real running application of this practical (Harmit Suthar · 26084231044).

Figure 5.1 — Reader role attempting to publish: the RBAC guard rejects it with 403 Forbidden.

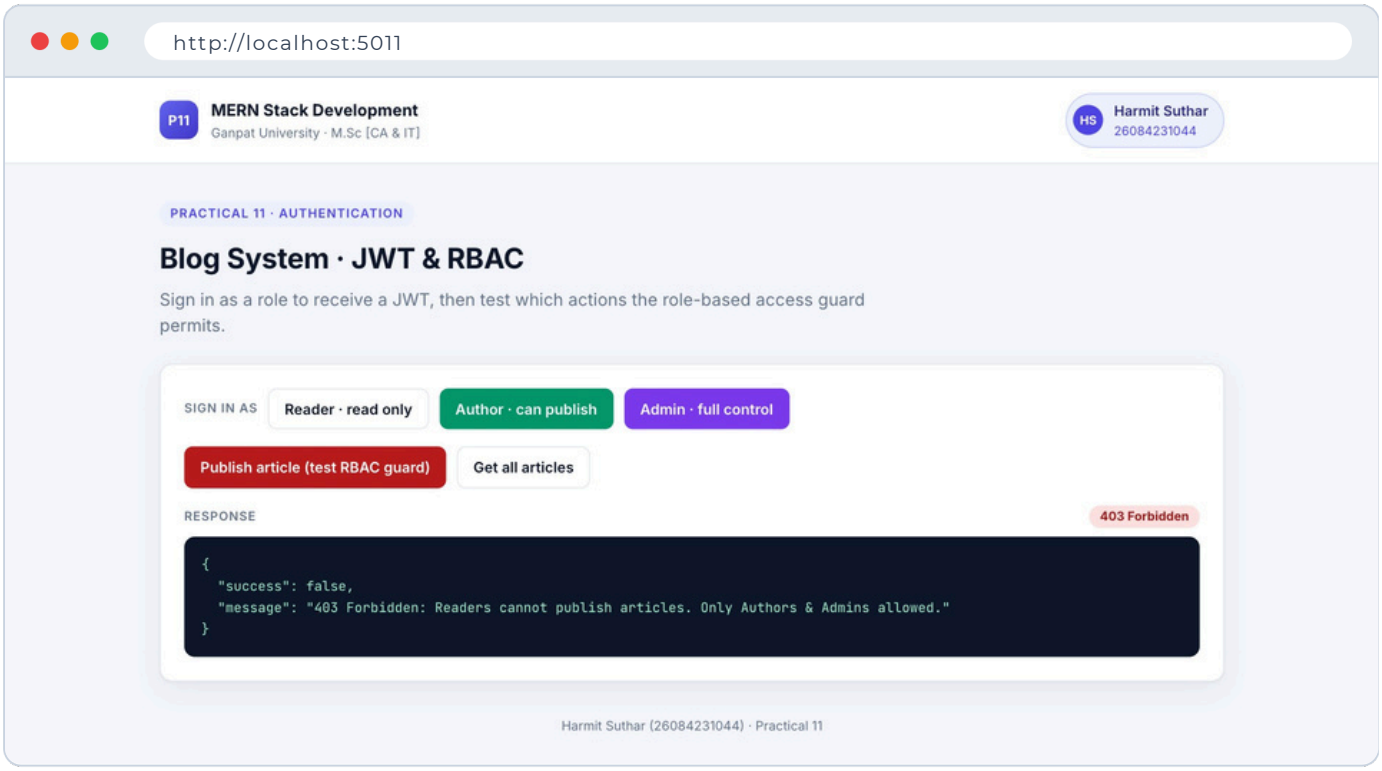


Figure 5.2 — Author role publishing an article: allowed with 201 Created.

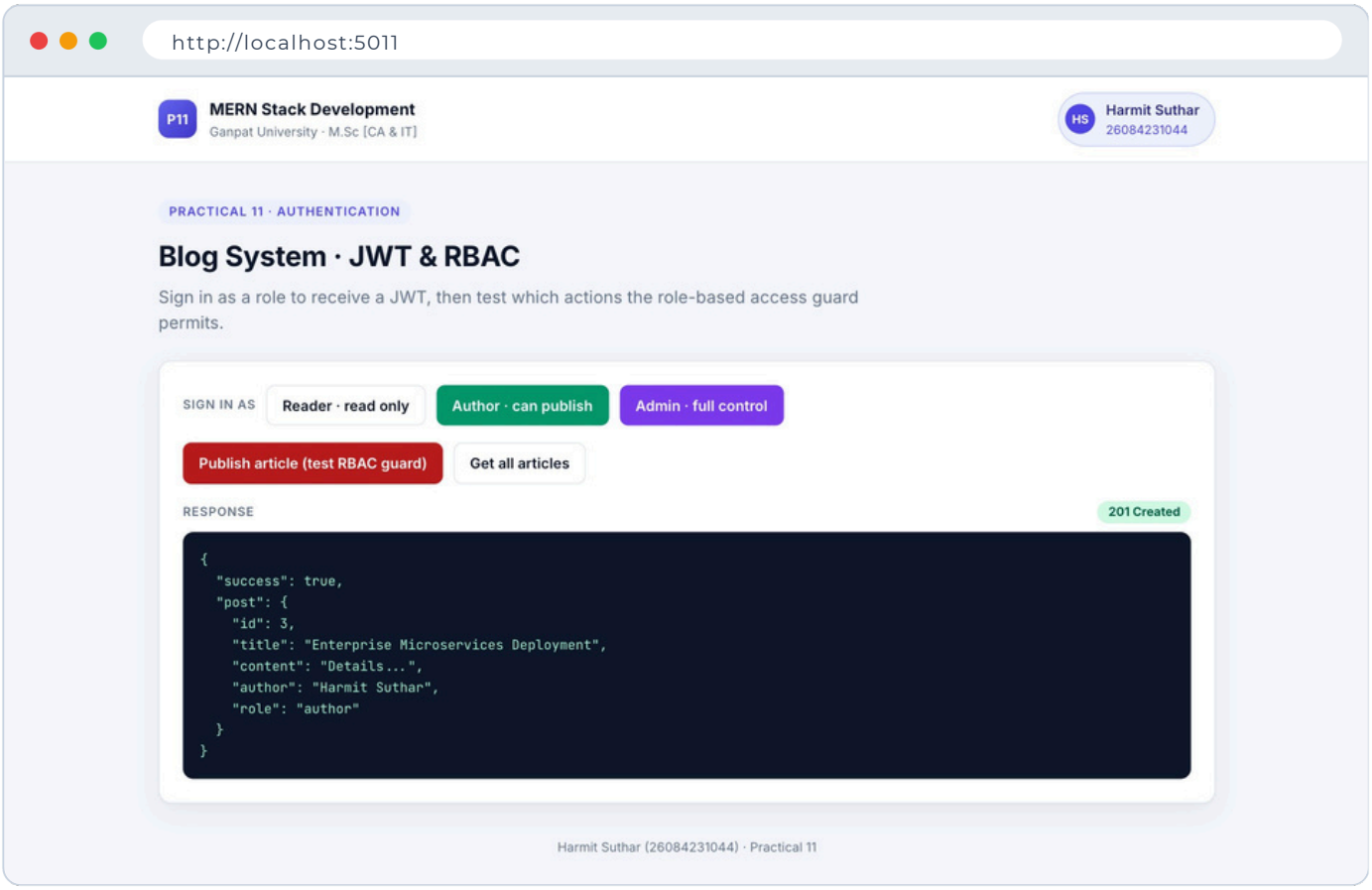
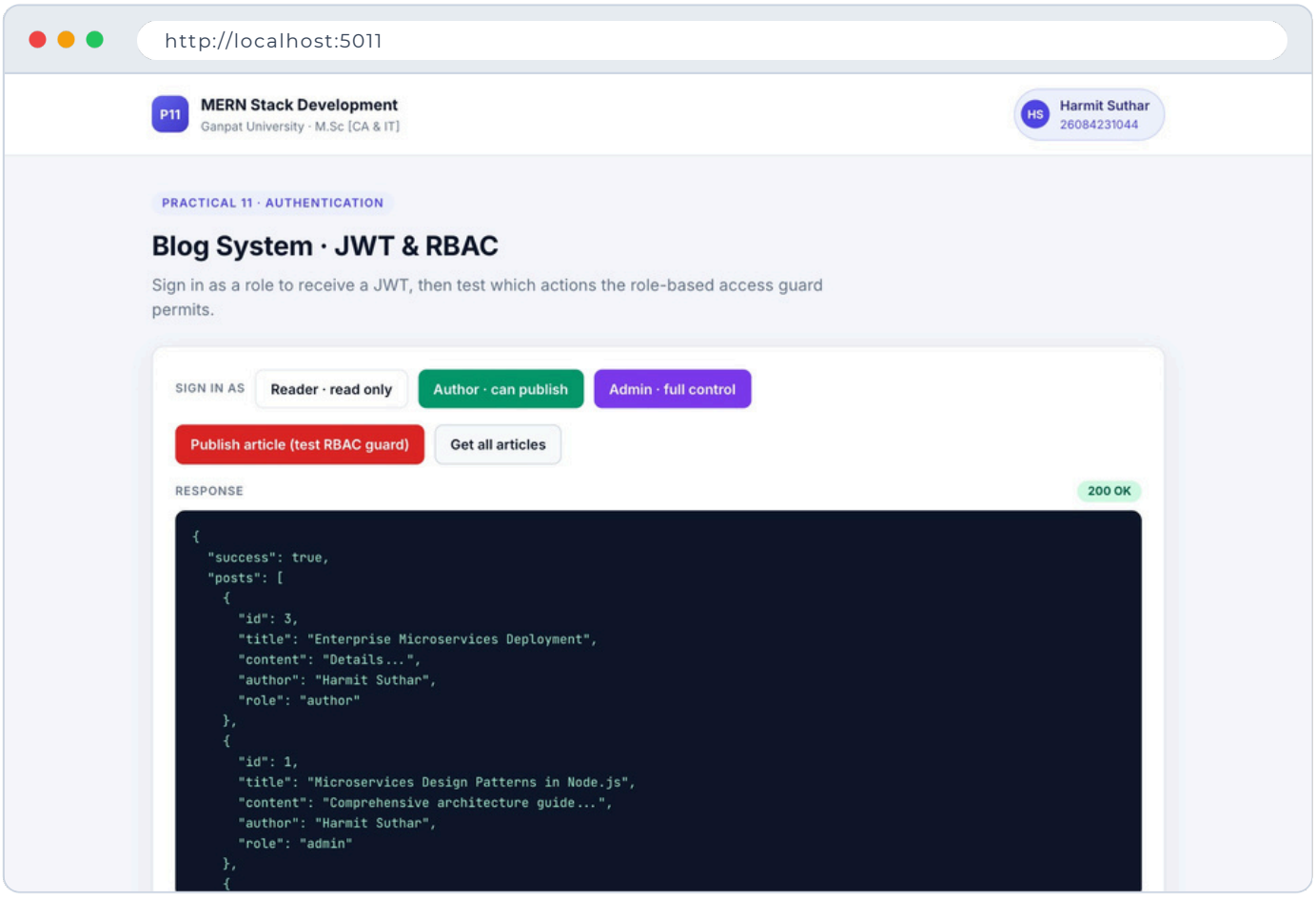


Figure 5.3 — GET /api/posts lists all articles including the newly published one.



🖥️ **Execution & Server Logs** **Port 5011**

```
$ npm run dev
> practical-11-blog-rbac@1.0.0 dev
> node --watch server.js

Practical 11 running on http://localhost:5011
```

PRACTICAL 12

Real-Time Systems / WebSockets & Socket.IO

Build a real-time chat application using Socket.IO and React

1. Practical Aim & Theoretical Definition

Aim & Theoretical Concept

The objective of this practical is to build a low-latency, bidirectional, event-driven Real-Time Chat Application using Socket.IO on a Node.js/Express backend and a React.js client. The system implements channel/room partitioning (socket.join), real-time message broadcasting, dynamic typing indicators, and connection lifecycle management.

Core Theoretical Concepts:

- **WebSockets vs HTTP Request-Response:** ***Traditional HTTP relies on client-initiated half-duplex polling. WebSockets establish a persistent, full-duplex TCP connection, allowing the server to push instantaneous data packets to connected clients with minimal overhead (2-byte framing).***
- **Socket.IO Architecture:** ***Built on top of Engine.IO, Socket.IO provides automated fallback to HTTP long-polling if WebSockets are blocked by proxies, automatic reconnection handling, and room abstraction.***
  - `io.on('connection', socket => ...)`***: Triggered when a new client connects.***
  - `socket.join(roomName)`***: Partitions clients into isolated communication channels.***
  - `socket.to(room).emit('receive_msg', payload)`***: Broadcasts messages to all clients in a room excluding the sender.***
  - `socket.on('typing', data)`***: Transmits debounced keystroke alerts to peers.***

Core Socket Events:

2. Step-by-Step Commands (Installation & Execution)

1. Navigate to the Practical Directory

```
$ cd practicals/p12
```

Action: *Switches to the Practical 12 project folder (contains server/ and client/).*

2. Install & Run Server + Client

```
$ npm run dev
```

Action: *Installs dependencies, starts the Socket.IO server (5012) and the React client (5179).*

3. Project Directory & File Structure

```
practical-12/
├── client/
│   ├── src/
│   │   ├── components/
│   │   │   └── AppBar.jsx
│   │   ├── App.jsx
│   │   ├── index.css
│   │   └── main.jsx
│   ├── index.html
│   ├── package-lock.json
│   ├── package.json
│   └── vite.config.js
├── server/
│   ├── package-lock.json
│   ├── package.json
│   └── server.js
├── package.json
├── README.md
└── run.bat
```

File / Directory Path	Architectural Purpose & Responsibility
client/src/components/AppBar.jsx	***Shared top bar showing the practical number and the student badge.***
client/src/App.jsx	***Root React component orchestrating state, layout and child components.***
client/src/index.css	***Global styles, design tokens and component styling.***
client/src/main.jsx	***Mounts the root React component into the DOM with ReactDOM.createRoot().***
client/index.html	***HTML entry point containing the #root mount node.***
client/package-lock.json	***Exact dependency tree lockfile generated by npm.***
client/package.json	***Project metadata, scripts and dependency declarations.***
client/vite.config.js	***Vite build and dev-server configuration.***
server/package-lock.json	***Exact dependency tree lockfile generated by npm.***
server/package.json	***Project metadata, scripts and dependency declarations.***
server/server.js	***Express server entry point: middleware setup, routes and listener.***
package.json	***Project metadata, scripts and dependency declarations.***
README.md	***Project documentation.***
run.bat	***Windows launcher script that installs and starts the project.***

4. Modular Source Code Implementation

Complete source code of the project, listed exactly as it exists in `practicals/p12/`

server/server.js

JAVASCRIPT

```
const express = require('express');
const http = require('http');
const { Server } = require('socket.io');
const cors = require('cors');

const app = express();
app.use(cors());
const server = http.createServer(app);
const io = new Server(server, { cors: { origin: "*" } });

io.on('connection', (socket) => {
  console.log("⚡ User Connected: " + socket.id);

  socket.on('join_room', (data) => {
    socket.join(data.room);
    console.log(`👤 ${data.username || socket.id} joined room: ${data.room}`);
    socket.to(data.room).emit('system_message', {
      text: `${data.username || 'A peer'} joined the channel`,
      time: new Date().toLocaleTimeString([], { hour: '2-digit', minute: '2-digit' })
    });
  });

  socket.on('send_message', (data) => {
    console.log(`💬 [Room ${data.room}] ${data.author}: ${data.message}`);
    socket.to(data.room).emit('receive_message', data);
  });

  socket.on('typing', (data) => {
    socket.to(data.room).emit('user_typing', data);
  });

  socket.on('disconnect', () => {
    console.log("❌ User Disconnected: " + socket.id);
  });
});

const PORT = 5012;
server.listen(PORT, () => console.log("Practical 12 Socket Server running on http://localhost:" + PORT));
```

client/src/main.jsx

JAVASCRIPT (JSX)

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode><App /></React.StrictMode>
);
```

client/src/App.jsx

JAVASCRIPT (JSX)

```
import React, { useState, useEffect } from 'react';
import io from 'socket.io-client';
import AppBar from './components/AppBar';

const socket = io.connect('http://localhost:5012');

function App() {
  const [username, setUsername] = useState('');
  const [room, setRoom] = useState('');
  const [inChat, setInChat] = useState(false);
  const [currentMessage, setCurrentMessage] = useState('');
  const [messageList, setMessageList] = useState([]);
```

```
  const [typingUser, setTypingUser] = useState('');

  const joinRoom = () => {
    if (username.trim()&& room.trim()){
      socket.emit('join_room', {username,room });
      setInChat(true);
    }
  };

  const sendMessage = async () => {
    if (currentMessage.trim()) {
      const messageData = {
        room,
        author:username,
        message:currentMessage,
        time:  newDate().toLocaleTimeString([],{      hour:  '2-digit', minute: '2-digit' })
      };

      await socket.emit('send_message',messageData);
      setMessageList((list) => [...list, messageData]);
      setCurrentMessage('');
      socket.emit('typing',    {room,   username:''    });   //Clear typing
    }
  };

  const handleTyping =(e)=>{
    setCurrentMessage(e.target.value);
    socket.emit('typing',    {room,   username:e.target.value ? username : '' });
  };

  useEffect(()=> {
    socket.on('receive_message', (data) => {
      setMessageList((list) => [...list, data]);
    });

    socket.on('system_message', (data)=> {
      setMessageList((list)    =>  [...list,       {  isSystem:  true, ...data }]);
    });

    socket.on('user_typing', (data)=> {
      setTypingUser(data.username);
    });

    return () => {
      socket.off('receive_message');
      socket.off('system_message');
      socket.off('user_typing');
    };
  }, []);

  return (
    <>
    <AppBar practical="12" />
    <div className="chat-app-wrapper">
      {!inChat ? (
        <div className="join-chat-card">
          <span className="eyebrow">Practical12· WebSockets</span>
          <h2>Real-TimeChatPortal</h2>
          <p>Instantbidirectional communicationwithSocket.IO.</p>
          <input
            type="text"
            placeholder="Screenname"
            value={username}
            onChange={(e) => setUsername(e.target.value)}
          />
          <input
            type="text"
            placeholder="Room code (e.g.MSD_LAB)"
            value={room}
            onChange={(e) => setRoom(e.target.value)}
          />
          <button onClick={joinRoom} className="join-btn">Join Chat Room</button>
```

```
          </div>
        ) : (
          <div className="chat-window">
            <div className="chat-header">
              <div>
                <h3>Room: #{room}</h3>
                <span className="live-status">● Live · WebSocket connected</span>
              </div>
              <span className="logged-user">{username}</span>
            </div>

            <div className="message-container">
              {messageList.map((msg, idx) => (
                msg.isSystem ? (
                  <div key={idx} className="system-bubble">{msg.text} ({msg.time})</div>
                ) : (
                  <div key={idx} className={`message-bubble ${msg.author === username ? 'self' : 'peer'}`}>
                    <div className="msg-author">{msg.author}</div>
                    <div className="msg-text">{msg.message}</div>
                    <div className="msg-time">{msg.time}</div>
                  </div>
              )
              ))}
              {typingUser && <div className="typing-indicator">{typingUser} is typing…</div>}
            </div>

            <div className="chat-input-bar">
              <input
                type="text"
                placeholder="Type your message..."
                value={currentMessage}
                onChange={handleTyping}
                onKeyPress={(e) => e.key === 'Enter' && sendMessage()}
              />
              <button onClick={sendMessage} className="send-btn">Send</button>
            </div>
          </div>
      )}
      </div>
      </>
  );


export default App;
```

client/src/components/AppBar.jsx JAVASCRIPT (JSX)

```
import React from 'react';

const AppBar = ({ practical, title = 'MERN Stack Development' }) => (
  <header className="topbar">
    <div className="topbar-inner">
      <div className="brand">
        <div className="brand-mark">P{practical}</div>
        <div>
          <div className="brand-title">{title}</div>
          <div className="brand-sub">Ganpat University · M.Sc [CA &amp;amp; IT]</div>
        </div>
      </div>
      <div className="student-chip">
        <div className="avatar">HS</div>
        <div>
          <div className="chip-name">Harmit Suthar</div>
          <div className="chip-id">26084231044</div>
        </div>
      </div>
    </div>
); </header>
```

```
export default AppBar;
```

client/src/index.css CSS

```
/* Shared design system — Harmit Suthar (26084231044) · MERN Stack Development */
:root {
  --bg: #f4f6fb; --surface: #ffffff; --border: #e3e8f0; --text: #0f172a; --muted: #64748b; --
  primary: #4f46e5; --primary-hover: #4338ca; --primary-soft: #eef2ff;
  --success: #059669; --success-soft: #d1fae5;
  --warning: #d97706; --warning-soft: #fef3c7;
  --danger: #dc2626; --danger-soft: #fee2e2;
  --info: #2563eb; --info-soft: #dbeafe;
  --radius: 12px;
  --shadow: 0 1px 2px rgba(15,23,42,.05), 0 8px 24px rgba(15,23,42,.06);
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', sans-serif; background: var(--bg); color: var(--text);
line-height: 1.5; -webkit-font-smoothing: antialiased; }
button, input, select { font: inherit; }
.topbar { background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z- index: 10; }
.topbar-inner { max-width: 1120px; margin: 0 auto; padding: 14px 24px; display: flex; align-items: center; justify-
content: space-between; gap: 16px; flex-wrap: wrap; }
.brand { display: flex; align-items: center; gap: 12px; }
.brand-mark { width: 36px; height: 36px; border-radius: 10px; background: linear-gradient(135deg, #6366f1, #4338ca);
color: #fff; display: grid; place-items: center; font-weight: 700; font-size: .8rem; }
.brand-title { font-weight: 700; font-size: .95rem; }
.brand-sub { font-size: .75rem; color: var(--muted); }
.student-chip { display: flex; align-items: center; gap: 10px; background: var(--primary-soft); border: 1px solid
#c7d2fe; padding: 6px 14px 6px 6px; border-radius: 999px; }
.avatar { width: 28px; height: 28px; border-radius: 50%; background: var(--primary); color: #fff; display: grid; place-
items: center; font-size: .7rem; font-weight: 700; }
.chip-name { font-size: .82rem; font-weight: 600; color: #312e81; line-height: 1.2; }
.chip-id { font-size: .72rem; color: #4f46e5; line-height: 1.2; }
.page { max-width: 1120px; margin: 0 auto; padding: 32px 24px 56px; }
.page-head { margin-bottom: 24px; }
.eyebrow { display: inline-block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter- spacing:
.08em; color: var(--primary); background: var(--primary-soft); padding: 3px 10px; border-radius: 999px; margin-
bottom: 10px; }
.page-head h1 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.page-head p { color: var(--muted); margin-top: 6px; max-width: 70ch; }
.panel { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box- shadow: var(--
shadow); padding: 22px; }
.btn { border: 1px solid transparent; border-radius: 8px; padding: 8px 14px; font-size: .85rem; font-weight: 600;
cursor: pointer; background: var(--primary); color: #fff; transition: background .15s, border-color .15s; }
.btn:hover { background: var(--primary-hover); }
.btn:focus-visible, input:focus-visible, select:focus-visible, button:focus-visible { outline: 3px solid #c7d2fe; outline-
offset: 1px; }
.btn.secondary { background: var(--surface); color: var(--text); border-color: var(--border); } .btn.secondary:hover {
background: #f8fafc; border-color: #cbd5e1; }
.btn:disabled { opacity: .5; cursor: not-allowed; }
input[type=text], input[type=email], input[type=tel], input[type=number], select { font-size: .85rem; padding: 9px
12px; border: 1px solid var(--border); border-radius: 8px; background: #fff; color: var(-- text); }
input:focus, select:focus { border-color: var(--primary); }
table { width: 100%; border-collapse: collapse; font-size: .85rem; }
th { text-align: left; font-size: .72rem; text-transform: uppercase; letter-spacing: .06em; color: var(-- muted);
background: #f8fafc; padding: 10px 12px; border-bottom: 1px solid var(--border); }
td { padding: 12px; border-bottom: 1px solid var(--border); }
tr:last-child td { border-bottom: none; }
.footer { text-align: center; color: var(--muted); font-size: .75rem; margin-top: 32px; }




body { min-height: 100vh; }
.chat-app-wrapper { width: 100%; max-width: 680px; margin: 0 auto; padding: 32px 24px 56px; }
.join-chat-card { background: var(--surface); border: 1px solid var(--border); box-shadow: var(--shadow);
padding: 32px; border-radius: 16px; }
.join-chat-card h2 { font-size: 1.35rem; margin-bottom: 4px; }
.join-chat-card p { color: var(--muted); margin-bottom: 22px; font-size: .9rem; }
```

```
.join-chat-card input { width: 100%; margin-bottom: 12px; padding: 11px 14px; }
.join-btn { width: 100%; background: var(--primary); color: #fff; border: none; padding: 11px; border-
radius: 8px; font-weight: 600; cursor: pointer; }
.join-btn:hover { background: var(--primary-hover); }
.chat-window { background: var(--surface); border: 1px solid var(--border); box-shadow: var(--shadow);
border-radius: 16px; overflow: hidden; height: 560px; display: flex; flex-direction: column; }
.chat-header { background: #f8fafc; padding: 14px 20px; display: flex; justify-content: space-between;
align-items: center; border-bottom: 1px solid var(--border); }
.chat-header h3 { font-size: 1rem; }
.live-status { color: var(--success); font-size: .75rem; font-weight: 500; }
.logged-user { font-size: .8rem; font-weight: 600; background: var(--primary-soft); color: #3730a3; padding:
4px 12px; border-radius: 999px; }
.message-container { flex: 1; padding: 20px; overflow-y: auto; display: flex; flex-direction: column; gap:
10px; background: #fbfcfe; }
.message-bubble { padding: 10px 14px; border-radius: 14px; max-width: 72%; font-size: .9rem; }
.message-bubble.self { align-self: flex-end; background: var(--primary); color: #fff; border-bottom-right-
radius: 4px; }
.message-bubble.peer { align-self: flex-start; background: #fff; border: 1px solid var(--border); border-
bottom-left-radius: 4px; }
.msg-author { font-size: .7rem; font-weight: 600; opacity: .75; margin-bottom: 2px; }
.msg-time { font-size: .65rem; opacity: .65; text-align: right; margin-top: 4px; }
.system-bubble { align-self: center; background: #e2e8f0; color: var(--muted); font-size: .75rem; padding:
4px 12px; border-radius: 999px; }
.typing-indicator { font-size: .8rem; color: var(--muted); font-style: italic; }
.chat-input-bar { display: flex; padding: 12px; background: var(--surface); border-top: 1px solid var(--
border); gap: 10px; }
.chat-input-bar input { flex: 1; }
.send-btn { background: var(--primary); color: #fff; border: none; padding: 9px 20px; border-radius: 8px;
font-weight: 600; cursor: pointer; }
.send-btn:hover { background: var(--primary-hover); }
```

5. Expected Output & Verification Results

All screenshots and logs below were captured from the real running application of this practical (Harmit Suthar · 26084231044).

Figure 5.1 — Join screen collecting screen name and room code.

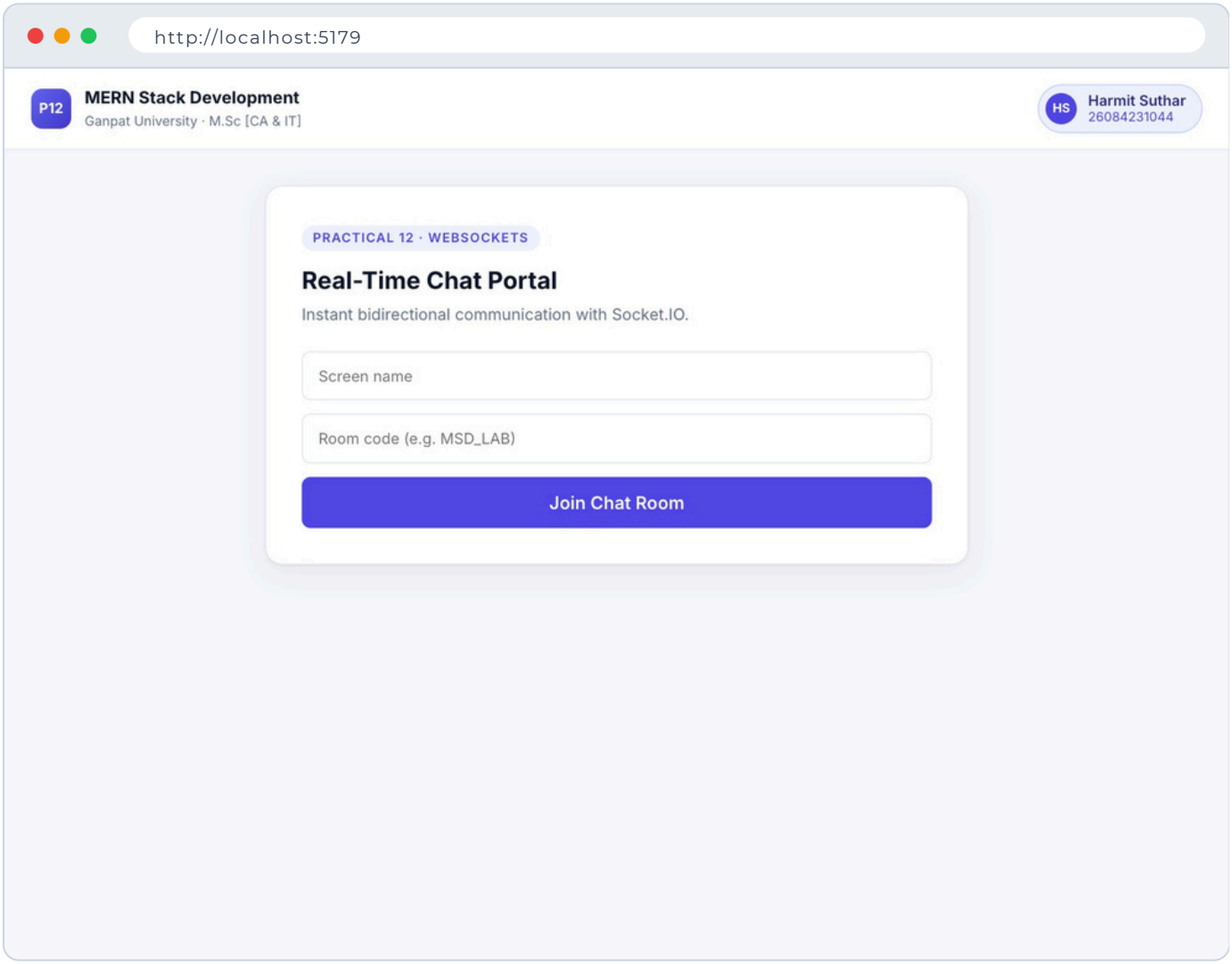


Figure 5.2 — Chat window of user "Harmit": messages are delivered in real time through Socket.IO.

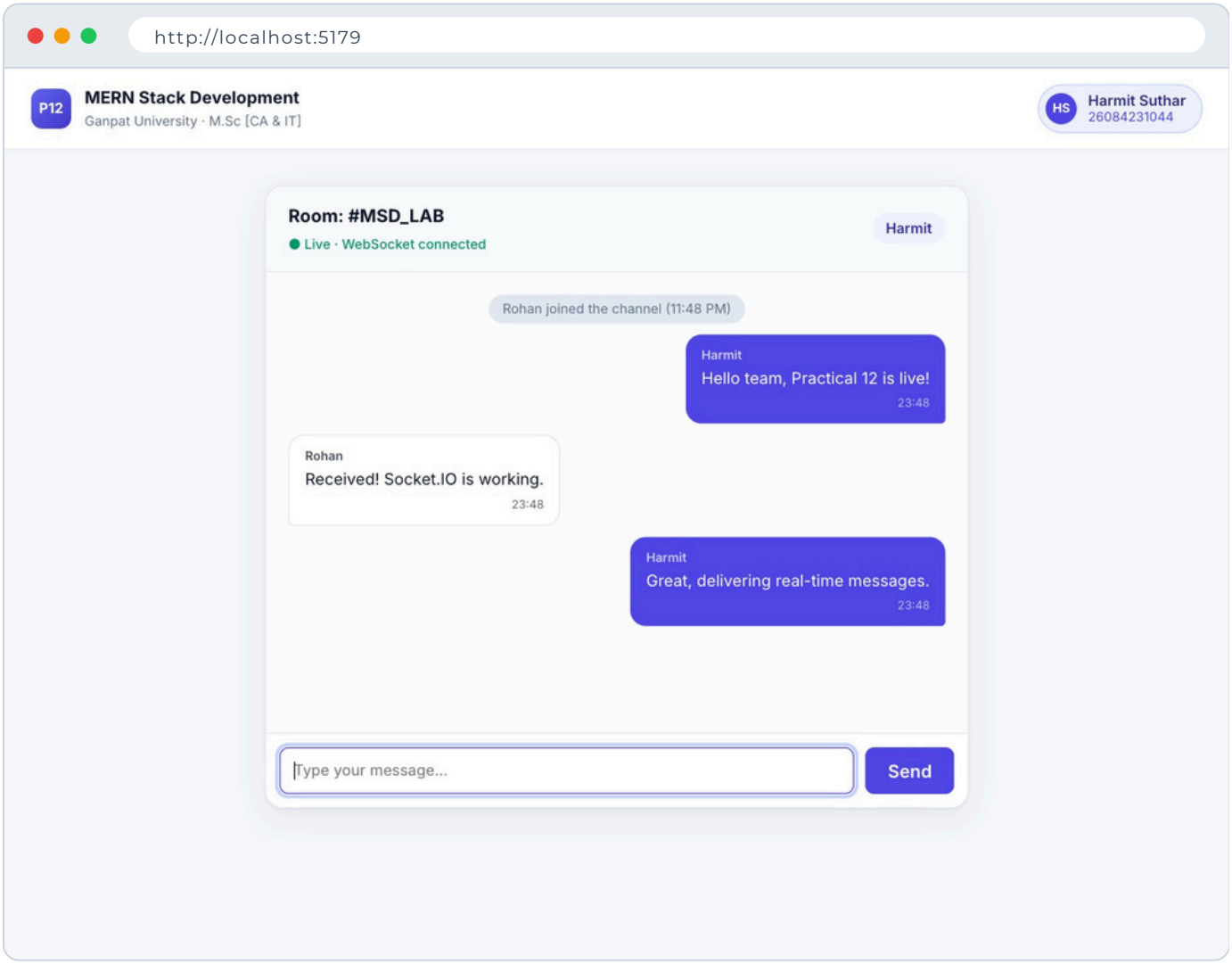
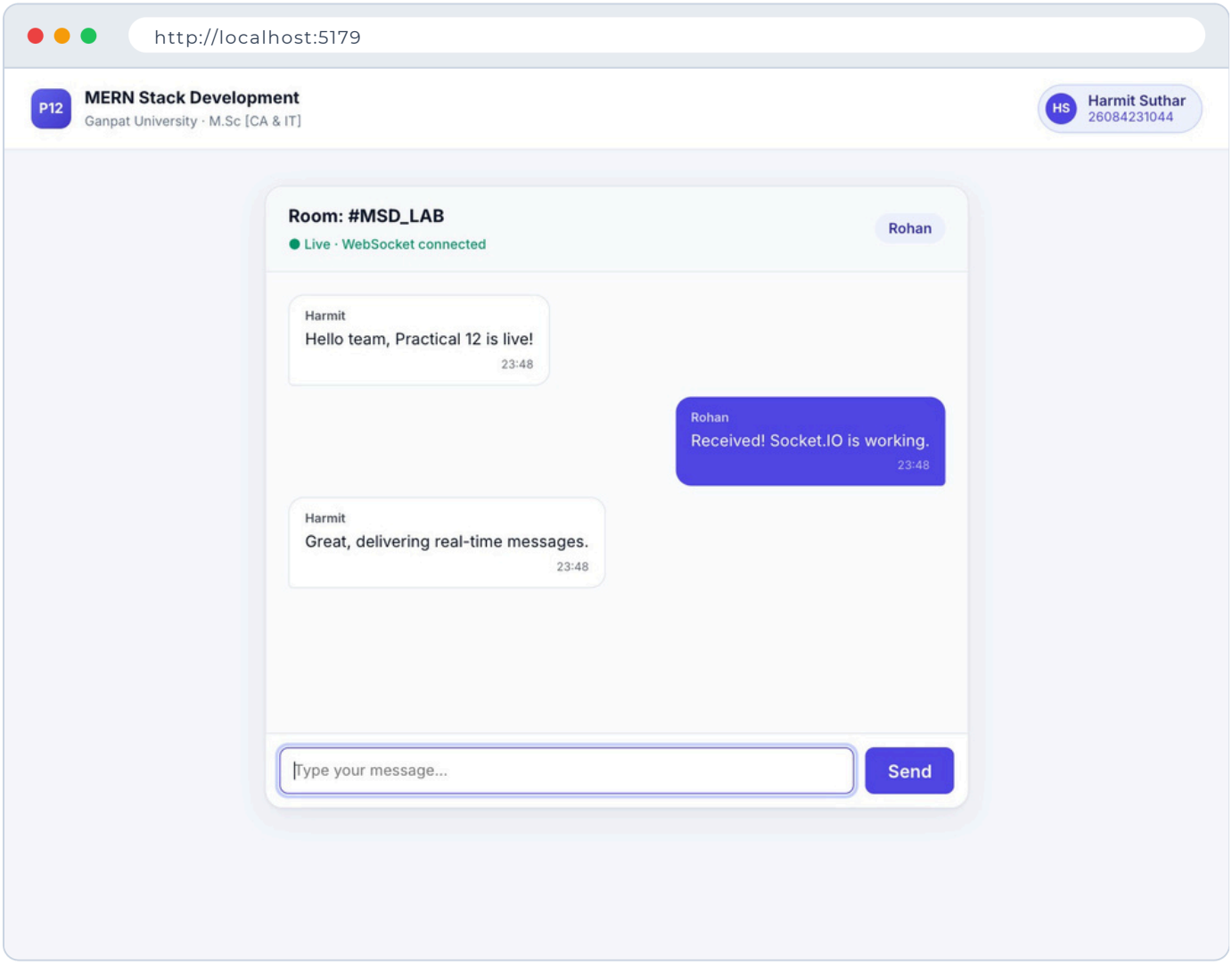


Figure 5.3 — Chat window of user "Rohan" in a second browser session, showing the same conversation.



Socket Server Logs — **Port 5012**

```
$ npm start (server)
> practical-12-server@1.0.0 start
> node server.js

Practical 12 Socket Server running on http://localhost:5012 ⚡
User Connected: G-t-RI44AUHQ7OiNAAAA
⚡ User Connected: EGxbJT7b1bN-4mgUAAAB
👤 Harmit joined room: MSD_LAB
👤 Rohan joined room: MSD_LAB
💬 [Room MSD_LAB] Harmit: Hello team, Practical 12 is live!
💬[Room MSD_LAB] Rohan: Received! Socket.IO is working.
💬 [Room MSD_LAB] Harmit: Great, delivering real-time messages.
```

React Client Logs **Port 5179**

```
$ npm run dev (client)
> practical-12-client@1.0.0 dev
> vite


  VITEv5.4.21 ready in 150 ms

  ➜  Local:   http://localhost:5179/
  ➜  Network: use --host to expose
```

PRACTICAL 13

Capstone Mini Project / Enterprise Full-Stack MERN

Develop an Enterprise Document Management System (DocuVault DMS) Capstone Mini Project

1. Practical Aim & Theoretical Definition

Aim & Theoretical Concept

The objective of this industry-grade Capstone Mini Project is to design, architect, and develop a production-ready Enterprise Document Management System (DocuVault DMS) leveraging the complete MERN stack.

Core Functional Modules & Enterprise Architecture:

- **Role-Based Access Control (RBAC):** ***Three user tiers: SuperAdmin (full system access, audit viewer), DepartmentManager (upload, approve, categorize), and Staff (view and download department docs).***
- **Secure Multi-Format File Management (Multer):** ***High-throughput uploads for PDF, DOCX, XLSX, and PNG files with unique hashing, file size limits (25MB), and sanitization.***
- **Document Metadata & Version Control:** ***Stores document title, department category (Finance, Legal, HR, Engineering), access level, active version number, and file storage path in MongoDB.***
- **Full-Text Search & Multi-Faceted Filtering:** ***Instant search by keywords, category dropdown, date range, and tags.***
- **Tamper-Evident Audit Logging:** ***Every interaction (UPLOAD, DOWNLOAD, PREVIEW, DELETE) automatically generates an immutable audit record capturing user identity, IP address, and timestamp.***
- **Modern Responsive React Dashboard:** ***Glassmorphism dashboard featuring KPI statistics cards, upload modal, document table with status badges, and interactive audit log viewer.***

2. Step-by-Step Commands (Installation & Execution)

1. Navigate to the Practical Directory

```
$ cd practicals/p13
```

Action: *Switches to the DocuVault DMS project folder (contains server/ and client/).*

2. Install & Run Server + Client

```
$ npm run dev
```

Action: *Installs dependencies, starts the Express API (5013) and the React dashboard (5173).*

3. Project Directory & File Structure

```
practical-13/
├── client/
│   ├── src/
│   │   ├── App.jsx
│   │   ├── index.css
│   │   └── main.jsx
│   ├── index.html
│   ├── package-lock.json
│   ├── package.json
│   └── vite.config.js
├── server/
│   ├── controllers/
│   │   └── docController.js
│   ├── models/
│   │   ├── AuditLog.js
│   │   └── Document.js
│   ├── package-lock.json
│   ├── package.json
│   └── server.js
├── package.json
├── README.md
├── run_backend.bat
├── run_frontend.bat
├── run_fullstack.bat
└── run.bat
```

File / Directory Path	Architectural Purpose & Responsibility
client/src/App.jsx	***Root React component orchestrating state, layout and child components.***
client/src/index.css	***Global styles, design tokens and component styling.***
client/src/main.jsx	***Mounts the root React component into the DOM with ReactDOM.createRoot().***
client/index.html	***HTML entry point containing the #root mount node.***
client/package-lock.json	***Exact dependency tree lockfile generated by npm.***
client/package.json	***Project metadata, scripts and dependency declarations.***
client/vite.config.js	***Vite build and dev-server configuration.***
server/controllers/docController.js	***Upload, search, download and delete handlers that also write audit logs.***
server/models/AuditLog.js	***Schema for the tamper-evident audit trail records.***
server/models/Document.js	***Schema for document metadata, category, access level and version.***
server/package-lock.json	***Exact dependency tree lockfile generated by npm.***
server/package.json	***Project metadata, scripts and dependency declarations.***
server/server.js	***Express server entry point: middleware setup, routes and listener.***
package.json	***Project metadata, scripts and dependency declarations.***
README.md	***Project documentation.***
run_backend.bat	***Windows launcher script that installs and starts the project.***

run_frontend.bat	***Windows launcher script that installs and starts the project.***
run_fullstack.bat	***Windows launcher script that installs and starts the project.***
run.bat	***Windows launcher script that installs and starts the project.***

4. Modular Source Code Implementation

Complete source code of the project, listed exactly as it exists in `practicals/p13 :`

server/server.js JAVASCRIPT

```
const express = require('express');
const cors = require('cors');
const path = require('path');
const fs = require('fs');

const app = express();
const PORT = process.env.PORT || 5013;

app.use(cors({ origin: '*' }));
app.use(express.json());

// In-memory initial documents
let documents = [
  {
    _id: "doc_101",
    title: "FY2026-Financial-Audit-Report.pdf",
    category: "Finance",
    accessLevel: "Restricted",
    size: "4.8 MB",
    uploadedBy: "Chief Financial Officer",
    updatedAt: "2026-10-23",
    version: 2
  },
  {
    _id: "doc_102",
    title: "Enterprise-Architecture-Guidelines.docx",
    category: "Engineering",
    accessLevel: "Internal",
    size: "1.2 MB",
    uploadedBy: "Lead Cloud Architect",
    updatedAt: "2026-10-22",
    version: 1
  },
  {
    _id: "doc_103",
    title: "Employee-NDA-Agreement-Template.pdf",
    category: "Legal & Compliance",
    accessLevel: "Internal",
    size: "820 KB",
    uploadedBy: "Legal Counsel",
    updatedAt: "2026-10-20",
    version: 3
];
  }


let auditLogs = [
  {
    action: "UPLOAD",
    documentTitle: "FY2026-Financial-Audit-Report.pdf",
    performedBy: "Chief Financial Officer",
    role: "SuperAdmin",
    time: "2026-10-23 11:20:00",
    ip: "192.168.1.10"
  },
```

```
    {
      action: "DOWNLOAD",
      documentTitle: "Enterprise-Architecture-Guidelines.docx",
      performedBy: "Lead Cloud Architect",
      role: "DepartmentManager",
      time: "2026-10-23 11:28:15",
      ip: "192.168.1.42"
    },
    {
      action: "VIEW",
      documentTitle: "Employee-NDA-Agreement-Template.pdf",
      performedBy: "HR Manager",
      role: "Staff",
      time: "2026-10-23 11:35:40",
      ip: "192.168.1.15"
  ]; }


  // GET: Retrieve documents with filter & search
  app.get('/api/documents', (req, res) => {
    const { category, search } = req.query;
    let results = [...documents];

    if (category && category !== 'All') {
      results = results.filter(d => d.category === category);
    }

    if (search) {
      const q = search.toLowerCase();
      results = results.filter(d => d.title.toLowerCase().includes(q));
    }

    res.json({ success: true, count: results.length, data: results });
  });

  // POST: Add / Upload Document
  app.post('/api/documents', (req, res) => {
    const { title, category, accessLevel, uploadedBy } = req.body;
    if (!title) return res.status(400).json({ success: false, message: "Title is required" });

    const newDoc = {
      _id: "doc_" + Date.now(),
      title,
      category: category || "Engineering",
      accessLevel: accessLevel || "Internal",
      size: "1.5 MB",
      uploadedBy: uploadedBy || "Harmit Suthar",
      updatedAt: new Date().toISOString().split('T')[0],
      version: 1
    };

    documents.unshift(newDoc);

    // Add audit log
    auditLogs.unshift({
    action: "UPLOAD",
    documentTitle: newDoc.title,
    performedBy: newDoc.uploadedBy,
    role: "SuperAdmin",
    time: new Date().toISOString().replace('T', ' ').substring(0, 19),
    ip: req.ip || "127.0.0.1"
    });

    res.status(201).json({ success: true, document: newDoc });
  });

  // DELETE: Remove document
  app.delete('/api/documents/:id', (req, res) => {
    const doc = documents.find(d => d._id === req.params.id);
    if (!doc) return res.status(404).json({ success: false, message: "Document not found" });

    documents = documents.filter(d => d._id !== req.params.id);
```

```
  auditLogs.unshift({
    action: "DELETE",
    documentTitle: doc.title,
    performedBy: "SuperAdmin",
    role: "SuperAdmin",
    time: new Date().toISOString().replace('T', ' ').substring(0, 19),
    ip: req.ip || "127.0.0.1"
  });

  res.json({ success: true, message: "Document deleted", deleted: doc });
});

// GET: Audit Logs
app.get('/api/audit-logs', (req, res) => {
  res.json({ success: true, count: auditLogs.length, data: auditLogs });
});

app.listen(PORT, () => {
  console.log("================================================");
  console.log(`🛡️ DocuVault DMS Backend running on http://localhost:${PORT}`);
  console.log(`📂 API Endpoints:`);
  console.log(`     - GET/POST /api/documents`);
  console.log(`     - DELETE    /api/documents/:id`);
  console.log(`     - GET       /api/audit-logs`);
  console.log("================================================");
});
```

client/src/main.jsx — JAVASCRIPT (JSX)

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

client/src/App.jsx — JAVASCRIPT (JSX)

```
import React, { useState, useEffect } from 'react';

const API_BASE = 'http://localhost:5013/api';

export default function App() {
  const [documents, setDocuments] = useState([
    {
      _id: "doc_101",
      title: "FY2026-Financial-Audit-Report.pdf",
      category: "Finance",
      accessLevel: "Restricted",
      size: "4.8 MB",
      uploadedBy: "Chief Financial Officer",
      updatedAt: "2026-10-23",
      version: 2
    },
    {
      _id: "doc_102",
      title:       "Enterprise-Architecture-Guidelines.docx",
      category: "Engineering",
      accessLevel: "Internal",
      size: "1.2 MB",
      uploadedBy: "Lead Cloud Architect",
      updatedAt: "2026-10-22",
```

```
        version: 1
      },
      {
        _id: "doc_103",
        title: "Employee-NDA-Agreement-Template.pdf",
        category: "Legal & Compliance",
        accessLevel: "Internal",
        size: "820 KB",
        uploadedBy: "Legal Counsel",
        updatedAt: "2026-10-20",
        version: 3
      }
    ]);

  const [auditLogs, setAuditLogs] = useState([
    { action: "UPLOAD", documentTitle: "FY2026-Financial-Audit-Report.pdf", performedBy: "Chief Financial Officer", role: "SuperAdmin", time: "2026-10-23 11:20:00", ip: "192.168.1.10" },
    { action: "DOWNLOAD", documentTitle: "Enterprise-Architecture-Guidelines.docx", performedBy: "Lead Cloud Architect", role: "DepartmentManager", time: "2026-10-23 11:28:15", ip: "192.168.1.42" },
    { action: "VIEW", documentTitle: "Employee-NDA-Agreement-Template.pdf", performedBy: "HR Manager", role: "Staff", time: "2026-10-23 11:35:40", ip: "192.168.1.15" }
  ]);

  const [backendOnline, setBackendOnline] = useState(false); const
  [loading, setLoading] = useState(false);
  const [search, setSearch] = useState('');
  const [filterCat, setFilterCat] = useState('All');
  const [showAudit, setShowAudit] = useState(false);
  const [showUploadModal, setShowUploadModal] =
  useState(false);

  // New Document Form
  const [newTitle, setNewTitle] = useState('');
  const [newCat, setNewCat] = useState('Engineering');
  const [newAccess, setNewAccess] = useState('Internal');

  // Load documents from backend
  const fetchDocuments = async () => {
    try {
      const res = await fetch(`${API_BASE}/documents`);
      if (res.ok) {
        const json = await res.json();
        if (json.data && json.data.length > 0) {
          setDocuments(json.data);
        }
        setBackendOnline(true);
      }
    } catch (e) {
      setBackendOnline(false);
    }
  };

  // Load audit logs from backend
  const fetchAuditLogs = async () => {
    try {
      const res = await fetch(`${API_BASE}/audit-logs`);
      if (res.ok) {
        const json = await res.json();
        if (json.data) setAuditLogs(json.data);
      }
    } catch (e) {
      // Keep sample fallback
    }
  };

  useEffect(() => {
    fetchDocuments();
    fetchAuditLogs();
    const interval = setInterval(fetchDocuments, 8000);
    return () => clearInterval(interval);
  }, []);

  // Handle Document Upload / Creation
```

```
  const handleUploadSubmit = async (e) => {
    e.preventDefault();
    if (!newTitle.trim()) return;

    setLoading(true);
    const newDoc = {
      title: newTitle.trim(),
      category: newCat,
      accessLevel: newAccess,
      uploadedBy: "Harmit Suthar"
    };

    try {
      const res = await fetch(`${API_BASE}/documents`, {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify(newDoc)
      });
      if (res.ok) {
        const json = await res.json();
        setDocuments(prev => [json.document, ...prev]);
        fetchAuditLogs();
      } else {
        throw new Error('Backend failed');
      }
    } catch (err) {
      // Local fallback
      const localDoc = {
        _id: "doc_" + Date.now(),
        title: newTitle.trim(),
        category: newCat,
        accessLevel: newAccess,
        size: "1.5 MB",
        uploadedBy: "Harmit Suthar",
        updatedAt: new Date().toISOString().split('T')[0],
        version: 1
      };
      setDocuments(prev => [localDoc, ...prev]);
    } finally {
      setNewTitle('');
      setShowUploadModal(false);
      setLoading(false);
    }
  };

  // Handle Document Deletion
  const handleDelete = async (id) => {
    if (!window.confirm("Are you sure you want to delete this document from the DMS archive?")) return;

    try {
      await fetch(`${API_BASE}/documents/${id}`, { method: 'DELETE' });
      fetchAuditLogs();
    } catch (err) {
      // Handled locally
    }
    setDocuments(prev => prev.filter(d => d._id !== id));
  };

  const filtered = documents.filter(d =>
    (filterCat === 'All' || d.category === filterCat) &&
    d.title.toLowerCase().includes(search.toLowerCase())
  );

  return (
    <div className="master-app">
      <header className="topbar" style={{ position: 'static', border: '1px solid var(--border)',
borderRadius: 'var(--radius)', marginBottom: 28, boxShadow: 'var(--shadow)' }}>
        <div className="topbar-inner" style={{ padding: '16px 22px' }}>
          <div className="brand">
            <div className="brand-mark">DV</div>
            <div>
              <div className="brand-title">Ganpat University · Department of Computer Science</div>
```

```
              <div className="brand-sub">M.Sc [CA &amp;amp; IT] – 1 · MERN Stack Development (P21A1MSD)</div>
            </div>
          </div>
          <div className="student-chip">
            <div className="avatar">HS</div>
            <div>
              <div className="chip-name">Harmit Suthar</div>
              <div className="chip-id">Enrollment No. 26084231044</div>
            </div>
          </div>
        </div>
      </header>

      <div className="section-intro">
        <div>
          <span className="eyebrow">Practical 13 · Capstone Mini Project</span>
          <h2>DocuVault Enterprise DMS</h2>
          <p>Full-stack MERN document management with role-based access control, Multer file streaming and
tamper-evident audit logs.</p>
        </div>
        <span className={`badge-pill ${backendOnline ? 'success' : 'warning'}`}>
          {backendOnline ? '● Backend online · port 5013' : '● Local mode · backend offline'}
        </span>
      </div>

      <div className="kpi-grid">
        <div className="card-box kpi"><span className="kpi-label">Total documents</span><span
className="kpi-value">{documents.length}</span><span className="kpi-note">Across all departments</span>
</div>
        <div className="card-box kpi"><span className="kpi-label">Storage consumed</span><span
className="kpi-value">14.8 GB</span><span className="kpi-note">Of 50 GB quota</span></div>
        <div className="card-box kpi"><span className="kpi-label">Security compliance</span><span
className="kpi-value">99.98%</span><span className="kpi-note">Audit-verified</span></div>
        <div className="card-box kpi"><span className="kpi-label">Departments</span><span className="kpi-
value">5</span><span className="kpi-note">RBAC partitioned</span></div>
      </div>

      {/* Action Bar */}
      <div className="card-box action-bar">
        <div className="grow">
          <input
            type="text"
            placeholder="Search documents by keyword or extension…"
            value={search}
            onChange={e => setSearch(e.target.value)}
            className="input-field"
        />
        </div>
        <div style={{ width: '200px' }}>
          <select value={filterCat} onChange={e => setFilterCat(e.target.value)} className="input-field">
            <option value="All">All Categories</option>
            <option value="Engineering">Engineering</option>
            <option value="Finance">Finance</option>
            <option value="Legal & Compliance">Legal & Compliance</option>
          </select>
        </div>
        <button className="btn-primary" onClick={() => setShowUploadModal(true)}>
          Upload document
        </button>
        <button className="btn-secondary" onClick={() => { fetchAuditLogs(); setShowAudit(true); }}>
          View audit trail ({auditLogs.length})
        </button>
      </div>

      {/* Documents Table */}
      <div className="card-box" style={{ overflowX: 'auto', padding: 0 }}>
        <table className="custom-table">
          <thead>
            <tr>
              <th>Document Title</th>
              <th>Category</th>
              <th>Access Level</th>
```

```
                <th>File          Size</th>
                <th>Uploaded   By</th>
                <th>Last  Modified</th>
                <th>Version</th>
                <th>Actions</th>
              </tr>
            </thead>
            <tbody>
              {filtered.length === 0 ? (
                <tr>
                  <td colSpan={8} className="empty-row">
                    No documents found{search ? <> matching &ldquo;{search}&rdquo;</> : null}.
                  </td>
                </tr>
              ) : (
                filtered.map(doc => (
                  <tr key={doc._id}>
                    <td>
                      <strong>{doc.title}</strong>
                    </td>
                    <td>
                      <span className="badge-pill info">{doc.category}</span>
                    </td>
                    <td>
'success'}`}>
                      <span className={`badge-pill ${doc.accessLevel === 'Restricted' ? 'danger'    :

                        {doc.accessLevel || 'Internal'}
                      </span>
                    </td>
                    <td>{doc.size || '1.5 MB'}</td>
                    <td className="muted-cell">{doc.uploadedBy || 'Harmit Suthar'}</td>
                    <td className="muted-cell">{doc.updatedAt || '2026-10-23'}</td>
                    <td>v{doc.version || 1}.0</td>
                    <td>
                      <div className="row-actions">
                        <button className="btn-icon" title="Download document" onClick={() =>
alert(`Downloading ${doc.title} via secure encrypted stream…`)}>Download</button>
                        <button className="btn-icon" title="View document details" onClick={() =>
alert(`DocuVault metadata\n\nTitle: ${doc.title}\nCategory: ${doc.category}\nAccess:
${doc.accessLevel}\nUploader: ${doc.uploadedBy}`)}>View</button>
                        <button className="btn-icon danger" title="Delete document" onClick={() =>
handleDelete(doc._id)}>Delete</button>
                      </div>
                    </td>
                  </tr>
                ))
              )}
            </tbody>
          </table>
        </div>

        {/* Upload Modal */}
        {showUploadModal && (
          <div className="modal-overlay" onClick={() => setShowUploadModal(false)}>
            <div className="modal-box" onClick={e => e.stopPropagation()}>
              <div className="modal-head">
                <h3>Upload document to DocuVault</h3>
                <button className="modal-close" onClick={() => setShowUploadModal(false)} aria-
label="Close">×</button>
              </div>
              <form onSubmit={handleUploadSubmit}>
                <div className="field">
                  <label className="field-label">Document Title / File Name:</label>
                  <input
                    type="text"
                    required
                    placeholder="e.g. Q3-Cloud-Security-Review.pdf"
                    value={newTitle}
                    onChange={e => setNewTitle(e.target.value)}
                    className="input-field"
                  />
                </div>
```

```
                <div className="field">
                  <label className="field-label">Department / Category:</label>
                  <select value={newCat} onChange={e => setNewCat(e.target.value)} className="input-field">
                    <option value="Engineering">Engineering</option>
                    <option value="Finance">Finance</option>
                    <option value="Legal & Compliance">Legal & Compliance</option>
                  </select>
                </div>
                <div className="field" style={{ marginBottom: 22 }}>
                  <label className="field-label">Access Level (RBAC Permission):</label>
                  <select value={newAccess} onChange={e => setNewAccess(e.target.value)} className="input-
field">

                    <option value="Internal">Internal (All Authenticated Staff)</option>
                    <option value="Restricted">Restricted (C-Suite & Admins Only)</option>
                  </select>
                </div>
                <div className="modal-actions">
                  <button type="button" className="btn-secondary" onClick={() =>
setShowUploadModal(false)}>Cancel</button>
                  <button type="submit" className="btn-primary" disabled={loading}>
                    {loading ? 'Uploading…' : 'Save & register document'}
                  </button>
                </div>
              </form>
            </div>
          </div>
        )}

        {/* Audit Trail Modal */}
        {showAudit && (
          <div className="modal-overlay" onClick={() => setShowAudit(false)}>
            <div className="modal-box wide" onClick={e => e.stopPropagation()}>
              <div className="modal-head">
                <h3>Tamper-evident security audit log</h3>
                <button className="modal-close" onClick={() => setShowAudit(false)} aria-
label="Close">×</button>
              </div>
              <div className="audit-list">
                {auditLogs.map((log, idx) => (
                  <div key={idx} className="audit-row">
                    <div>
                      <span className={`badge-pill ${log.action === 'UPLOAD' ? 'success' : log.action ===
'DELETE' ? 'danger' : 'info'}`} style={{ marginRight: '8px' }}>
                        {log.action}
                      </span>
                      <strong>{log.documentTitle}</strong>
                      <div className="audit-meta">
                        By: {log.performedBy} ({log.role})  |  IP: {log.ip || '127.0.0.1'}
                      </div>
                    </div>
                    <span className="audit-time">{log.time}</span>
                  </div>
                ))}
              </div>
            </div>
          </div>
        )}
        <p className="footer">Harmit Suthar (26084231044) · Practical 13 · DocuVault DMS</p>
      </div>
} );
```

client/src/index.css CSS

```
/* Shared design system — Harmit Suthar (26084231044) · MERN Stack Development */
:root {
  --bg: #f4f6fb; --surface: #ffffff; --border: #e3e8f0; --text: #0f172a; --muted: #64748b; --
  primary: #4f46e5; --primary-hover: #4338ca; --primary-soft: #eef2ff;
  --success: #059669; --success-soft: #d1fae5;
  --warning: #d97706; --warning-soft: #fef3c7;
```

```
  --danger: #dc2626; --danger-soft: #fee2e2;
  --info: #2563eb; --info-soft: #dbeafe;
  --radius: 12px;
  --shadow: 0 1px 2px rgba(15,23,42,.05), 0 8px 24px rgba(15,23,42,.06);
}
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: 'Inter', system-ui, -apple-system, 'Segoe UI', sans-serif; background: var(--bg); color: var(--text);
line-height: 1.5; -webkit-font-smoothing: antialiased; }
button, input, select { font: inherit; }
.topbar { background: var(--surface); border-bottom: 1px solid var(--border); position: sticky; top: 0; z- index: 10; }
.topbar-inner { max-width: 1120px; margin: 0 auto; padding: 14px 24px; display: flex; align-items: center; justify-
content: space-between; gap: 16px; flex-wrap: wrap; }
.brand { display: flex; align-items: center; gap: 12px; }
.brand-mark { width: 36px; height: 36px; border-radius: 10px; background: linear-gradient(135deg, #6366f1, #4338ca);
color: #fff; display: grid; place-items: center; font-weight: 700; font-size: .8rem; }
.brand-title { font-weight: 700; font-size: .95rem; }
.brand-sub { font-size: .75rem; color: var(--muted); }
.student-chip { display: flex; align-items: center; gap: 10px; background: var(--primary-soft); border: 1px solid
#c7d2fe; padding: 6px 14px 6px 6px; border-radius: 999px; }
.avatar { width: 28px; height: 28px; border-radius: 50%; background: var(--primary); color: #fff; display: grid; place-
items: center; font-size: .7rem; font-weight: 700; }
.chip-name { font-size: .82rem; font-weight: 600; color: #312e81; line-height: 1.2; }
.chip-id { font-size: .72rem; color: #4f46e5; line-height: 1.2; }
.page { max-width: 1120px; margin: 0 auto; padding: 32px 24px 56px; }
.page-head { margin-bottom: 24px; }
.eyebrow { display: inline-block; font-size: .72rem; font-weight: 600; text-transform: uppercase; letter- spacing:
.08em; color: var(--primary); background: var(--primary-soft); padding: 3px 10px; border-radius: 999px; margin-
bottom: 10px; }
.page-head h1 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.page-head p { color: var(--muted); margin-top: 6px; max-width: 70ch; }
.panel { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box- shadow: var(--
shadow); padding: 22px; }
.btn { border: 1px solid transparent; border-radius: 8px; padding: 8px 14px; font-size: .85rem; font-weight: 600;
cursor: pointer; background: var(--primary); color: #fff; transition: background .15s, border-color .15s; }
.btn:hover { background: var(--primary-hover); }
.btn:focus-visible, input:focus-visible, select:focus-visible, button:focus-visible { outline: 3px solid #c7d2fe; outline-
offset: 1px; }
.btn.secondary { background: var(--surface); color: var(--text); border-color: var(--border); } .btn.secondary:hover {
background: #f8fafc; border-color: #cbd5e1; }
.btn:disabled { opacity: .5; cursor: not-allowed; }
input[type=text], input[type=email], input[type=tel], input[type=number], select { font-size: .85rem; padding: 9px
12px; border: 1px solid var(--border); border-radius: 8px; background: #fff; color: var(-- text); }
input:focus, select:focus { border-color: var(--primary); }
table { width: 100%; border-collapse: collapse; font-size: .85rem; }
th { text-align: left; font-size: .72rem; text-transform: uppercase; letter-spacing: .06em; color: var(-- muted);
background: #f8fafc; padding: 10px 12px; border-bottom: 1px solid var(--border); }
td { padding: 12px; border-bottom: 1px solid var(--border); }
tr:last-child td { border-bottom: none; }
.footer { text-align: center; color: var(--muted); font-size: .75rem; margin-top: 32px; }
```

```
.master-app { max-width: 1180px; margin: 0 auto; padding: 28px 24px 56px; }
.page-head { margin-bottom: 0; }
.section-intro { display: flex; justify-content: space-between; align-items: flex-end; gap: 16px; flex-wrap:
wrap; margin-bottom: 24px; }
.section-intro h2 { font-size: 1.65rem; font-weight: 700; letter-spacing: -.01em; }
.section-intro p { color: var(--muted); margin-top: 6px; max-width: 70ch; font-size: .95rem; }
.kpi-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(210px, 1fr)); gap: 16px; margin-
bottom: 20px; }
.kpi { display: flex; flex-direction: column; gap: 2px; }
.kpi-label { font-size: .72rem; text-transform: uppercase; letter-spacing: .06em; font-weight: 600; color:
var(--muted); }
.kpi-value { font-size: 1.8rem; font-weight: 700; letter-spacing: -.02em; }
.kpi-note { font-size: .78rem; color: var(--muted); }
.card-box { background: var(--surface); border: 1px solid var(--border); border-radius: var(--radius); box-
shadow: var(--shadow); padding: 22px; }
.action-bar { display: flex; gap: 12px; align-items: center; margin-bottom: 20px; flex-wrap: wrap; }
.action-bar .grow { flex: 1; min-width: 240px; }
.badge-pill { display: inline-block; padding: 2px 10px; border-radius: 999px; font-size: .72rem; font-
weight: 600; white-space: nowrap; }
```

```
.badge-pill.success { background: var(--success-soft); color: #065f46; }
.badge-pill.info { background: var(--info-soft); color: #1e40af; }
.badge-pill.warning { background: var(--warning-soft); color: #92400e; }
.badge-pill.danger { background: var(--danger-soft); color: #991b1b; }
.btn-primary { background: var(--primary); color: #fff; border: none; padding: 9px 16px; border-radius: 8px;
font-weight: 600; cursor: pointer; font-size: .85rem; }
.btn-primary:hover { background: var(--primary-hover); }
.btn-secondary { background: var(--surface); color: var(--text); border: 1px solid var(--border); padding:
9px 16px; border-radius: 8px; font-weight: 600; cursor: pointer; font-size: .85rem; }
.btn-secondary:hover { background: #f8fafc; border-color: #cbd5e1; }
.btn-icon { background: var(--surface); border: 1px solid var(--border); color: var(--text); padding: 4px
10px; border-radius: 8px; cursor: pointer; font-size: .78rem; font-weight: 600; }
.btn-icon:hover { background: #f8fafc; border-color: #cbd5e1; }
.btn-icon.danger { color: var(--danger); border-color: #fecaca; }
.btn-icon.danger:hover { background: var(--danger-soft); }
.field { margin-bottom: 14px; }
.field-label { display: block; font-size: .72rem; color: var(--muted); margin-bottom: 5px; font-weight: 600;
text-transform: uppercase; letter-spacing: .05em; }
.input-field { width: 100%; }
.custom-table td.strong { font-weight: 600; }
.custom-table .muted-cell { color: var(--muted); }
.row-actions { display: flex; gap: 6px; }
.empty-row { text-align: center; padding: 32px !important; color: var(--muted); }
.modal-overlay { position: fixed; inset: 0; background: rgba(15,23,42,.45); backdrop-filter: blur(3px);
display: flex; align-items: center; justify-content: center; z-index: 999; padding: 16px; }
.modal-box { background: var(--surface); border-radius: 16px; padding: 24px; width: 100%; max-width: 560px;
box-shadow: 0 24px 64px rgba(15,23,42,.25); }
.modal-box.wide { max-width: 720px; }
.modal-head { display: flex; justify-content: space-between; align-items: center; border-bottom: 1px solid
var(--border); padding-bottom: 14px; margin-bottom: 18px; }
.modal-head h3 { font-size: 1.05rem; }
.modal-close { background: none; border: none; color: var(--muted); font-size: 1.1rem; cursor: pointer;
padding: 4px 8px; border-radius: 6px; }
.modal-close:hover { background: #f1f5f9; }
.modal-actions { display: flex; gap: 10px; justify-content: flex-end; }
.audit-list { display: flex; flex-direction: column; gap: 10px; max-height: 360px; overflow-y: auto; }
.audit-row { display: flex; justify-content: space-between; align-items: center; gap: 12px; background:
#f8fafc; border: 1px solid var(--border); padding: 12px 14px; border-radius: 10px; font-size: .85rem; }
.audit-meta { font-size: .75rem; color: var(--muted); margin-top: 2px; }
.audit-time { font-size: .75rem; color: var(--muted); white-space: nowrap; }
```

server/controllers/docController.js JAVASCRIPT

```
const Document = require('../models/Document');
const AuditLog = require('../models/AuditLog');

// 1. Upload & Register Document
exports.uploadDocument = async (req, res) => {
  try {
    if (!req.file) {
      return res.status(400).json({ success: false, message: "No document attached." });
    }

    const { title, description, category, tags, accessLevel, uploadedBy, userRole } = req.body;

    const newDoc = await Document.create({
      title,
      description,
      category,
      tags: tags ? tags.split(',').map(t => t.trim()) : [],
      fileUrl: `/uploads/${req.file.filename}`,
      fileName: req.file.filename,
      originalName: req.file.originalname,
      mimeType: req.file.mimetype,
      sizeBytes: req.file.size,
      accessLevel: accessLevel || "Internal",
      uploadedBy: uploadedBy || "Admin"

    });

    // Create Audit Log Entry
```

```
    await AuditLog.create({
      action: "UPLOAD",
      documentId: newDoc._id,
      documentTitle: newDoc.title,
      performedBy: uploadedBy || "Admin",
      userRole: userRole || "SuperAdmin",
      ipAddress: req.ip,
      details: `Uploaded version 1 (${(req.file.size / 1024).toFixed(1)} KB)`
    });

    res.status(201).json({
      success: true,
      message: "Document uploaded and audit record registered successfully",
      document: newDoc
    });
  } catch (error) {
    res.status(500).json({ success: false, error: error.message });
  }
};

// 2. Search & Filter Documents
exports.getDocuments = async (req, res) => {
  try {
    const { category, search, access } = req.query;
    const filter = {};

    if (category && category !== 'All') filter.category = category;
    if (access && access !== 'All') filter.accessLevel = access;
    if (search) {
      filter.$or = [
        { title: { $regex: search, $options: 'i' } },
        { description: { $regex: search, $options: 'i' } },
        { tags: { $in: [new RegExp(search, 'i')] } }
      ];
    }

    const docs = await Document.find(filter).sort({ createdAt: -1 });
    res.json({ success: true, count: docs.length, documents: docs });
  } catch (error) {
    res.status(500).json({ success: false, error: error.message });
  }
};
```

server/models/AuditLog.js — JAVASCRIPT

```
const mongoose = require('mongoose');

const auditLogSchema = new mongoose.Schema(
  {
    action: {
      type: String,
      required: true,
      enum: ["UPLOAD", "DOWNLOAD", "VIEW", "UPDATE", "DELETE"]
    },
    documentId: { type: mongoose.Schema.Types.ObjectId, ref: 'Document' },
    documentTitle: { type: String, required: true },
    performedBy: { type: String, required: true },
    userRole: { type: String, required: true },
    ipAddress: { type: String, default: "127.0.0.1" },
    status: { type: String, enum: ["SUCCESS", "FAILED"], default: "SUCCESS" },
    details: { type: String }

  },
); { timestamps: true }


module.exports = mongoose.model('AuditLog', auditLogSchema);
```

server/models/Document.js JAVASCRIPT

```
const mongoose = require('mongoose');

const documentSchema = new mongoose.Schema(
  {
    title: { type: String, required: true, trim: true },
    description: { type: String, default: "" },
    category: {
      type: String,
      required: true,
      enum: ["Engineering", "Legal & Compliance", "Finance", "Human Resources", "Executive"]
    },
    tags: [String],
    fileUrl: { type: String, required: true },
    fileName: { type: String, required: true },
    originalName: { type: String, required: true },
    mimeType: { type: String, required: true },
    sizeBytes: { type: Number, required: true },
    version: { type: Number, default: 1 },
    accessLevel: {
      type: String,
      enum: ["Public", "Internal", "Restricted"],
      default: "Internal"
    },
    uploadedBy: {
      type: String,
      default: "System User"
    }
); },
  { timestamps: true }


module.exports = mongoose.model('Document', documentSchema);
```

5. Expected Output & Verification Results

All screenshots and logs below were captured from the real running application of this practical (Harmit Suthar · 26084231044).

Figure 5.1 — DocuVault dashboard: KPI cards, search, category filter and the documents table.

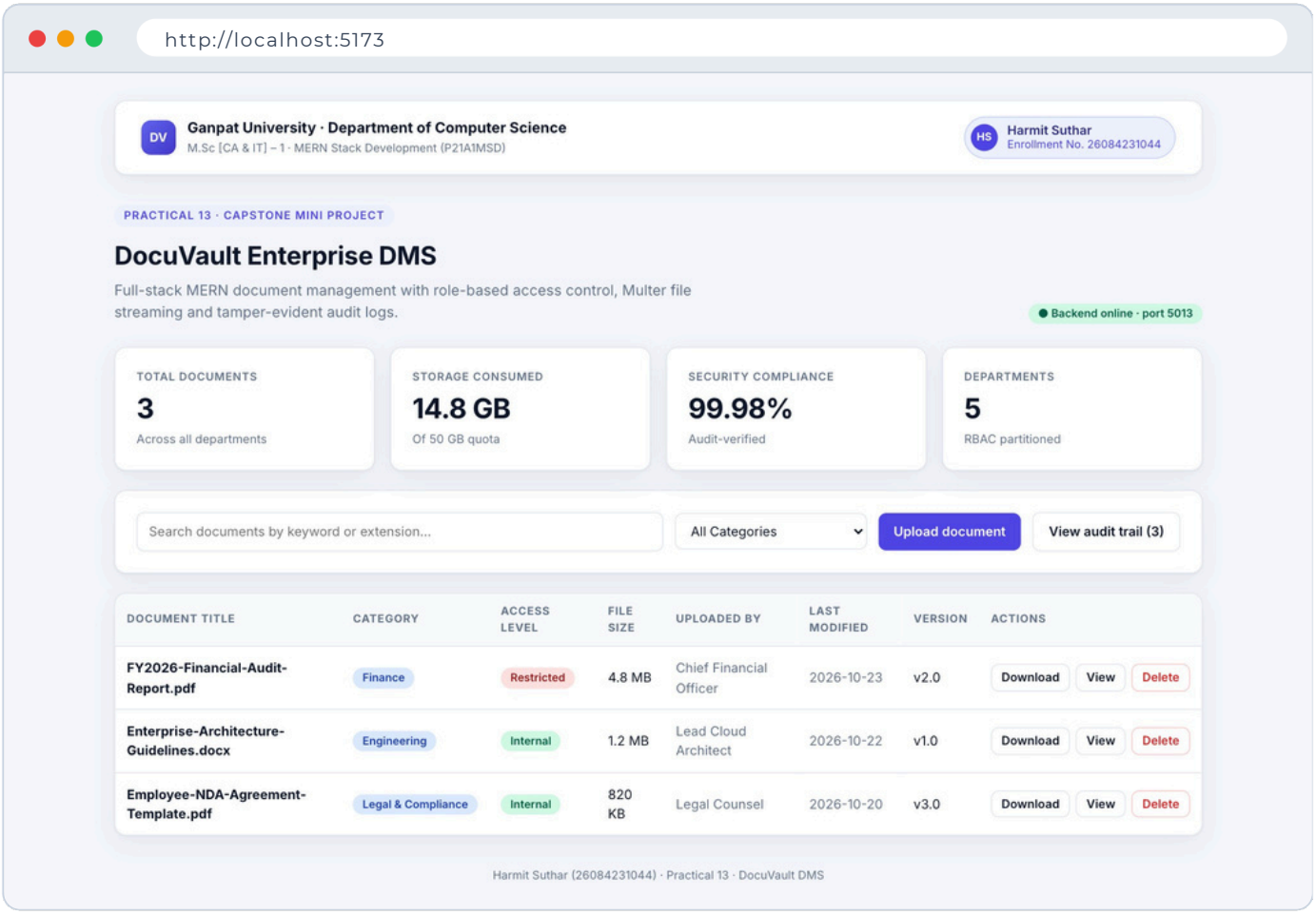


Figure 5.2 — Upload dialog capturing title, department and RBAC access level.

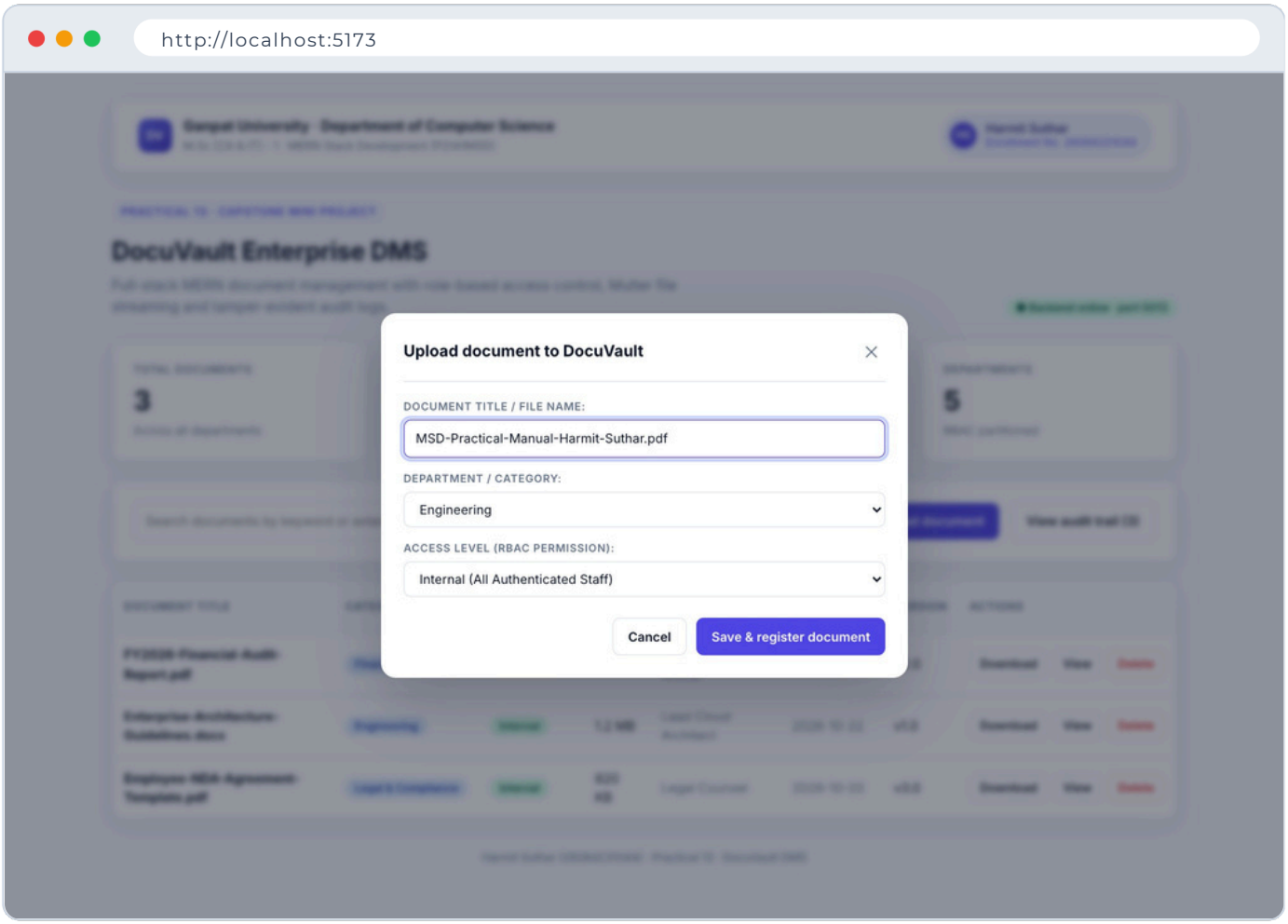


Figure 5.3 — The new document registered through the REST API and listed at the top of the table.

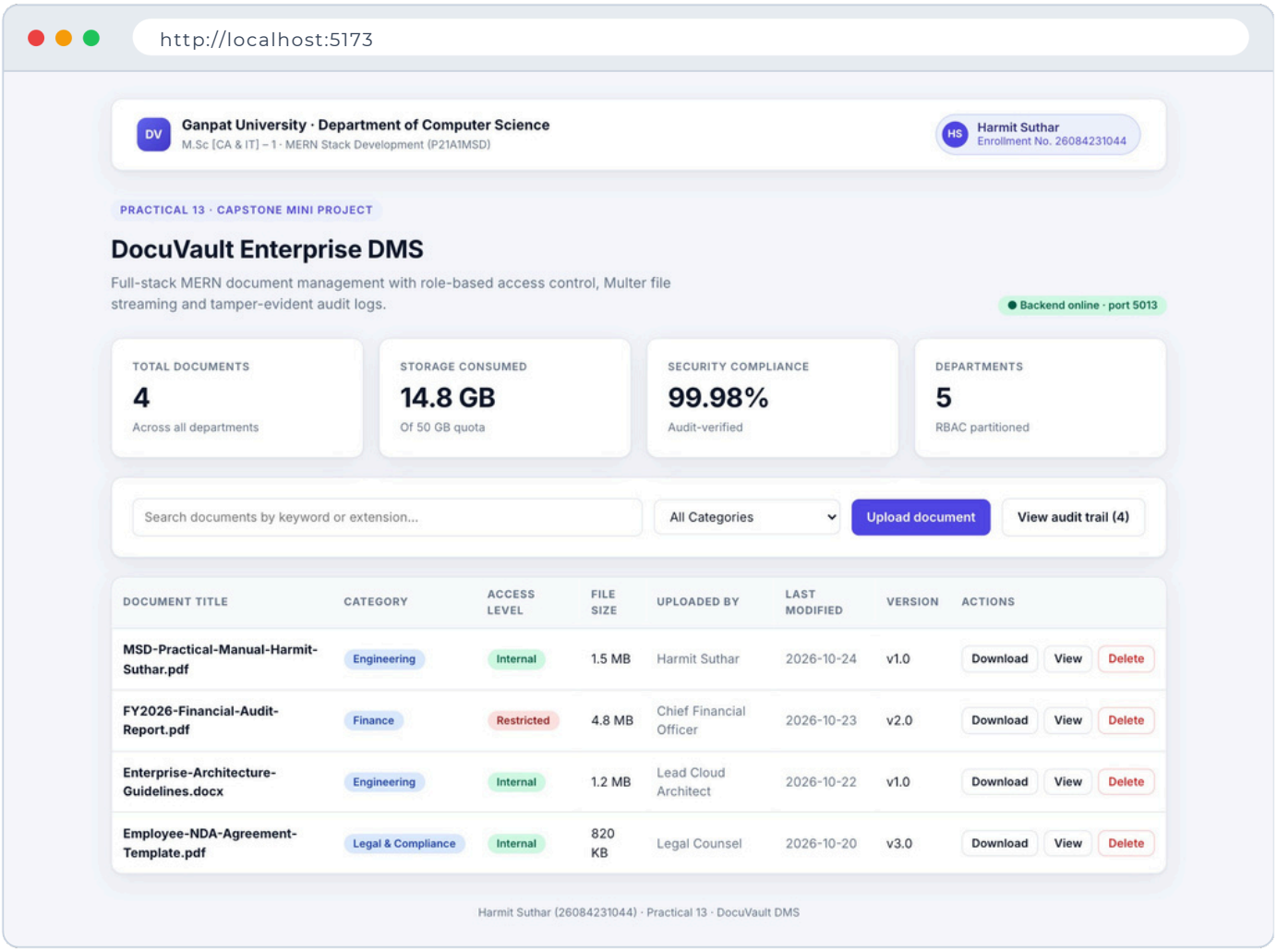
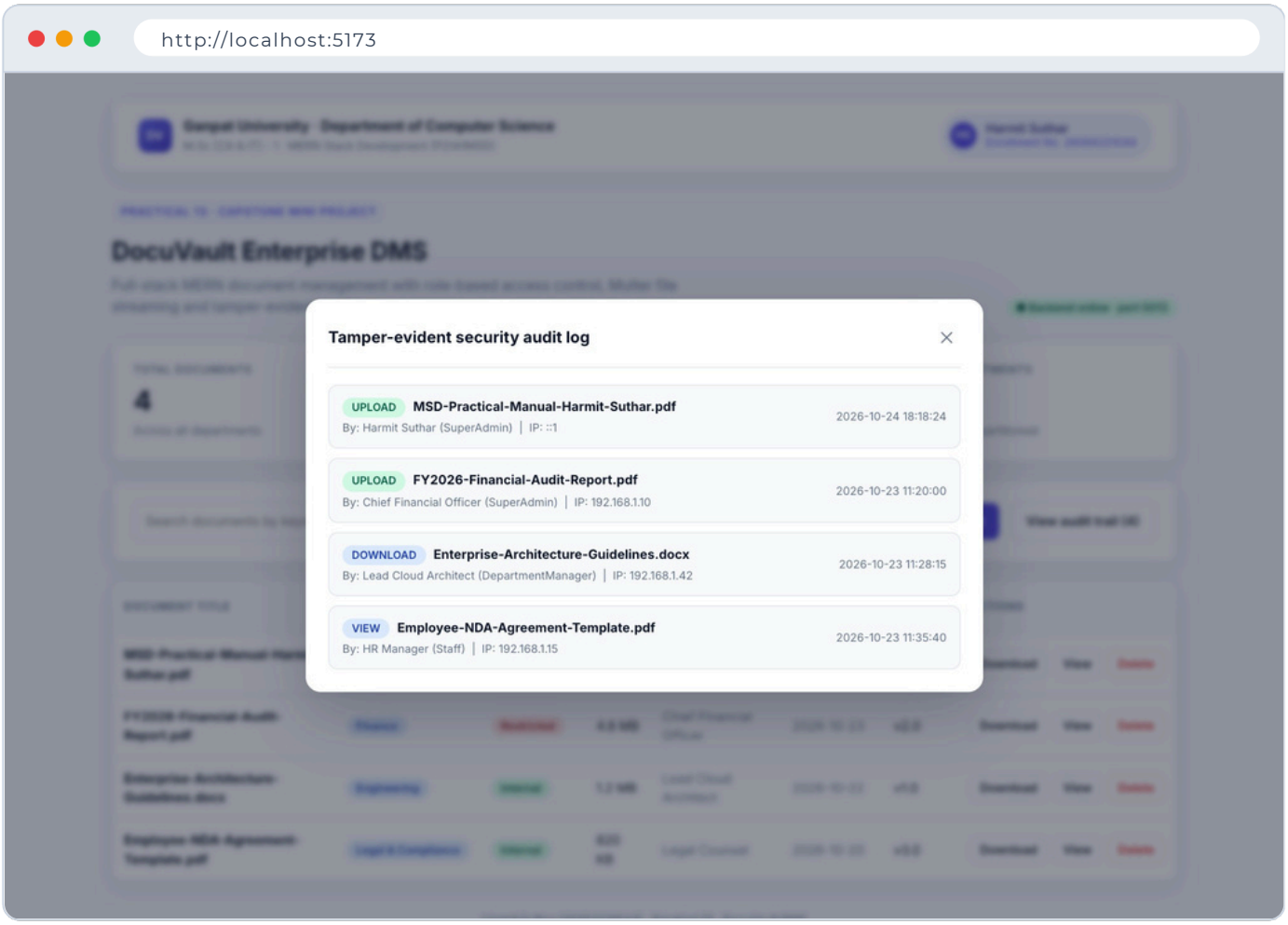


Figure 5.4 — Tamper-evident audit trail recording the UPLOAD action with user, role, IP and timestamp.



🖥️ DocuVault Backend Logs **Port 5013**

```
$ npm start (server)
> docuvault-server@1.0.0 start
> node server.js

================================================
🛡️ DocuVault DMS Backend running on http://localhost:5013
📂API Endpoints:
   - GET/POST /api/documents
   - DELETE   /api/documents/:id
   - GET      /api/audit-logs
================================================
```

🖥️ React Client Logs **Port 5173**

```
$ npm run dev (client)
> docuvault-client@1.0.0 dev
> vite


  VITEv5.4.21 ready in 147 ms

  ➜ Local:   http://localhost:5173/
  ➜ Network: use --host to expose
```

API Verification (curl)

```
$ curl -i http://localhost:5013/api/documents
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 856
Date: Sat, 24 Oct 2026 18:18:27 GMT

{
  "success": true,
  "count": 4,
  "data": [
    {
      "_id": "doc_1792865904032",
      "title":     "MSD-Practical-Manual-Harmit-Suthar.pdf",
      "category": "Engineering",
      "accessLevel": "Internal",
      "size": "1.5 MB",
      "uploadedBy": "Harmit Suthar",
      "updatedAt": "2026-10-24",
      "version": 1
    },
    {
      "_id": "doc_101",
      "title": "FY2026-Financial-Audit-Report.pdf",
      "category": "Finance",
      "accessLevel": "Restricted",
      "size": "4.8 MB",
      "uploadedBy": "Chief Financial Officer",
      "updatedAt": "2026-10-23",
      "version": 2
    },
    {
      "_id": "doc_102",
      "title":        "Enterprise-Architecture-Guidelines.docx",
      "category": "Engineering",
      "accessLevel": "Internal",
      "size": "1.2 MB",
      "uploadedBy": "Lead Cloud Architect",
      "updatedAt": "2026-10-22",
      "version": 1
    },
    {
      "_id": "doc_103",
      "title": "Employee-NDA-Agreement-Template.pdf",
      "category": "Legal & Compliance",
      "accessLevel": "Internal",
      "size": "820 KB",
      "uploadedBy": "Legal Counsel",
      "updatedAt": "2026-10-20",
      "version": 3
}
    }
  ]
```